1 Introduction

In this study, we numerically address an optimal control problem employing the Hamiltonian framework to derive the necessary conditions for optimality. The Hamiltonian formulation provides a systematic approach for deriving the governing equations.

Consider the following optimal control problem:

$$\begin{aligned}
\min \quad & J(v,u) = \int_0^T \left(\frac{v^2}{2} + \frac{\alpha}{2} u^2 \right) dt, \\
& \dot{v} = g + \frac{u}{m} - \frac{k}{m} v, \\
& v(0) = v_0.
\end{aligned} \tag{1}$$

The Hamiltonian, defined as

$$\mathcal{H}(v,u,\lambda) = \frac{v^2}{2} + \frac{\alpha}{2} u^2 + \lambda \left(g + \frac{u}{m} - \frac{k}{m} v \right),$$

is utilized to write the necessary optimality conditions for the above optimal control problem as follows:

$$\begin{aligned}
& \dot{v} = \frac{\partial \mathcal{H}}{\partial \lambda}(v,u,\lambda) = g + \frac{u}{m} - \frac{k}{m} v, \\
& \dot{\lambda} = -\frac{\partial \mathcal{H}}{\partial v}(v,u,\lambda) = -v + \frac{k}{m} \lambda, \\
& 0 = \frac{\partial \mathcal{H}}{\partial u}(v,u,\lambda) = \alpha u + \frac{k}{m} \lambda \\
& v(0) = v_0, \ \lambda(T) = 0.
\end{aligned} \tag{2}$$

To solve the resulting system of equations, we apply four numerical techniques, specifically the $SE1_1, SE1_2, SE2_1$ and $SE2_2$ methods. Each method begins with three initial points $\mathbf{z}_1 = (v_1(0), \lambda_1(0)), \mathbf{z}_2 = (v_2(0), \lambda_2(0))$ and $\mathbf{z}_3 = (v_3(0), \lambda_3(0))$, which form a triangle in the phase space. The area of this triangle is computed at the initial time $t = 0$, and the methods are applied, respectively, to determine the area of the triangle at the final time $t = T$. So, it allows us to identify which method effectively preserves the geometric properties of the triangle, particularly its area.

Preserving the triangle's area is important in this context, as it serves as a test for identifying symplectic algorithms. Symplectic algorithms are numerical methods designed to preserve the symplectic structure of Hamiltonian systems, which is directly related to the conservation of geometric properties like area in phase space.

By applying the $SE1_1, SE1_2, SE2_1$ and $SE2_2$ methods to (**??**), respectively, the resulting equations are as follows:

$$\begin{aligned}
SE1_1 &= \begin{cases} \frac{\Delta v}{\Delta t} = \frac{v_{n+1} - v_n}{\Delta t} = \frac{\partial \mathcal{H}}{\partial \lambda}(v_{n+1}, u_n, \lambda_n) \\ \frac{\Delta \lambda}{\Delta t} = \frac{\lambda_{n+1} - \lambda_n}{\Delta t} = -\frac{\partial \mathcal{H}}{\partial v}(v_{n+1}, u_n, \lambda_n) \\ 0 = \frac{\partial \mathcal{H}}{\partial u}(v_n, u_n, \lambda_n) = \alpha u_n + \frac{k}{m} \lambda_n \\ v(0) = v_0, \ \lambda(0) = \lambda_0. \end{cases} \\
SE1_2 &= \begin{cases} \frac{\Delta v}{\Delta t} = \frac{v_{n+1} - v_n}{\Delta t} = \frac{\partial \mathcal{H}}{\partial \lambda}(v_{n+1}, u_{n+1}, \lambda_n) \\ \frac{\Delta \lambda}{\Delta t} = \frac{\lambda_{n+1} - \lambda_n}{\Delta t} = -\frac{\partial \mathcal{H}}{\partial v}(v_{n+1}, u_{n+1}, \lambda_n) \\ 0 = \frac{\partial \mathcal{H}}{\partial u}(v_n, u_n, \lambda_n) = \alpha u_n + \frac{k}{m} \lambda_n \\ v(0) = v_0, \ \lambda(0) = \lambda_0. \end{cases} \\
SE2_1 &= \begin{cases} \frac{\Delta v}{\Delta t} = \frac{v_{n+1} - v_n}{\Delta t} = \frac{\partial \mathcal{H}}{\partial \lambda}(v_n, u_{n+1}, \lambda_{n+1}) \\ \frac{\Delta \lambda}{\Delta t} = \frac{\lambda_{n+1} - \lambda_n}{\Delta t} = -\frac{\partial \mathcal{H}}{\partial v}(v_n, u_{n+1}, \lambda_{n+1}) \\ 0 = \frac{\partial \mathcal{H}}{\partial u}(v_n, u_n, \lambda_n) = \alpha u_n + \frac{k}{m} \lambda_n \\ v(0) = v_0, \ \lambda(0) = \lambda_0. \end{cases} \\
SE2_2 &= \begin{cases} \frac{\Delta v}{\Delta t} = \frac{v_{n+1} - v_n}{\Delta t} = \frac{\partial \mathcal{H}}{\partial \lambda}(v_n, u_n, \lambda_{n+1}) \\ \frac{\Delta \lambda}{\Delta t} = \frac{\lambda_{n+1} - \lambda_n}{\Delta t} = -\frac{\partial \mathcal{H}}{\partial v}(v_n, u_n, \lambda_{n+1}) \\ 0 = \frac{\partial \mathcal{H}}{\partial u}(v_n, u_n, \lambda_n) = \alpha u_n + \frac{k}{m} \lambda_n \\ v(0) = v_0, \ \lambda(0) = \lambda_0. \end{cases}
\end{aligned} \tag{3}$$

Before presenting numerical results, we first demonstrate that the numerical methods $SE1_1$ and $SE2_2$ are symplectic integrators, which means that they exactly preserve the symplectic condition. In contrast, $SE1_2$ and $SE2_2$ are not symplectic integrators.

SE1₁ :

$$v_{n+1} = v_v + \Delta t g + \frac{\Delta t}{m} u_n - \frac{k\Delta t}{m} v_{n+1} \Rightarrow \frac{m + k\Delta t}{m} v_{n+1} = v_n + \Delta t g + \frac{\Delta t}{m} u_n \Rightarrow$$

$$\boxed{v_{n+1} = \frac{m}{m + k\Delta t} v_n + \frac{m\Delta t}{m + k\Delta t} g + \frac{-k\Delta t}{m\alpha(m + k\Delta t)} \lambda_n.} \tag{4}$$

$$\lambda_{n+1} = \lambda_n - \Delta t v_{n+1} + \frac{k\Delta t}{m} \lambda_n \Rightarrow \boxed{\lambda_{n+1} = \frac{m + k\Delta t}{m} \lambda_n - \Delta t v_{n+1}.} \tag{5}$$

Taking differentials of both sides of equations (**??**) and (**??**) and using the linearity of wedge product, we have

$$\begin{aligned}
&dv_{n+1} = \frac{m}{m + k\Delta t} dv_n + \frac{-k\Delta t}{m\alpha(m + k\Delta t)} d\lambda_n, \\
&d\lambda_{n+1} = \frac{m + k\Delta t}{m} d\lambda_n - \Delta t dv_{n+1}, \\
&d\lambda_{n+1} \wedge dv_{n+1} = \frac{m + k\Delta t}{m} d\lambda_n \wedge dv_{n+1} = \frac{m + k\Delta t}{m} d\lambda_n \wedge \left(\frac{m}{m + k\Delta t} dv_n + \frac{-k\Delta t}{m\alpha(m + k\Delta t)} d\lambda_n \right) \Rightarrow \\
&\boxed{d\lambda_{n+1} \wedge dv_{n+1} = \frac{m + k\Delta t}{m} d\lambda_n \wedge \frac{m}{m + k\Delta t} dv_n = d\lambda_n \wedge dv_n}.
\end{aligned}$$

SE2₁ :

$$v_{n+1} = \frac{m - k\Delta t}{m} v_n + \Delta t g + \frac{\Delta t}{m} u_{n+1} \Rightarrow$$

$$\boxed{v_{n+1} = \frac{m - k\Delta t}{m} v_n + \Delta t g + \frac{-k\Delta t}{m^2\alpha} \lambda_{n+1}.} \tag{6}$$

$$\boxed{\lambda_{n+1} = \frac{m}{m - \Delta t k} (-\Delta t) v_n + \frac{m}{m - \Delta t k} \lambda_n.} \tag{7}$$

Similar to $SE1_1$, by taking differentials of both sides of equations (**??**) and (**??**) and using wedge product, we have

$$\begin{aligned}
&dv_{n+1} = \frac{m - k\Delta t}{m} dv_n + \frac{-k\Delta t}{m^2\alpha} d\lambda_{n+1}, \\
&d\lambda_{n+1} = \frac{m}{m - \Delta t k} (-\Delta t) dv_n + \frac{m}{m - \Delta t k} d\lambda_n, \\
&\boxed{dv_{n+1} \wedge d\lambda_{n+1} = \frac{m - k\Delta t}{m} dv_n \wedge d\lambda_{n+1} = dv_n \wedge d\lambda_n.}
\end{aligned}$$

SE1₂ :

$$v_{n+1} = \frac{m}{m + k\Delta t} v_n + \frac{m}{m + k\Delta t} \Delta t g + \frac{\Delta t}{m + k\Delta t} u_{n+1} \Rightarrow$$

$$\boxed{v_{n+1} = \frac{m}{m + k\Delta t} v_n + \frac{m}{m + k\Delta t} \Delta t g + \frac{-k\Delta t}{m\alpha(m + k\Delta t)} \lambda_{n+1}.} \tag{8}$$

$$\boxed{\lambda_{n+1} = \frac{m + k\Delta t}{m} \lambda_n + (-\Delta t) v_{n+1}.} \tag{9}$$

Now by taking differentials of both sides of equations (**??**) and (**??**), we have

$$dv_{n+1} = \frac{m}{m + k\Delta t} dv_n + \frac{-k\Delta t}{m\alpha(m + k\Delta t)} d\lambda_{n+1}, \tag{10}$$

$$d\lambda_{n+1} = \frac{m + k\Delta t}{m} d\lambda_n + (-\Delta t) dv_{n+1}. \tag{11}$$

From equations (**??**) and (**??**), dv_{n+1} and $d\lambda_{n+1}$ can be expressed in terms of dv_n and $d\lambda_n$ so we have:

$$dv_{n+1} = \frac{m^2\alpha}{m\alpha(m+k\Delta t) - k(\Delta t)^2} dv_n + \frac{-k\Delta t(m+k\Delta t)}{m^2\alpha(m+k\Delta t) - km(\Delta t)^2} d\lambda_n \tag{12}$$

$$d\lambda_{n+1} = \frac{\alpha(m+k\Delta t)^2}{m\alpha(m+k\Delta t) - k(\Delta t)^2} d\lambda_n + \frac{-m^2\alpha\Delta t}{m\alpha(m+k\Delta t) - k(\Delta t)^2} dv_n \tag{13}$$

If we take the wedge product of (**??**) with (**??**), we get

$$\boxed{dv_{n+1} \wedge d\lambda_{n+1} = \left(1 + \frac{k(\Delta t)^2}{m\alpha(m+k\Delta t) - k(\Delta t)^2}\right) dv_n \wedge d\lambda_n.} \tag{14}$$

$\mathbf{SE2_2}$:

$$v_{n+1} = g\Delta t + (\frac{m - k\Delta t}{m})v_n + (\frac{-k\Delta t}{m^2\alpha})\lambda_n, \tag{15}$$

$$\lambda_{n+1} = (\frac{-m\Delta t}{m - k\Delta t})v_n + (\frac{m}{m - k\Delta t})\lambda_n. \tag{16}$$

Taking differentials of both sides of equations (**??**) and (**??**) and using wedge product, we have

$$dv_{n+1} \wedge d\lambda_{n+1} = \left(1 - \frac{k(\Delta t)^2}{m\alpha(m - k\Delta t)}\right) dv_n \wedge d\lambda_n. \tag{17}$$

1.1 Linear optimal control problem

Let us consider the following linear optimal control problem:

$$\begin{aligned} J(\boldsymbol{v},\boldsymbol{u}) &= \frac{1}{2}\int_a^b \boldsymbol{v}(t)^T\boldsymbol{Q}\boldsymbol{v}(t) + \boldsymbol{u}(t)^T\boldsymbol{R}\boldsymbol{u}(t)\, dt \\ \dot{\boldsymbol{v}}(t) &= \boldsymbol{A}\boldsymbol{v}(t) + \boldsymbol{B}\boldsymbol{u}(t), \\ \boldsymbol{v}(a) &= \boldsymbol{v}_0, \end{aligned} \tag{18}$$

where $\boldsymbol{Q}$ is a real symmetric positive semi-definite matrix and $\boldsymbol{R}$ is a real symmetric positive definite matrix.

The Hamiltonian is

$$\mathcal{H}(\boldsymbol{v},\boldsymbol{u},\boldsymbol{\lambda}) = \frac{1}{2}\boldsymbol{v}(t)^T\boldsymbol{Q}\boldsymbol{v}(t) + \frac{1}{2}\boldsymbol{u}(t)^T\boldsymbol{R}\boldsymbol{u}(t) + \boldsymbol{\lambda}^T\boldsymbol{A}\boldsymbol{v}(t) + \boldsymbol{\lambda}^T\boldsymbol{B}\boldsymbol{u}(t), \tag{19}$$

and necessary conditions for optimality are

$$\dot{\boldsymbol{v}}(t) = \frac{\partial\mathcal{H}(\boldsymbol{v},\boldsymbol{u},\boldsymbol{\lambda})}{\boldsymbol{\partial\lambda}} = \boldsymbol{A}\boldsymbol{v}(t) + \boldsymbol{B}\boldsymbol{u}(t), \tag{20}$$

$$\dot{\boldsymbol{\lambda}}(t) = -\frac{\partial\mathcal{H}(\boldsymbol{v},\boldsymbol{u},\boldsymbol{\lambda})}{\boldsymbol{\partial v}} = -\boldsymbol{Q}\boldsymbol{v}(t) - \mathbf{A}^T\boldsymbol{\lambda}(t), \tag{21}$$

$$\mathbf{0} = \frac{\partial\mathcal{H}(\boldsymbol{v},\boldsymbol{u},\boldsymbol{\lambda})}{\boldsymbol{\partial u}} = \boldsymbol{R}\boldsymbol{u}(t) + \boldsymbol{B}^T\boldsymbol{\lambda}(t). \tag{22}$$

Equation (**??**) can be solved for $\mathbf{u}$ to give

$$\boldsymbol{u}(t) = -\boldsymbol{R}^{-1}\boldsymbol{B}^T\boldsymbol{\lambda}(t), \tag{23}$$

the existence of $\boldsymbol{R}^{-1}$ is assured, since $\boldsymbol{R}$ is a positive definite matrix.

Let us apply $SE11$ and $SE21$ schemes to equations (**??**)-(**??**), respectively. we want to show that both are symplectic integrators, that is, they preserve the symplectic condition, i.e. $d\boldsymbol{v}_{n+1} \wedge d\boldsymbol{\lambda}_{n+1} = d\boldsymbol{v}_n \wedge d\boldsymbol{\lambda}_n$ or $det(\frac{\partial(\boldsymbol{v}_{n+1},\boldsymbol{\lambda}_{n+1})}{\partial(\boldsymbol{v}_n,\boldsymbol{\lambda}_n)}) = 1$.

$\mathbf{SE1_1}$:

$$\begin{aligned} &\frac{\boldsymbol{v}_{n+1} - \boldsymbol{v}_n}{\Delta t} = \boldsymbol{A}\boldsymbol{v}_{n+1} + \boldsymbol{B}\boldsymbol{u}_n = \boldsymbol{A}\boldsymbol{v}_{n+1} - \boldsymbol{B}\boldsymbol{R}^{-1}\boldsymbol{B}^T\boldsymbol{\lambda}_n \Rightarrow \\ &\boldsymbol{v}_{n+1} = (\boldsymbol{I} - \Delta t\boldsymbol{A})^{-1}\boldsymbol{v}_n - \Delta t(\boldsymbol{I} - \Delta t\boldsymbol{A})^{-1}\boldsymbol{B}\boldsymbol{R}^{-1}\boldsymbol{B}^T\boldsymbol{\lambda}_n, \end{aligned} \tag{24}$$

$$\begin{aligned} &\frac{\boldsymbol{\lambda}_{n+1} - \boldsymbol{\lambda}_n}{\Delta t} = -\mathbf{Q}\mathbf{v}_{n+1} - \mathbf{A}^T\boldsymbol{\lambda}_n \Rightarrow \\ &\boldsymbol{\lambda}_{n+1} = -\Delta t\boldsymbol{Q}\boldsymbol{v}_{n+1} + (\boldsymbol{I} - \Delta t\boldsymbol{A}^T)\boldsymbol{\lambda}_n, \end{aligned} \tag{25}$$

Taking differentials of (??) and (??), respectively, we obtain the following implicit system of equations:

$$d\boldsymbol{v}_{n+1} = (\boldsymbol{I} - \Delta t\boldsymbol{A})^{-1} d\boldsymbol{v}_n - \Delta t(\boldsymbol{I} - \Delta t\boldsymbol{A})^{-1}\boldsymbol{B}\boldsymbol{R}^{-1}\boldsymbol{B}^T d\boldsymbol{\lambda}_n, \tag{26}$$

$$d\boldsymbol{\lambda}_{n+1} = -\Delta t\boldsymbol{Q} d\boldsymbol{v}_{n+1} + (\boldsymbol{I} - \Delta t\boldsymbol{A}^T) d\boldsymbol{\lambda}_n. \tag{27}$$

By taking the wedge product of (??) with (??) from the right, we have

$$\begin{aligned} d\boldsymbol{v}_{n+1} \wedge d\boldsymbol{\lambda}_{n+1} =& (\boldsymbol{I} - \Delta t\boldsymbol{A})^{-1} d\boldsymbol{v}_n \wedge (-\Delta t\boldsymbol{Q} d\boldsymbol{v}_{n+1}) + (\boldsymbol{I} - \Delta t\boldsymbol{A})^{-1} d\boldsymbol{v}_n \wedge (\boldsymbol{I} - \Delta t\boldsymbol{A}^T) d\boldsymbol{\lambda}_n + \\ & \left(-\Delta t(\boldsymbol{I} - \Delta t\boldsymbol{A})^{-1}\boldsymbol{B}\boldsymbol{R}^{-1}\boldsymbol{B}^T d\boldsymbol{\lambda}_n\right) \wedge (-\Delta t\boldsymbol{Q} d\boldsymbol{v}_{n+1}). \end{aligned} \tag{28}$$

Upon plugging (??) into (??) and using $d\boldsymbol{a} \wedge d(\boldsymbol{M} d\boldsymbol{b}) = (\boldsymbol{M}^T d\boldsymbol{a}) \wedge d\boldsymbol{b}$, we obtain the following equality

$$\begin{aligned} d\boldsymbol{v}_{n+1} \wedge d\boldsymbol{\lambda}_{n+1} =& (\boldsymbol{I} - \Delta t\boldsymbol{A})^{-1} d\boldsymbol{v}_n \wedge \Delta t^2 \boldsymbol{Q}(\boldsymbol{I} - \Delta t\boldsymbol{A})^{-1}\boldsymbol{B}\boldsymbol{R}^{-1}\boldsymbol{B}^T d\boldsymbol{\lambda}_n + d\boldsymbol{v}_n \wedge d\boldsymbol{\lambda}_n + \\ & - \Delta t(\boldsymbol{I} - \Delta t\boldsymbol{A})^{-1}\boldsymbol{B}\boldsymbol{R}^{-1}\boldsymbol{B}^T d\boldsymbol{\lambda}_n \wedge -\Delta t\boldsymbol{Q}(\boldsymbol{I} - \Delta t\boldsymbol{A})^{-1} d\boldsymbol{v}_n \\ & = d\boldsymbol{v}_n \wedge d\boldsymbol{\lambda}_n - \Delta t^2 (\boldsymbol{I} - \Delta t A)^{-T}\boldsymbol{Q}(\boldsymbol{I} - \Delta t\boldsymbol{A})^{-1}\boldsymbol{B}\boldsymbol{R}^{-1}\boldsymbol{B}^T d\boldsymbol{\lambda}_n \wedge d\boldsymbol{v}_n + \\ & \Delta t^2 (\boldsymbol{I} - \Delta t A)^{-T}\boldsymbol{Q}^T(\boldsymbol{I} - \Delta t\boldsymbol{A})^{-1}\boldsymbol{B}\boldsymbol{R}^{-1}\boldsymbol{B}^T d\boldsymbol{\lambda}_n \wedge d\boldsymbol{v}_n \\ & = d\boldsymbol{v}_n \wedge d\boldsymbol{\lambda}_n. \end{aligned} \tag{29}$$

SE2$_1$:

$$\boldsymbol{v}_{n+1} = (\boldsymbol{I} + \Delta t\boldsymbol{A})\boldsymbol{v}_n - \Delta t\boldsymbol{B}\boldsymbol{R}^{-1}\boldsymbol{B}^T\boldsymbol{\lambda}_{n+1}, \tag{30}$$

$$\boldsymbol{\lambda}_{n+1} = -\Delta t(\boldsymbol{I} + \Delta t A^T)^{-1}\boldsymbol{Q}\boldsymbol{v}_n + (\boldsymbol{I} + \Delta t\boldsymbol{A}^T)^{-1}\boldsymbol{\lambda}_n, \tag{31}$$

$$d\boldsymbol{v}_{n+1} = (\boldsymbol{I} + \Delta t\boldsymbol{A}) d\boldsymbol{v}_n - \Delta t\boldsymbol{B}\boldsymbol{R}^{-1}\boldsymbol{B}^T d\boldsymbol{\lambda}_{n+1}, \tag{32}$$

$$d\boldsymbol{\lambda}_{n+1} = -\Delta t(\boldsymbol{I} + \Delta t A^T)^{-1}\boldsymbol{Q} d\boldsymbol{v}_n + (\boldsymbol{I} + \Delta t\boldsymbol{A}^T)^{-1} d\boldsymbol{\lambda}_n. \tag{33}$$

By taking the wedge product of (??) with (??) we obtain

$$\begin{aligned} d\boldsymbol{\lambda}_{n+1} \wedge d\boldsymbol{v}_{n+1} =& \left(-\Delta t(\boldsymbol{I} + \Delta t A^T)^{-1}\boldsymbol{Q} d\boldsymbol{v}_n\right) \wedge \left(-\Delta t\boldsymbol{B}\boldsymbol{R}^{-1}\boldsymbol{B}^T d\boldsymbol{\lambda}_{n+1}\right) + \\ & (\boldsymbol{I} + \Delta t\boldsymbol{A}^T)^{-1} d\boldsymbol{\lambda}_n \wedge (\boldsymbol{I} + \Delta t\boldsymbol{A}) d\boldsymbol{v}_n + \\ & (\boldsymbol{I} + \Delta t\boldsymbol{A}^T)^{-1} d\boldsymbol{\lambda}_n \wedge \left(-\Delta t\boldsymbol{B}\boldsymbol{R}^{-1}\boldsymbol{B}^T d\boldsymbol{\lambda}_{n+1}\right). \end{aligned} \tag{34}$$

Upon plugging (??) into (??), we obtain the desired equality (??).

$$\begin{aligned} d\boldsymbol{\lambda}_{n+1} \wedge d\boldsymbol{v}_{n+1} =& d\boldsymbol{\lambda}_n \wedge d\boldsymbol{v}_n + \\ & - \Delta t^2 \left((\boldsymbol{I} + \Delta t\boldsymbol{A}^T)^{-1}\boldsymbol{Q}\right)^T \boldsymbol{B}\boldsymbol{R}^{-1}\boldsymbol{B}^T(\boldsymbol{I} + \Delta t\boldsymbol{A}^T)^{-1} d\boldsymbol{\lambda}_n \wedge d\boldsymbol{v}_n + \\ & \Delta t^2 \left(\boldsymbol{B}\boldsymbol{R}^{-1}\boldsymbol{B}^T(\boldsymbol{I} + \Delta t\boldsymbol{A}^T)^{-1}\boldsymbol{Q}\right)^T (\boldsymbol{I} + \Delta t\boldsymbol{A}^T)^{-1} d\boldsymbol{\lambda}_n \wedge d\boldsymbol{v}_n = \\ & d\boldsymbol{\lambda}_n \wedge d\boldsymbol{v}_n. \end{aligned} \tag{35}$$

2 Numerical Results (linear case)

In this section, we present numerical results (summarized in Table ??) that compare the initial and final areas computed by each method. Figures ??-?? depict the area of the triangles calculated in discrete time steps ($t = i\Delta t,\ \Delta t = 0.1$) with different α over the interval $[0, 10]$. This stepwise analysis highlights how the area varies in the time range for each approach.

3 Optimal control problem

Consider the following basic optimal control problem,

$$\begin{aligned} & \min \int_a^b j(\boldsymbol{y}, \boldsymbol{u})\, dt \\ & \dot{\boldsymbol{y}} = \boldsymbol{f}(\boldsymbol{y}, \boldsymbol{u}) \\ & \boldsymbol{y}(0) = \boldsymbol{y}_0 \end{aligned} \tag{36}$$

Method	α	A_0	A_T
$SE1_1$	10	0.1000	0.1000
	7	0.1000	0.1000
	5	0.1000	0.1000
	2	0.1000	0.1000
	1	0.1000	0.1000
	0.5	0.1000	0.0991
$SE1_2$	10	0.1000	0.1095
	7	0.1000	0.1139
	5	0.1000	0.1200
	2	0.1000	0.1577
	1	0.1000	0.2493
	0.5	0.1000	0.4609
$SE2_1$	10	0.1000	0.1000
	7	0.1000	0.1000
	5	0.1000	0.1000
	2	0.1000	0.1000
	1	0.1000	0.1001
	0.5	0.1000	0.2500
$SE2_2$	10	0.1000	0.0895
	7	0.1000	0.0853
	5	0.1000	0.0801
	2	0.1000	0.0573
	1	0.1000	0.0327
	0.5	0.1000	0.0117

Table 1: Application of methods $SE1_1, SE1_2, SE2_1$ and $SE2_2$ to system (**??**) on the interval $[0,10]$ with three initial points $\mathbf{z}_1 = (10, 0.1)^T, \mathbf{z}_2 = (11, 0.1)^T$ and $\mathbf{z}_3 = (10, -0.1)^T$, in addition, $k = 1, m = 1, \Delta t = 0.1, T = 10$. A_0 and A_T denote the areas of the triangle at time $t = 0$ and $t = T = 10$, respectively.

The necessary conditions from the Hamiltonian H, which is defined as follows are generated

$$H(\boldsymbol{y}, \boldsymbol{u}, \boldsymbol{\lambda}) = j(\boldsymbol{y}, \boldsymbol{u}) + \boldsymbol{\lambda}^T \boldsymbol{f}(\boldsymbol{y}, \boldsymbol{u}) \tag{37}$$

The necessary optimality conditions can be written in term of the Hamiltonian:

$$\dot{\boldsymbol{y}} = H_{\boldsymbol{\lambda}} = \frac{\partial H}{\partial \boldsymbol{\lambda}}(\boldsymbol{y}, \boldsymbol{u}, \boldsymbol{\lambda}) \tag{38}$$

$$\dot{\boldsymbol{\lambda}} = -H_{\boldsymbol{y}} = -\frac{\partial H}{\partial \boldsymbol{y}}(\boldsymbol{y}, \boldsymbol{u}, \boldsymbol{\lambda}) \tag{39}$$

$$\mathbf{0} = H_{\boldsymbol{u}} = \frac{\partial H}{\partial \boldsymbol{u}}(\boldsymbol{y}, \boldsymbol{u}, \boldsymbol{\lambda}). \tag{40}$$

We want to show that $SE11$ and $SE21$ schemes are simplistic integrators, that is $d\boldsymbol{y}_{n+1} \wedge d\boldsymbol{\lambda}_{n+1} = d\boldsymbol{y}_n \wedge d\boldsymbol{\lambda}_n$ if $H_{\boldsymbol{yu}} = \frac{\partial^2 H}{\partial \boldsymbol{u} \partial \boldsymbol{y}} = \mathbf{0}$.

$\mathbf{SE1_1}$: Let $\boldsymbol{u}$ be a function of $\boldsymbol{y}$ and $\boldsymbol{\lambda}$, i.e. $\boldsymbol{u} = \boldsymbol{g}(\boldsymbol{y}, \boldsymbol{\lambda})$ so from (**??**) we have

$$H_{\boldsymbol{uy}} + H_{\boldsymbol{uu}} \boldsymbol{g}_{\boldsymbol{y}} = 0 \Rightarrow \boldsymbol{g}_{\boldsymbol{y}} = -H_{\boldsymbol{uu}}^{-1} H_{\boldsymbol{uy}} \tag{41}$$

$$H_{\boldsymbol{u\lambda}} + H_{\boldsymbol{uu}} \boldsymbol{g}_{\boldsymbol{\lambda}} = 0 \Rightarrow \boldsymbol{g}_{\boldsymbol{\lambda}} = -H_{\boldsymbol{uu}}^{-1} H_{\boldsymbol{u\lambda}}, \tag{42}$$

$$d\boldsymbol{u} = -H_{\boldsymbol{uu}}^{-1} H_{\boldsymbol{uy}} d\boldsymbol{y} - H_{\boldsymbol{uu}}^{-1} H_{\boldsymbol{u\lambda}} d\boldsymbol{\lambda} \tag{43}$$

where $\boldsymbol{g}_{\boldsymbol{y}} = \frac{\partial \boldsymbol{g}}{\partial \boldsymbol{y}}$ and $\boldsymbol{g}_{\boldsymbol{\lambda}} = \frac{\partial \boldsymbol{g}}{\partial \boldsymbol{\lambda}}$, respectively.

$$\boldsymbol{y}_{n+1} = \boldsymbol{y}_n + \Delta t H_{\boldsymbol{\lambda}}(\boldsymbol{y}_{n+1}, \boldsymbol{u}_n, \boldsymbol{\lambda}_n). \tag{44}$$

Taking differential of (**??**) and plugging (**??**)-(**??**) into (**??**) we obtain the equation:

$$d\boldsymbol{y}_{n+1} = d\boldsymbol{y}_n + \Delta t (H_{\boldsymbol{\lambda y}} d\boldsymbol{y}_{n+1} + H_{\boldsymbol{\lambda u}} d\boldsymbol{u}_n + H_{\boldsymbol{\lambda\lambda}} d\boldsymbol{\lambda}_n), \tag{45}$$

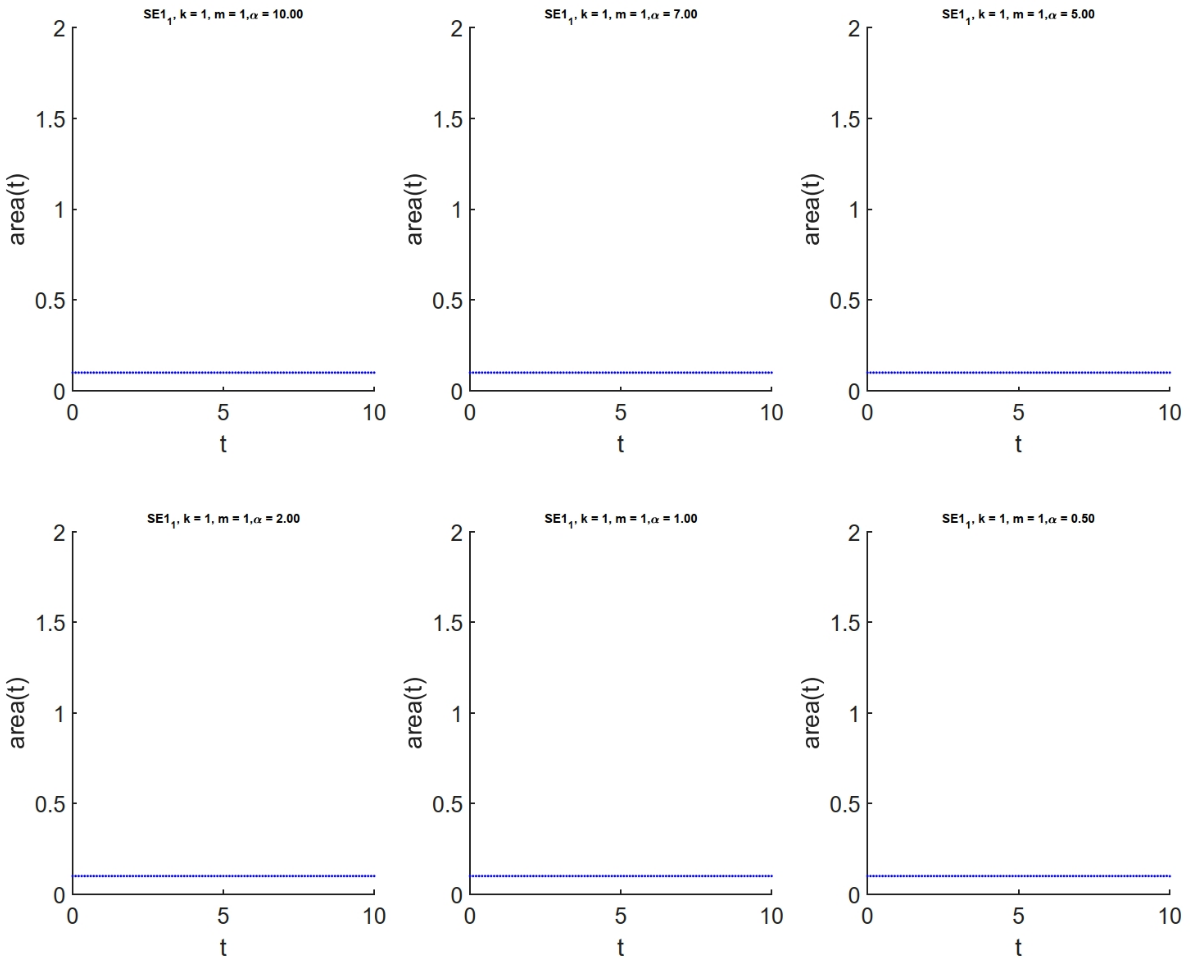


Figure 1: $SE1_1$ method. Triangle area vs. time, computed at discrete steps, $t = i\Delta t$, $\Delta t = 0.1$, over interval $[0, 10]$.

$$d\boldsymbol{y}_{n+1} = d\boldsymbol{y}_n + \Delta t H_{\boldsymbol{\lambda y}} d\boldsymbol{y}_{n+1} - \Delta t H_{\boldsymbol{\lambda u}} H_{\boldsymbol{uu}}^{-1} H_{\boldsymbol{uy}} d\boldsymbol{y}_n - \Delta t H_{\boldsymbol{\lambda u}} H_{\boldsymbol{uu}}^{-1} H_{\boldsymbol{u\lambda}} d\boldsymbol{\lambda}_n + \Delta t H_{\boldsymbol{\lambda\lambda}} d\boldsymbol{\lambda}_n \tag{46}$$

$$\boxed{d\boldsymbol{y}_{n+1} = (\boldsymbol{I} - \Delta t H_{\boldsymbol{\lambda y}})^{-1}(\boldsymbol{I} - \Delta t \boldsymbol{A}) d\boldsymbol{y}_n + \Delta t (\boldsymbol{I} - \Delta t H_{\boldsymbol{\lambda y}})^{-1}(-\boldsymbol{B} + H_{\boldsymbol{\lambda\lambda}}) d\boldsymbol{\lambda}_n}, \tag{47}$$

where $\boldsymbol{A} = H_{\boldsymbol{\lambda u}} H_{\boldsymbol{uu}}^{-1} H_{\boldsymbol{uy}}$ and $\boldsymbol{B} = H_{\boldsymbol{\lambda u}} H_{\boldsymbol{uu}}^{-1} H_{\boldsymbol{u\lambda}}$.

$$\boldsymbol{\lambda}_{n+1} = \boldsymbol{\lambda}_n - \Delta t H_{\boldsymbol{y}}(\boldsymbol{y}_{n+1}, \boldsymbol{u}_n, \boldsymbol{\lambda}_n) \tag{48}$$

$$d\boldsymbol{\lambda}_{n+1} = d\boldsymbol{\lambda}_n - \Delta t (H_{\boldsymbol{yy}} d\boldsymbol{y}_{n+1} + H_{\boldsymbol{yu}} d\boldsymbol{u}_n + H_{\boldsymbol{y\lambda}} d\boldsymbol{\lambda}_n) \tag{49}$$

Upon plugging (**??**)-(**??**) into (**??**), we have

$$d\boldsymbol{\lambda}_{n+1} = d\boldsymbol{\lambda}_n - \Delta t H_{\boldsymbol{yy}} d\boldsymbol{y}_{n+1} + \Delta t H_{\boldsymbol{yu}} H_{\boldsymbol{uu}}^{-1} H_{\boldsymbol{uy}} d\boldsymbol{y}_n + \Delta t H_{\boldsymbol{yu}} H_{\boldsymbol{uu}}^{-1} H_{\boldsymbol{u\lambda}} d\boldsymbol{\lambda}_n - \Delta t H_{\boldsymbol{y\lambda}} d\boldsymbol{\lambda}_n \tag{50}$$

$$d\boldsymbol{\lambda}_{n+1} = (I - \Delta t H_{\boldsymbol{y\lambda}} + \Delta t \boldsymbol{A}^T) d\boldsymbol{\lambda}_n + \Delta t \boldsymbol{D} d\boldsymbol{y}_n - \Delta t H_{\boldsymbol{yy}} d\boldsymbol{y}_{n+1} \tag{51}$$

where $D = H_{\boldsymbol{yu}} H_{\boldsymbol{uu}}^{-1} H_{\boldsymbol{uy}}$.

By taking the wedge product of (**??**) with $d\boldsymbol{y}_{n+1}$ we obtain

$$\begin{aligned} d\boldsymbol{\lambda}_{n+1} \wedge d\boldsymbol{y}_{n+1} = &(\boldsymbol{I} - \Delta t H_{\boldsymbol{y\lambda}} + \Delta t \boldsymbol{A}^T) d\boldsymbol{\lambda}_n \wedge (\boldsymbol{I} - \Delta t H_{\boldsymbol{\lambda y}})^{-1}(\boldsymbol{I} - \Delta t \boldsymbol{A}) d\boldsymbol{y}_n + \\ &\Delta t (\boldsymbol{I} - \Delta t H_{\boldsymbol{\lambda y}})^{-1}(\boldsymbol{B} - H_{\boldsymbol{\lambda\lambda}}) d\boldsymbol{\lambda}_n \wedge \Delta t \boldsymbol{D} d\boldsymbol{y}_n. \end{aligned}$$

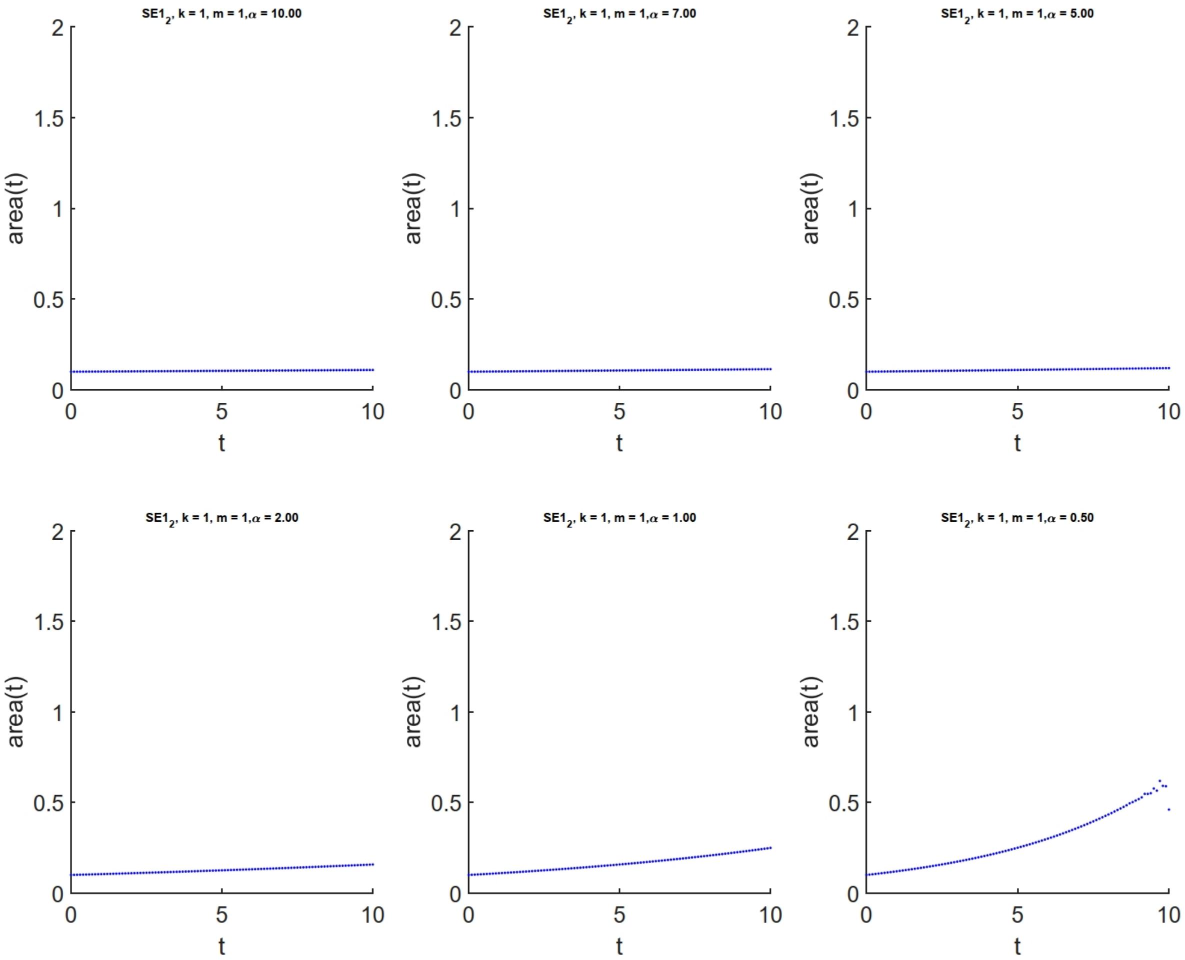


Figure 2: $SE1_2$ method. Triangle area vs. time, computed at discrete steps, $t = i\Delta t$, $\Delta t = 0.1$, over interval $[0, 10]$.

If $H_{\boldsymbol{yu}} = \mathbf{0}$, then $\boldsymbol{A} = \mathbf{0}$, $\boldsymbol{A}^T = \mathbf{0}$, $\boldsymbol{D} = \mathbf{0}$ so we have

$$
\begin{aligned}
d\boldsymbol{\lambda}_{n+1} \wedge d\boldsymbol{y}_{n+1} =& (\boldsymbol{I} - \Delta t H_{\boldsymbol{y\lambda}}) d\boldsymbol{\lambda}_n \wedge (\boldsymbol{I} - \Delta t H_{\boldsymbol{\lambda y}})^{-1} d\boldsymbol{y}_n = && (52)\\
& (\boldsymbol{I} - \Delta t H_{\boldsymbol{\lambda y}})^{-T} (\boldsymbol{I} - \Delta t H_{\boldsymbol{y\lambda}}) d\boldsymbol{\lambda}_n \wedge d\boldsymbol{y}_n = && (53)\\
& (\underbrace{\boldsymbol{I} - \Delta t H_{\boldsymbol{\lambda y}}^T}_{H_{\boldsymbol{y\lambda}} = H_{\boldsymbol{\lambda y}}^T})^{-1} (\boldsymbol{I} - \Delta t H_{\boldsymbol{y\lambda}}) d\boldsymbol{\lambda}_n \wedge d\boldsymbol{y}_n = && (54)\\
& (\boldsymbol{I} - \Delta t H_{\boldsymbol{y\lambda}})^{-1} (\boldsymbol{I} - \Delta t H_{\boldsymbol{y\lambda}}) d\boldsymbol{\lambda}_n \wedge d\boldsymbol{y}_n && (55)\\
& d\boldsymbol{\lambda}_n \wedge d\boldsymbol{y}_n. && (56)
\end{aligned}
$$

SE2$_\mathbf{1}$:

$$
\boldsymbol{y}_{n+1} = \boldsymbol{y}_n + \Delta t H_{\boldsymbol{\lambda}}(\boldsymbol{y}_n, \boldsymbol{u}_{n+1}, \boldsymbol{\lambda}_{n+1}) \tag{57}
$$

$$
\begin{aligned}
d\boldsymbol{y}_{n+1} =& d\boldsymbol{y}_n + \Delta t H_{\boldsymbol{\lambda y}} d\boldsymbol{y}_n - \Delta t H_{\boldsymbol{\lambda u}} H_{\boldsymbol{uu}}^{-1} H_{\boldsymbol{uy}} d\boldsymbol{y}_{n+1} - \\
& \Delta t H_{\boldsymbol{\lambda u}} H_{\boldsymbol{uu}}^{-1} H_{\boldsymbol{u\lambda}} d\boldsymbol{\lambda}_{n+1} + \Delta t H_{\boldsymbol{\lambda\lambda}} d\boldsymbol{\lambda}_{n+1} \\
=& (\boldsymbol{I} + \Delta t H_{\boldsymbol{\lambda u}} H_{\boldsymbol{uu}}^{-1} H_{\boldsymbol{uy}})^{-1} (\boldsymbol{I} + \Delta t H_{\lambda y}) d\boldsymbol{y}_n + \\
& (\boldsymbol{I} + \Delta t H_{\boldsymbol{\lambda u}} H_{\boldsymbol{uu}}^{-1} H_{\boldsymbol{uy}})^{-1} (-\Delta t H_{\boldsymbol{\lambda u}} H_{\boldsymbol{uu}} H_{\boldsymbol{u\lambda}} + \Delta t H_{\lambda\lambda}) d\boldsymbol{\lambda}_{n+1},
\end{aligned}
\tag{58}
$$

$$
\boldsymbol{\lambda}_{n+1} = \boldsymbol{\lambda}_n - \Delta t H_{\boldsymbol{y}}(\boldsymbol{y}_n, \boldsymbol{u}_{n+1}, \boldsymbol{\lambda}_{n+1}) \tag{59}
$$

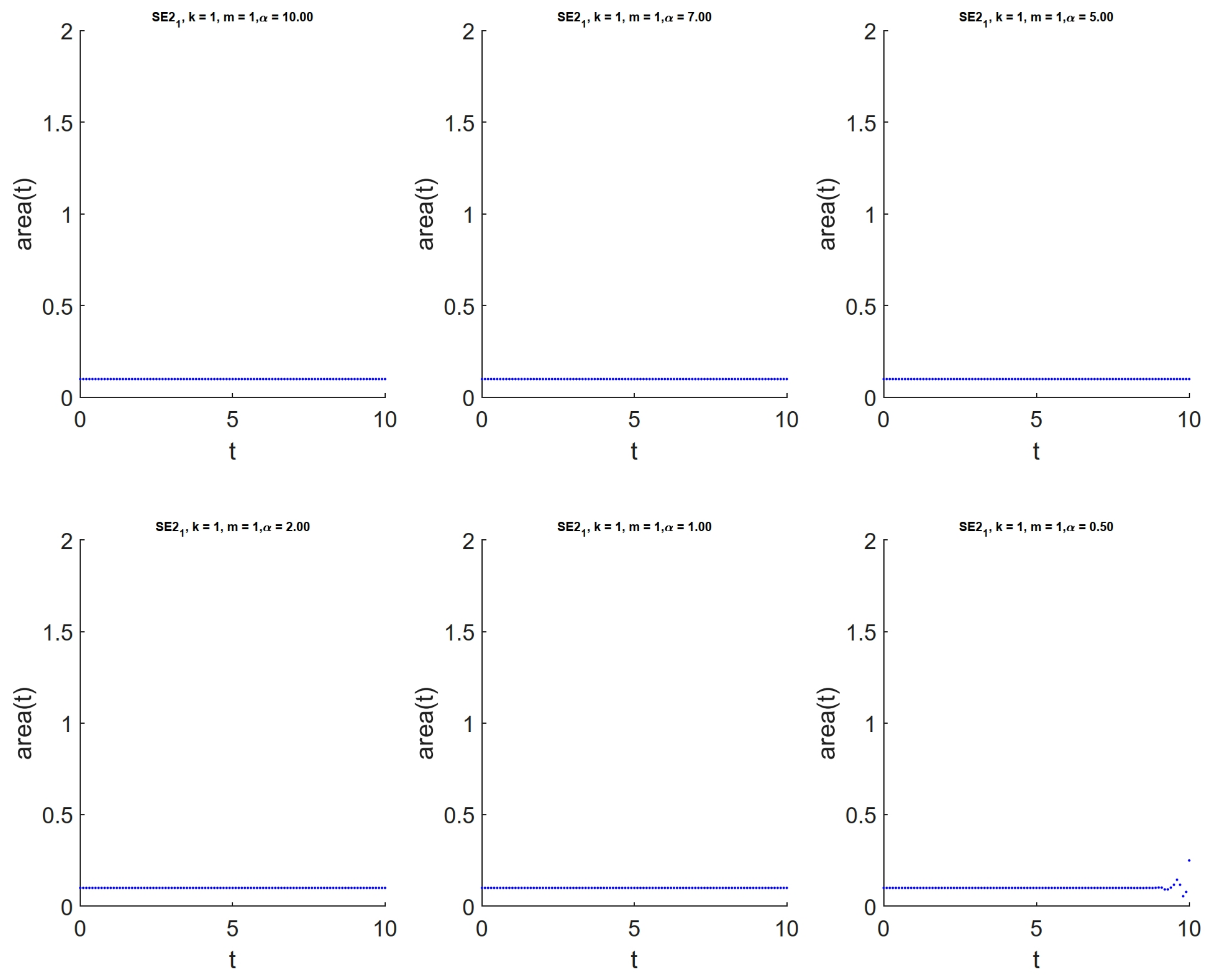


Figure 3: $SE2_1$ method. Triangle area vs. time, computed at discrete steps, $t = i\Delta t$, $\Delta t = 0.1$, over interval $[0, 10]$.

$$
\begin{aligned}
d\boldsymbol{\lambda}_{n+1} =& d\boldsymbol{\lambda}_n - \Delta t H_{\boldsymbol{yy}} d\boldsymbol{y}_n + \Delta t H_{\boldsymbol{yu}} H_{\boldsymbol{uu}}^{-1} H_{\boldsymbol{uy}} d\boldsymbol{y}_{n+1} + \\
& \Delta t H_{\boldsymbol{yu}} H_{\boldsymbol{uu}}^{-1} H_{\boldsymbol{u\lambda}} d\boldsymbol{\lambda}_{n+1} - \Delta t H_{\boldsymbol{y\lambda}} d\boldsymbol{\lambda}_{n+1} \\
& = (\boldsymbol{I} - \Delta t H_{\boldsymbol{yu}} H_{\boldsymbol{uu}}^{-1} H_{\boldsymbol{u\lambda}} + \Delta t H_{\boldsymbol{y\lambda}})^{-1} d\boldsymbol{\lambda}_n - \\
& \Delta t (\boldsymbol{I} - \Delta t H_{\boldsymbol{yu}} H_{\boldsymbol{uu}}^{-1} H_{\boldsymbol{u\lambda}} + \Delta t H_{\boldsymbol{y\lambda}})^{-1} H_{\boldsymbol{yy}} d\boldsymbol{y}_n + \\
& \Delta t (\boldsymbol{I} - \Delta t H_{\boldsymbol{yu}} H_{\boldsymbol{uu}}^{-1} H_{\boldsymbol{u\lambda}} + \Delta t H_{\boldsymbol{y\lambda}})^{-1} H_{\boldsymbol{yu}} H_{\boldsymbol{uu}}^{-1} H_{\boldsymbol{uy}} d\boldsymbol{y}_{n+1}.
\end{aligned}
\tag{60}
$$

If $H_{\boldsymbol{uy}} = \mathbf{0}$, the equations (**??**) and (**??**) simplify to

$$
d\boldsymbol{y}_{n+1} = (\boldsymbol{I} + \Delta t H_{\boldsymbol{\lambda y}}) d\boldsymbol{y}_n + (-\Delta t H_{\boldsymbol{\lambda u}} H_{\boldsymbol{uu}}^{-1} H_{\boldsymbol{u\lambda}} + \Delta t H_{\boldsymbol{\lambda\lambda}}) d\boldsymbol{\lambda}_{n+1} \tag{61}
$$

$$
d\boldsymbol{\lambda}_{n+1} = (\boldsymbol{I} + \Delta t H_{\boldsymbol{y\lambda}})^{-1} d\boldsymbol{y}_n - \Delta t (\boldsymbol{I} + \Delta t H_{\boldsymbol{y\lambda}})^{-1} H_{\boldsymbol{yy}} d\boldsymbol{y}_n \tag{62}
$$

Taking the wedge product of (**??**) with $d\boldsymbol{\lambda}_{n+1}$ and substituting equation (**??**) into the resulting expression, we obtain:

$$
\begin{aligned}
d\boldsymbol{y}_{n+1} \wedge d\boldsymbol{\lambda}_{n+1} =& (\boldsymbol{I} + \Delta t H_{\boldsymbol{\lambda y}}) d\boldsymbol{y}_n \wedge (\boldsymbol{I} + \Delta t H_{\boldsymbol{y\lambda}})^{-1} d\boldsymbol{\lambda}_n \\
& = \underbrace{(\boldsymbol{I} + \Delta t H_{\boldsymbol{y\lambda}}^T)}_{H_{\boldsymbol{y\lambda}}^T = H_{\boldsymbol{\lambda y}}}{}^{-1} (\boldsymbol{I} + \Delta t H_{\boldsymbol{\lambda y}}) d\boldsymbol{y}_n \wedge d\boldsymbol{\lambda}_n \\
& = d\boldsymbol{y}_n \wedge d\boldsymbol{\lambda}_n.
\end{aligned}
\tag{63}
$$

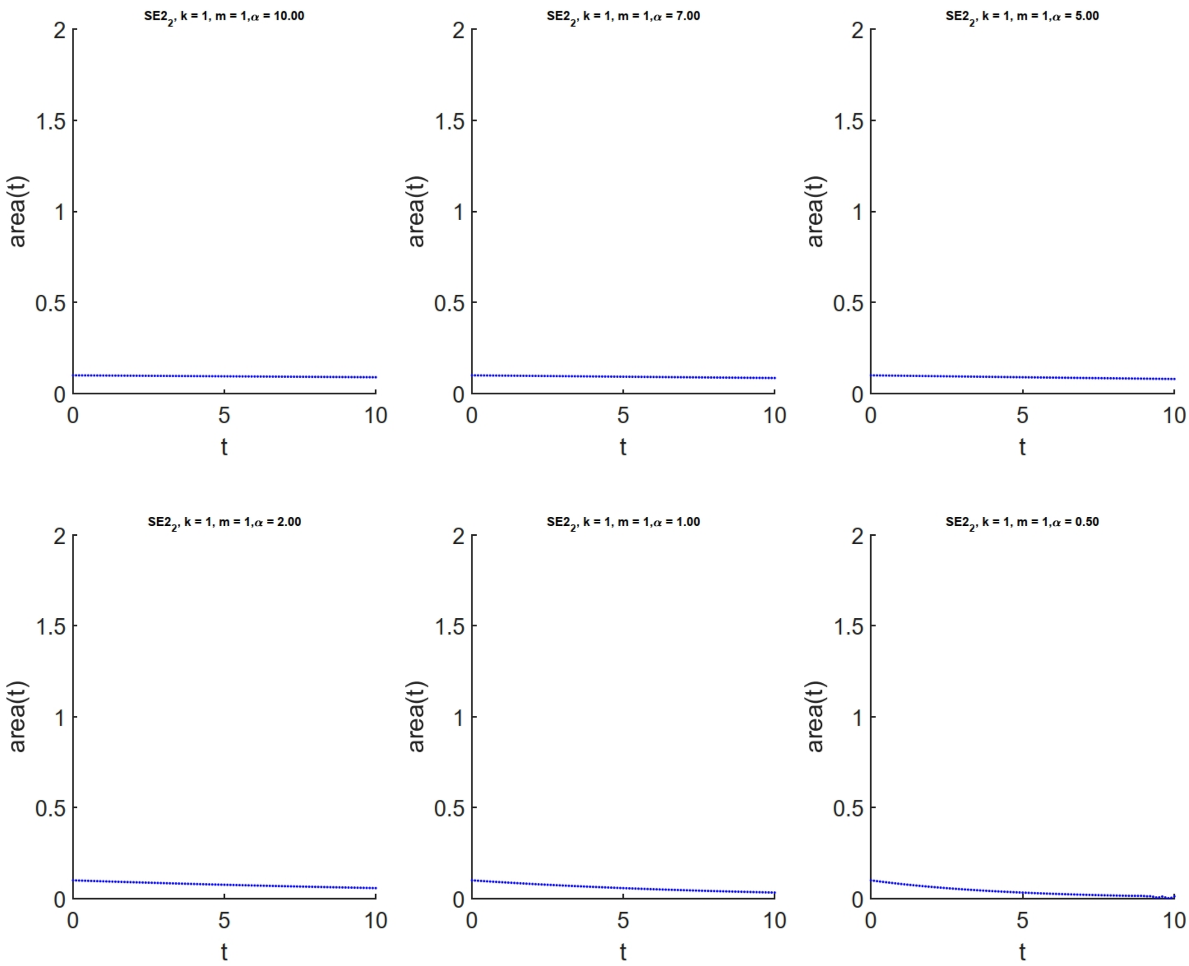


Figure 4: $SE2_2$ method.Triangle area vs. time, computed at discrete steps, $t = i\Delta t$, $\Delta t = 0.1$, over interval $[0, 10]$.

Modified Mid Point: (M.M.P)

$$\boldsymbol{y}_{n+1} = \boldsymbol{y}_n + \Delta t H_{\boldsymbol{\lambda}}(\boldsymbol{y}_{\boldsymbol{n+\frac{1}{2}}}, \boldsymbol{u}(\boldsymbol{y}_{\boldsymbol{n+\frac{1}{2}}}, \boldsymbol{\lambda}_{\boldsymbol{n+\frac{1}{2}}}), \boldsymbol{\lambda}_{\boldsymbol{n+\frac{1}{2}}}),$$

$$d\boldsymbol{y}_{n+1} = d\boldsymbol{y}_n + \frac{\Delta t}{2} H_{\boldsymbol{\lambda y}}(d\boldsymbol{y}_{n+1} + d\boldsymbol{y}_n) + \frac{\Delta t}{2} H_{\boldsymbol{\lambda u}} d\boldsymbol{u} + \underbrace{\frac{\Delta t}{2} H_{\boldsymbol{\lambda\lambda}}(d\boldsymbol{\lambda}_{n+1} + d\boldsymbol{\lambda}_n)}_{\boldsymbol{0}},$$

$$d\boldsymbol{y}_{n+1} = d\boldsymbol{y}_n + \frac{\Delta t}{2} H_{\boldsymbol{\lambda y}}(d\boldsymbol{y}_{n+1} + d\boldsymbol{y}_n) + \frac{\Delta t}{2} H_{\boldsymbol{\lambda u}}\left(-H_{\boldsymbol{uu}}^{-1} H_{\boldsymbol{uy}} d(d\boldsymbol{y}_{n+1} + d\boldsymbol{y}_n) - H_{\boldsymbol{uu}}^{-1} H_{\boldsymbol{u\lambda}}(d\boldsymbol{\lambda}_{n+1} + d\boldsymbol{\lambda}_n)\right),$$

$$\boxed{d\boldsymbol{y}_{n+1} = (\boldsymbol{I} - \frac{\Delta t}{2}\boldsymbol{A})^{-1}(\boldsymbol{I} + \frac{\Delta t}{2}\boldsymbol{A})d\boldsymbol{y}_n + (-\frac{\Delta t}{2})(\boldsymbol{I} - \frac{\Delta t}{2}\boldsymbol{A})^{-1}\boldsymbol{B}d\boldsymbol{\lambda}_n + (-\frac{\Delta t}{2})(\boldsymbol{I} - \frac{\Delta t}{2}\boldsymbol{A})^{-1}\boldsymbol{B}d\boldsymbol{\lambda}_{n+1},} \quad (64)$$

where $\boldsymbol{A} = H_{\boldsymbol{\lambda y}} - H_{\boldsymbol{\lambda u}} H_{\boldsymbol{uu}}^{-1} H_{\boldsymbol{uy}}$, $\boldsymbol{B} = H_{\boldsymbol{\lambda u}} H_{\boldsymbol{uu}}^{-1} H_{\boldsymbol{u\lambda}}$.

Applying a similar procedure, we obtain:

$$\boxed{d\boldsymbol{\lambda}_{n+1} = (\boldsymbol{I} + \frac{\Delta t}{2}\boldsymbol{A}^T)^{-1}(\boldsymbol{I} - \frac{\Delta t}{2}\boldsymbol{A}^T)d\boldsymbol{\lambda}_n + (-\frac{\Delta t}{2})(\boldsymbol{I} + \frac{\Delta t}{2}\boldsymbol{A}^T)^{-1}\boldsymbol{C}d\boldsymbol{y}_{n+1} + (-\frac{\Delta t}{2})(\boldsymbol{I} + \frac{\Delta t}{2}\boldsymbol{A}^T)^{-1}\boldsymbol{C}d\boldsymbol{y}_n,} \quad (65)$$

where $\boldsymbol{C} = H_{\boldsymbol{yy}} - H_{\boldsymbol{yu}} H_{\boldsymbol{uu}}^{-1} H_{\boldsymbol{uy}}$ and

$$\boldsymbol{\lambda}_{n+1} = \boldsymbol{\lambda}_n - \Delta t H_{\boldsymbol{y}}(\boldsymbol{y}_{\boldsymbol{n+\frac{1}{2}}}, \boldsymbol{u}(\boldsymbol{y}_{\boldsymbol{n+\frac{1}{2}}}, \boldsymbol{\lambda}_{\boldsymbol{n+\frac{1}{2}}}), \boldsymbol{\lambda}_{\boldsymbol{n+\frac{1}{2}}}).$$

By taking the wedge product of (??) with $d\boldsymbol{\lambda}_{n+1}$ we have the following:

$$\begin{aligned}
d\boldsymbol{y}_{n+1} \wedge d\boldsymbol{\lambda}_{n+1} =&(\boldsymbol{I} - \frac{\Delta t}{2}\boldsymbol{A})^{-1}(\boldsymbol{I} + \frac{\Delta t}{2}\boldsymbol{A})d\boldsymbol{y}_n \wedge d\boldsymbol{\lambda}_{n+1}+\\
&(-\frac{\Delta t}{2})(\boldsymbol{I} - \frac{\Delta t}{2}\boldsymbol{A})^{-1}\boldsymbol{B}d\boldsymbol{\lambda}_n \wedge d\boldsymbol{\lambda}_{n+1} =\\
&(\boldsymbol{I} - \frac{\Delta t}{2}\boldsymbol{A})^{-1}(\boldsymbol{I} + \frac{\Delta t}{2}\boldsymbol{A})d\boldsymbol{y}_n \wedge (\boldsymbol{I} + \frac{\Delta t}{2}\boldsymbol{A}^T)^{-1}(\boldsymbol{I} - \frac{\Delta t}{2}\boldsymbol{A}^T)d\boldsymbol{\lambda}_n+\\
&(\boldsymbol{I} - \frac{\Delta t}{2}\boldsymbol{A})^{-1}(\boldsymbol{I} + \frac{\Delta t}{2}\boldsymbol{A})d\boldsymbol{y}_n \wedge (-\frac{\Delta t}{2})(\boldsymbol{I} + \frac{\Delta t}{2}\boldsymbol{A}^T)^{-1}\boldsymbol{C}d\boldsymbol{y}_{n+1}+\\
&(-\frac{\Delta t}{2})(\boldsymbol{I} - \frac{\Delta t}{2}\boldsymbol{A})^{-1}\boldsymbol{B}d\boldsymbol{\lambda}_n \wedge (-\frac{\Delta t}{2})(\boldsymbol{I} + \frac{\Delta t}{2}\boldsymbol{A}^T)^{-1}\boldsymbol{C}d\boldsymbol{y}_{n+1}+\\
&(-\frac{\Delta t}{2})(\boldsymbol{I} - \frac{\Delta t}{2}\boldsymbol{A})^{-1}\boldsymbol{B}d\boldsymbol{\lambda}_n \wedge (-\frac{\Delta t}{2})(\boldsymbol{I} + \frac{\Delta t}{2}\boldsymbol{A}^T)^{-1}\boldsymbol{C}d\boldsymbol{y}_n.
\end{aligned}\tag{66}$$

Using the facts $d\boldsymbol{a} \wedge d\boldsymbol{c} = -d\boldsymbol{c} \wedge d\boldsymbol{a}$ and $d\boldsymbol{a} \wedge \boldsymbol{M}d\boldsymbol{c} = \boldsymbol{M}^T d\boldsymbol{a} \wedge d\boldsymbol{c}$, equation (??) can be written as:

$$\begin{aligned}
d\boldsymbol{y}_{n+1} \wedge d\boldsymbol{\lambda}_{n+1} =&(\boldsymbol{I} - \frac{\Delta t}{2}\boldsymbol{A})(\boldsymbol{I} + \frac{\Delta t}{2}\boldsymbol{A})^{-1}(\boldsymbol{I} - \frac{\Delta t}{2}\boldsymbol{A})^{-1}(\boldsymbol{I} + \frac{\Delta t}{2}\boldsymbol{A})d\boldsymbol{y}_n \wedge d\boldsymbol{\lambda}_n+\\
&(\boldsymbol{I} - \frac{\Delta t}{2}\boldsymbol{A})^{-1}(\boldsymbol{I} + \frac{\Delta t}{2}\boldsymbol{A})^{-1}d\boldsymbol{y}_n \wedge (-\frac{\Delta t}{2})(\boldsymbol{I} + \frac{\Delta t}{2}\boldsymbol{A}^T)^{-1}\boldsymbol{C}d\boldsymbol{y}_{n+1}+\\
&(-\frac{\Delta t}{2})(\boldsymbol{I} - \frac{\Delta t}{2}\boldsymbol{A})^{-1}\boldsymbol{B}d\boldsymbol{\lambda}_n \wedge (-\frac{\Delta t}{2})(\boldsymbol{I} + \frac{\Delta t}{2}\boldsymbol{A}^T)^{-1}\boldsymbol{C}d\boldsymbol{y}_{n+1}+\\
&(\frac{\Delta t}{2})(\boldsymbol{I} + \frac{\Delta t}{2}\boldsymbol{A}^T)^{-1}\boldsymbol{C}d\boldsymbol{y}_n \wedge (-\frac{\Delta t}{2})(\boldsymbol{I} - \frac{\Delta t}{2}\boldsymbol{A})^{-1}\boldsymbol{B}d\boldsymbol{\lambda}_n.
\end{aligned}\tag{67}$$

Since $(\boldsymbol{I}-\frac{\Delta t}{2}\boldsymbol{A})(\boldsymbol{I}+\frac{\Delta t}{2}\boldsymbol{A}) = (\boldsymbol{I}+\frac{\Delta t}{2}\boldsymbol{A})(\boldsymbol{I}-\frac{\Delta t}{2}\boldsymbol{A})$, it follows that $(\boldsymbol{I}-\frac{\Delta t}{2}\boldsymbol{A})^{-1}(\boldsymbol{I}+\frac{\Delta t}{2}\boldsymbol{A})^{-1} = (\boldsymbol{I}+\frac{\Delta t}{2}\boldsymbol{A})^{-1}(\boldsymbol{I}-\frac{\Delta t}{2}\boldsymbol{A})^{-1}$, So the equation (??) can be rewritten as:

$$\begin{aligned}
d\boldsymbol{y}_{n+1} \wedge d\boldsymbol{\lambda}_{n+1} =&d\boldsymbol{y}_n \wedge d\boldsymbol{\lambda}_n+\\
&(\boldsymbol{I} - \frac{\Delta t}{2}\boldsymbol{A})^{-1}(\boldsymbol{I} + \frac{\Delta t}{2}\boldsymbol{A})^{-1}d\boldsymbol{y}_n \wedge (-\frac{\Delta t}{2})(\boldsymbol{I} + \frac{\Delta t}{2}\boldsymbol{A}^T)^{-1}\boldsymbol{C}d\boldsymbol{y}_{n+1}+\\
&(-\frac{\Delta t}{2})(\boldsymbol{I} - \frac{\Delta t}{2}\boldsymbol{A})^{-1}\boldsymbol{B}d\boldsymbol{\lambda}_n \wedge (-\frac{\Delta t}{2})(\boldsymbol{I} + \frac{\Delta t}{2}\boldsymbol{A}^T)^{-1}\boldsymbol{C}d\boldsymbol{y}_{n+1}+\\
&(\frac{\Delta t}{2})(\boldsymbol{I} + \frac{\Delta t}{2}\boldsymbol{A}^T)^{-1}\boldsymbol{C}d\boldsymbol{y}_n \wedge (-\frac{\Delta t}{2})(\boldsymbol{I} - \frac{\Delta t}{2}\boldsymbol{A})^{-1}\boldsymbol{B}d\boldsymbol{\lambda}_n.
\end{aligned}\tag{68}$$

$$\begin{aligned}
d\boldsymbol{y}_{n+1} \wedge d\boldsymbol{\lambda}_{n+1} =&\boldsymbol{I} - \frac{\Delta t^2}{4}\boldsymbol{B}^T(\boldsymbol{I} - \frac{\Delta t}{2}\boldsymbol{A}^T)^{-1}(\boldsymbol{I} + \frac{\Delta t}{2}\boldsymbol{A}^T)^{-1}\boldsymbol{C}d\boldsymbol{y}_n \wedge d\boldsymbol{\lambda}_n+\\
&(\boldsymbol{I} - \frac{\Delta t}{2}\boldsymbol{A})^{-1}(\boldsymbol{I} + \frac{\Delta t}{2}\boldsymbol{A})^{-1}d\boldsymbol{y}_n \wedge (-\frac{\Delta t}{2})(\boldsymbol{I} + \frac{\Delta t}{2}\boldsymbol{A}^T)^{-1}\boldsymbol{C}d\boldsymbol{y}_{n+1}+\\
&(-\frac{\Delta t}{2})(\boldsymbol{I} - \frac{\Delta t}{2}\boldsymbol{A})^{-1}\boldsymbol{B}d\boldsymbol{\lambda}_n \wedge (-\frac{\Delta t}{2})(\boldsymbol{I} + \frac{\Delta t}{2}\boldsymbol{A}^T)^{-1}\boldsymbol{C}d\boldsymbol{y}_{n+1}.
\end{aligned}\tag{69}$$

By taking the wedge product of $d\boldsymbol{y}_{n+1}$ with $(-\frac{\Delta t}{2})(\boldsymbol{I} + \frac{\Delta t}{2}\boldsymbol{A}^T)^{-1}\boldsymbol{C}d\boldsymbol{y}_{n+1}$ we have the following:

$$\begin{aligned}
\underbrace{d\boldsymbol{y}_{n+1} \wedge (-\frac{\Delta t}{2})(\boldsymbol{I} + \frac{\Delta t}{2}\boldsymbol{A}^T)^{-1}\boldsymbol{C}d\boldsymbol{y}_{n+1}}_{\boldsymbol{0}} =&(\boldsymbol{I} - \frac{\Delta t}{2}\boldsymbol{A})^{-1}(\boldsymbol{I} + \frac{\Delta t}{2}\boldsymbol{A})d\boldsymbol{y}_n \wedge (-\frac{\Delta t}{2})(\boldsymbol{I} + \frac{\Delta t}{2}\boldsymbol{A}^T)^{-1}\boldsymbol{C}d\boldsymbol{y}_{n+1}+\\
&(-\frac{\Delta t}{2})(\boldsymbol{I} - \frac{\Delta t}{2}\boldsymbol{A})^{-1}\boldsymbol{B}d\boldsymbol{\lambda}_n \wedge (-\frac{\Delta t}{2})(\boldsymbol{I} + \frac{\Delta t}{2}\boldsymbol{A}^T)^{-1}\boldsymbol{C}d\boldsymbol{y}_{n+1}+\\
&(-\frac{\Delta t}{2})(\boldsymbol{I} - \frac{\Delta t}{2}\boldsymbol{A})^{-1}\boldsymbol{B}d\boldsymbol{\lambda}_{n+1} \wedge (-\frac{\Delta t}{2})(\boldsymbol{I} + \frac{\Delta t}{2}\boldsymbol{A}^T)^{-1}\boldsymbol{C}d\boldsymbol{y}_{n+1}.
\end{aligned}\tag{70}$$

Rewriting the equation (??) by isolating $(-\frac{\Delta t}{2})(\boldsymbol{I} - \frac{\Delta t}{2}\boldsymbol{A})^{-1}\boldsymbol{B}d\boldsymbol{\lambda}_{n+1} \wedge (-\frac{\Delta t}{2})(\boldsymbol{I} + \frac{\Delta t}{2}\boldsymbol{A}^T)^{-1}\boldsymbol{C}d\boldsymbol{y}_{n+1}$, we obtain the following equivalent form:

$$(-\frac{\Delta t}{2})(\boldsymbol{I}+\frac{\Delta t}{2}\boldsymbol{A}^T)^{-1}\boldsymbol{C}d\boldsymbol{y}_{n+1}\wedge(-\frac{\Delta t}{2})(\boldsymbol{I}-\frac{\Delta t}{2}\boldsymbol{A})^{-1}\boldsymbol{B}d\boldsymbol{\lambda}_{n+1}=$$
$$(\boldsymbol{I}-\frac{\Delta t}{2}\boldsymbol{A})^{-1}(\boldsymbol{I}+\frac{\Delta t}{2}\boldsymbol{A})d\boldsymbol{y}_n\wedge(-\frac{\Delta t}{2})(\boldsymbol{I}+\frac{\Delta t}{2}\boldsymbol{A}^T)^{-1}\boldsymbol{C}d\boldsymbol{y}_{n+1}+$$
$$(-\frac{\Delta t}{2})(\boldsymbol{I}-\frac{\Delta t}{2}\boldsymbol{A})^{-1}\boldsymbol{B}d\boldsymbol{\lambda}_n\wedge(-\frac{\Delta t}{2})(\boldsymbol{I}+\frac{\Delta t}{2}\boldsymbol{A}^T)^{-1}\boldsymbol{C}d\boldsymbol{y}_{n+1}. \quad (71)$$

Upon plugging (**??**) into (**??**), we get:

$$d\boldsymbol{y}_{n+1}\wedge d\boldsymbol{\lambda}_{n+1}=\boldsymbol{I}-\frac{\Delta t^2}{4}\boldsymbol{B}^T(\boldsymbol{I}-\frac{\Delta t}{2}\boldsymbol{A}^T)^{-1}(\boldsymbol{I}+\frac{\Delta t}{2}\boldsymbol{A}^T)^{-1}\boldsymbol{C}d\boldsymbol{y}_n\wedge d\boldsymbol{\lambda}_n$$
$$+\frac{\Delta t^2}{4}\boldsymbol{B}^T(\boldsymbol{I}-\frac{\Delta t}{2}\boldsymbol{A}^T)^{-1}(\boldsymbol{I}+\frac{\Delta t}{2}\boldsymbol{A}^T)^{-1}\boldsymbol{C}d\boldsymbol{y}_{n+1}\wedge d\boldsymbol{\lambda}_{n+1} \quad (72)$$

By solving (**??**) for $d\boldsymbol{y}_{n+1}\wedge d\boldsymbol{\lambda}_{n+1}$, we have:

$$\left(\boldsymbol{I}-\frac{\Delta t^2}{4}\boldsymbol{B}^T(\boldsymbol{I}-\frac{\Delta t}{2}\boldsymbol{A}^T)^{-1}(\boldsymbol{I}+\frac{\Delta t}{2}\boldsymbol{A}^T)^{-1}\boldsymbol{C}\right)d\boldsymbol{y}_{n+1}\wedge d\boldsymbol{\lambda}_{n+1}=$$
$$\left(\boldsymbol{I}-\frac{\Delta t^2}{4}\boldsymbol{B}^T(\boldsymbol{I}-\frac{\Delta t}{2}\boldsymbol{A}^T)^{-1}(\boldsymbol{I}+\frac{\Delta t}{2}\boldsymbol{A}^T)^{-1}\boldsymbol{C}\right)d\boldsymbol{y}_n\wedge d\boldsymbol{\lambda}_n. \quad (73)$$

3.1 Numerical results (nonlinear case)

We consider the following nonlinear optimal control problems and apply $SE11, SE21$, and implicit midpoint methods. As analytically proven, condition $H_{yu}=0$ in Problem (**??**) ensures the symplecticity of $SE11$ and $SE21$, thus guaranteeing the preservation of the phase-space area. This property is numerically validated in Table **??**, which compares the absolute relative errors in the preserved area at the initial and final times. In contrast, problem (**??**) does not satisfy the condition $H_{yu}=0$, and as shown in Table **??**, the $SE11$ and $SE21$ methods lose their symplectic property in this case, leading to nonpreservation of the phase-space area.

These simulations employ three distinct initial points to construct a triangular shape at the initial time see Figure **??**. To compute the area of this triangle, we split it into four smaller non-overlapping triangles, calculate their individual areas, and add them to determine the total area.

$$\begin{aligned}&\min\int_0^2 \frac{1}{2}v^2+\frac{\alpha}{2}u^2\,dt\\ &\dot v = g+\frac{u}{m}-\frac{k}{m}v^2\\ &v(0)=v_0\end{aligned} \quad (74)$$

$$\begin{aligned}&\min\int_0^2 \frac{1}{2}v^2+\frac{\alpha}{2}u^2\,dt\\ &\dot v = g+\frac{u}{m}-\frac{k}{m}uv\\ &v(0)=v_0\end{aligned} \quad (75)$$

Necessary optimality conditions for (**??**) and (**??**), respectively,

$$\begin{aligned}&H(v,u,\lambda)=\frac{1}{2}v^2+\frac{\alpha}{2}u^2+\lambda(g+\frac{1}{m}u-\frac{k}{m}v^2)\\ &\begin{cases}\dot v = H_\lambda(v,u,\lambda)=g+\frac{u}{m}-\frac{k}{m}v^2\\ \dot\lambda=-H_v(v,u,\lambda)=-v+\frac{2k}{m}\lambda v\\ 0=H_u(v,u,\lambda)=\alpha u+\frac{1}{m}\lambda\end{cases}\\ &H_{vu}=0.\end{aligned} \quad (76)$$

$$\begin{aligned}&H(v,u,\lambda)=\frac{1}{2}v^2+\frac{\alpha}{2}u^2+\lambda(g+\frac{1}{m}u-\frac{k}{m}uv)\\ &\begin{cases}\dot v = H_\lambda(v,u,\lambda)=g+\frac{u}{m}-\frac{k}{m}uv\\ \dot\lambda=-H_v(v,u,\lambda)=-v+\frac{k}{m}\lambda u\\ 0=H_u(v,u,\lambda)=\alpha u+\frac{1}{m}\lambda-\frac{k}{m}\lambda v\end{cases}\\ &H_{vu}\neq 0.\end{aligned} \quad (77)$$

$\alpha = 0.5,\ N.T = 1553$	Δt	$\lvert A_0 - A_T\rvert, H_{vu} = 0$	$\frac{\lvert A_0 - A_T\rvert}{A_0},\ H_{vu} = 0$	$\lvert A_0 - A_T\rvert, H_{vu} \neq 0$	$\frac{\lvert A_0 - A_T\rvert}{A_0},\ H_{vu} \neq 0$
	0.02	3.0978×10^{-5}	2.478×10^{-2}	1.1293×10^{-3}	0.9032
$SE1_1$	0.01	2.2191×10^{-5}	1.7753×10^{-2}	1.1883×10^{-3}	0.9506
	0.02	1.1769×10^{-5}	9.415×10^{-3}	1.2610×10^{-4}	0.1008
$SE2_1$	0.01	1.3989×10^{-5}	1.1191×10^{-2}	1.5809×10^{-4}	0.1264
	0.02	1.7307×10^{-5}	1.3845×10^{-2}	2.0385×10^{-4}	0.1630
Mid Point	0.01	1.7222×10^{-5}	1.3777×10^{-2}	1.9555×10^{-4}	0.1564

Table 2: Application of $SE1_1, SE1_2$ and implicit mid point methods to systems (**??**) and (**??**) on the interval $[0, 2]$ with three initial points $\mathbf{z}_1 = (0, 1)^T, \mathbf{z}_2 = (0.05, 1)^T$ and $\mathbf{z}_3 = (0, 1.05)^T$, where, $k = 1, m = 20, T = 2$. $A_0 = 0.0013$ and A_T denote the areas of the triangle at time $t = 0$ and $t = T = 2$, respectively.

Method, $\alpha = 0.5$	Δt	$\lVert A_0^T J A_0 - J\rVert,\ H_{uv} \neq 0$
$SE1_1$	0.5	0.0174
	0.2	0028
	0.02	2.7721×10^{-5}
$SE2_1$	0.5	0.0166
	0.2	0028
	0.02	2.7712×10^{-5}
Mid Point	0.5	0.0079
	0.2	2.7735×10^{-4}
	0.02	1.5254×10^{-7}
Modified $SE1_1$	0.5	2.5931×10^{-8}
	0.2	3.6732×10^{-9}
	0.02	4.9304×10^{-10}
Modified $SE2_1$	0.5	1.0876×10^{-8}
	0.2	3.6732×10^{-9}
	0.02	4.9304×10^{-10}
Modified Mid Point	0.5	2.4509×10^{-8}
	0.2	5.0484×10^{-9}
	0.02	5.1423×10^{-10}

Table 3: Numerical evaluation of symplecticity for $SE1_1, SE2_1, M.P, M.SE1_1, M.SE2_1$, and $M.M.P$ methods (applied to system (**??**)) with different time step sizes (Δt). $k = 1, m = 20$. $A_0 = \frac{\partial z_{n+1}}{\partial z_n}(v_n, \lambda_n)$.

4 Nonlinear optimal control problems(Inverted pendulum problem)

Consider the optimal control problem of the inverted pendulum:

$$\begin{aligned} &\min_u \frac{1}{2}\int_0^T \lVert \boldsymbol{x} - \boldsymbol{x}_d \rVert^2 \, dt + \frac{\alpha}{2}\int_0^T u^2 dt \\ &m\ddot{\boldsymbol{x}} + \frac{\partial U}{\partial \boldsymbol{x}} = \mathbf{0}, \\ &\boldsymbol{x}(0) = \boldsymbol{x}_0, \quad \dot{\boldsymbol{x}}(0) = \boldsymbol{v}_0, \end{aligned}$$

where

$$\begin{aligned} U &= \frac{1}{2}k(L - l_0)^2 - \boldsymbol{a}^T\boldsymbol{x},\ L = \lVert \boldsymbol{x} - (u, 0)^T \rVert, \\ \frac{\partial U}{\partial \boldsymbol{x}} &= k\left(\frac{L - l_0}{L}\right)(\boldsymbol{x} - (u, 0)^T) - \boldsymbol{a} \end{aligned}$$

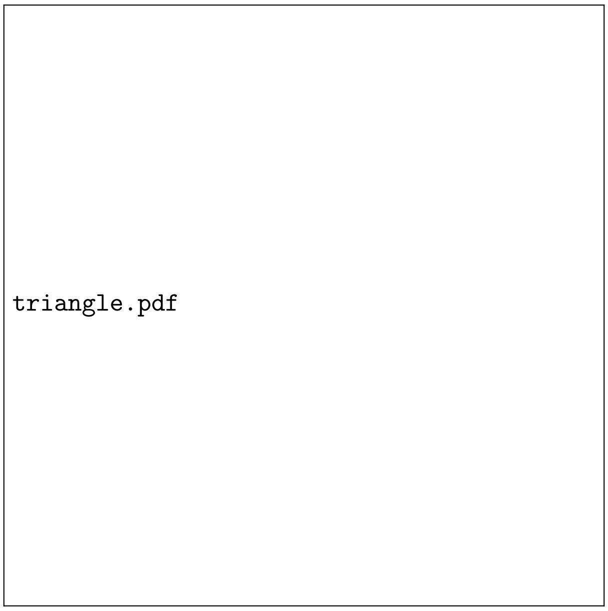


Figure 5: Splitting the triangle with vertices A, B and C into 1553 smaller non-overlapping triangles

By introducing an auxiliary variable $\boldsymbol{v}$ and set it to $\dot{\boldsymbol{x}}$, the second order differential equations is converted into a first order system of differential equations so the control problem is formulated as follows:

$$\begin{aligned}
&\min_{\boldsymbol{x},\boldsymbol{v},u} J(\boldsymbol{x},u)\\
&\qquad \dot{\boldsymbol{x}} - \boldsymbol{v} = \boldsymbol{0},\\
&\qquad \dot{\boldsymbol{v}} + \frac{1}{m}\frac{\partial U}{\partial \boldsymbol{x}} = \boldsymbol{0},\\
&\qquad \boldsymbol{x}(0) = \boldsymbol{x}_0,\ \boldsymbol{v}(0) = \boldsymbol{v}_0.
\end{aligned} \tag{78}$$

To derive the necessary conditions for optimality, we introduce the Hamiltonian function H:

$$H(\boldsymbol{v},\boldsymbol{x},u,\boldsymbol{\lambda},\boldsymbol{\mu}) = f(\boldsymbol{x},u) + \boldsymbol{\lambda}^T(\frac{-1}{m})\frac{\partial U}{\partial \boldsymbol{x}} + \boldsymbol{\mu}^T\boldsymbol{v}$$

where $f(\boldsymbol{x},u) = r(\boldsymbol{x}) + q(u)$. By applying the Pontryagin's maximum principle to the Hamiltonian function, the necessary conditions for optimality are given as follows:

$$\begin{aligned}
&\text{Adjoint Equations: } \begin{cases} \dot{\boldsymbol{\lambda}} = -\frac{\partial H}{\partial \boldsymbol{v}}, & \boldsymbol{\lambda}(T) = \boldsymbol{0}\\ \dot{\boldsymbol{\mu}} = -\frac{\partial H}{\partial \boldsymbol{x}}, & \boldsymbol{\mu}(T) = \boldsymbol{0}\end{cases}\\
&\text{Control Equation: } \left\{ \frac{\partial H}{\partial u} = 0 \right.\\
&\text{State Equations: } \begin{cases} \dot{\boldsymbol{v}} = \frac{\partial H}{\partial \boldsymbol{\lambda}}, & \boldsymbol{v}(\boldsymbol{0}) = \boldsymbol{v}_0\\ \dot{\boldsymbol{x}} = \frac{\partial H}{\partial \boldsymbol{\mu}}, & \boldsymbol{x}(\boldsymbol{0}) = \boldsymbol{x}_0\end{cases}
\end{aligned}$$

This system of equations forms a two-point boundary value problem (TPBVP), with boundary conditions specified at both the initial and terminal times.

$$\begin{aligned}
&\begin{cases} \dot{\boldsymbol{\lambda}} + \boldsymbol{\mu} = \boldsymbol{0}, & \boldsymbol{\lambda}(T) = \boldsymbol{0},\\ \dot{\boldsymbol{\mu}} + (\boldsymbol{x} - \boldsymbol{x}_d) - \left(\frac{kl_0}{mL^3}(\boldsymbol{x} - (u,0)^T)(\boldsymbol{x} - (u,0)^T)^T + \frac{k}{m}(\frac{L-l_0}{L})\boldsymbol{I}\right)\boldsymbol{\lambda} = \boldsymbol{0}, & \boldsymbol{\mu}(T) = \boldsymbol{0},\end{cases}\\
&\alpha u - \boldsymbol{\lambda}^T\left(\left[\frac{-kl_0}{mL^3}\boldsymbol{e}_1^T(\boldsymbol{x} - (u,0)^T)\right](\boldsymbol{x} - (u,0)^T) + \frac{-k}{m}(\frac{L-l_0}{L})\boldsymbol{e}_1\right) = 0,\\
&\begin{cases} \dot{\boldsymbol{v}} + \frac{k}{m}(\frac{L-l_0}{L})(\boldsymbol{x} - (u,0)^T) - \frac{1}{m}\boldsymbol{a} = \boldsymbol{0}, & \boldsymbol{v}(0) = \boldsymbol{v}_0,\\ \dot{\boldsymbol{x}} - \boldsymbol{v} = \boldsymbol{0}, & \boldsymbol{x}(0) = \boldsymbol{x}_0\end{cases}
\end{aligned} \tag{79}$$

To solve (??), six numerical schemes are employed as follows:

$$\text{Set } \boldsymbol{y} := (\boldsymbol{v}, \boldsymbol{x})^T, \quad \boldsymbol{p} := (\boldsymbol{\lambda}, \boldsymbol{\mu})^T,$$

$$\begin{array}{ll}
SE1: \begin{cases} \dfrac{\Delta \boldsymbol{y}}{\Delta t} = \dfrac{\partial H}{\partial \boldsymbol{p}}(\boldsymbol{y}_{n+1}, u_n, \boldsymbol{p}_n) \\ \frac{\Delta \boldsymbol{p}}{\Delta t} = -\frac{\partial H}{\partial \boldsymbol{y}}(\boldsymbol{y}_{n+1}, u_n, \boldsymbol{p}_n) \\ H_u(\boldsymbol{y}_n, u_n, \boldsymbol{p}_n) = 0 \end{cases} &
M.SE1: \begin{cases} \frac{\Delta \boldsymbol{y}}{\Delta t} = \frac{\partial H}{\partial \boldsymbol{p}}(\boldsymbol{y}_{n+1}, u_n, \boldsymbol{p}_n) \\ \frac{\Delta \boldsymbol{p}}{\Delta t} = -\frac{\partial H}{\partial \boldsymbol{y}}(\boldsymbol{y}_{n+1}, u_n, \boldsymbol{p}_n) \\ H_u(\boldsymbol{y}_{n+1}, u_n, \boldsymbol{p}_n) = 0 \end{cases} \\
SE2: \begin{cases} \frac{\Delta \boldsymbol{y}}{\Delta t} = \frac{\partial H}{\partial \boldsymbol{p}}(y_n, u_{n+1}, \boldsymbol{p}_{n+1}) \\ \frac{\Delta \boldsymbol{p}}{\Delta t} = -\frac{\partial H}{\partial \boldsymbol{y}}(\boldsymbol{y}_n, u_{n+1}, \boldsymbol{p}_{n+1}) \\ H_u(\boldsymbol{y}_n, u_n, \boldsymbol{p}_n) = 0 \end{cases} &
M.SE2: \begin{cases} \frac{\Delta \boldsymbol{y}}{\Delta t} = \frac{\partial H}{\partial \boldsymbol{p}}(\boldsymbol{y}_n, u_n, \boldsymbol{p}_{n+1}) \\ \frac{\Delta \boldsymbol{p}}{\Delta t} = -\frac{\partial H}{\partial \boldsymbol{y}}(\boldsymbol{y}_n, u_n, \boldsymbol{p}_{n+1}) \\ H_u(\boldsymbol{y}_n, u_n, \boldsymbol{p}_{n+1}) = 0 \end{cases} \\
M.P: \begin{cases} \frac{\Delta \boldsymbol{y}}{\Delta t} = \frac{\partial H}{\partial \boldsymbol{p}}(\boldsymbol{y}_{n+0.5}, u_{n+0.5}, \boldsymbol{p}_{n+0.5}) \\ \frac{\Delta \boldsymbol{p}}{\Delta t} = -\frac{\partial H}{\partial \boldsymbol{y}}(\boldsymbol{y}_{n+0.5}, u_{n+0.5}, \boldsymbol{p}_{n+0.5}) \\ H_u(\boldsymbol{y}_n, u_n, \boldsymbol{p}_n) = 0 \end{cases} &
M.M.P \begin{cases} \frac{\Delta \boldsymbol{y}}{\Delta t} = \frac{\partial H}{\partial \boldsymbol{p}}(\boldsymbol{y}_{n+0.5}, u_n, \boldsymbol{p}_{n+0.5}) \\ \frac{\Delta \boldsymbol{p}}{\Delta t} = -\frac{\partial H}{\partial \boldsymbol{y}}(\boldsymbol{y}_{n+0.5}, u_n, \boldsymbol{p}_{n+0.5}) \\ H_u(\boldsymbol{y}_{n+0.5}, u_n, \boldsymbol{p}_{n+0.5}) = 0 \end{cases}
\end{array} \tag{80}$$

Table **??** presents the numerical results obtained by applying the six numerical schemes for different values of α.

Method	$\alpha = 10.0$		$\alpha = 5.0$		$\alpha = 1.0$		$\alpha = 0.5$	
	$\Delta t = 0.1$	$\Delta t = 0.025$	$\Delta t = 0.1$	$\Delta t = 0.025$	$\Delta t = 0.1$	$\Delta t = 0.025$	$\Delta t = 0.1$	$\Delta t = 0.025$
SE1	6	6	6	6	8	14	-	-
M. SE1	6	6	6	6	7	15	55	104
SE2	6	6	6	6	77	38	84	113
M. SE2	6	6	6	6	26	28	29	41
MidPoint	6	6	6	6	25	24	67	51
M. MidPoint	6	6	6	6	19	19	29	61

Table 4: Number of iterations (N.I.) for different methods and values of α, with $\Delta t = 0.1, \Delta = 0.025, k = 1, m = 1, \boldsymbol{x}_0 = (1,1)^T, \boldsymbol{x}_d = (0,2)^T, \boldsymbol{v}_0 = (0,0)^T, l_0 = \sqrt{2}, T = [0,10]$.

Table 5: Numerical evaluation of symplecticity for SE1, SE2, M.P, M.SE1, M.SE2, and M.M.P methods with different time step sizes Δt. $||\mathbf{A}_0^T \mathbf{J} \mathbf{A}_0 - \mathbf{J}||$, where $\mathbf{A}_0 = \frac{\partial(\mathbf{y}_{n+1}, \mathbf{p}_{n+1})}{\partial(\mathbf{y}_n, \mathbf{p}_n)}$.

Method	$\alpha = 10.0$			$\alpha = 5.0$			$\alpha = 1.0$		
	$\Delta t = 1$	$\Delta t = 0.1$	$\Delta t = 0.025$	$\Delta t = 1$	$\Delta t = 0.1$	$\Delta t = 0.025$	$\Delta t = 1$	$\Delta t = 0.1$	$\Delta t = 0.025$
SE1	0.0614853	0.0030918	1.4210×10^{-4}	0.1148578	0.0031728	1.0397×10^{-4}	0.3654304	0.0045715	1.7756×10^{-4}
M. SE1	3.7994×10^{-10}	4.7898×10^{-10}	6.8444×10^{-10}	3.8148×10^{-10}	3.9087×10^{-10}	6.7330×10^{-10}	3.9091×10^{-10}	4.9181×10^{-10}	4.0767×10^{-10}
SE2	81.5165	5.5104×10^{-4}	8.2952×10^{-5}	29.9389	0.0021965	3.2923×10^{-5}	25.2454	0.1528280	5.3776×10^{-5}
M. SE2	4.1198×10^{-8}	1.2966×10^{-9}	5.8667×10^{-10}	4.3593×10^{-8}	6.6569×10^{-10}	4.4809×10^{-10}	1.9479×10^{-8}	5.4068×10^{-10}	7.0251×10^{-10}
MidPoint	0.1192	0.0020301	1.3219×10^{-4}	0.1403	0.0026914	1.7456×10^{-4}	0.4556	0.0121969	7.6003×10^{-4}
M. MidPoint	1.3738×10^{-9}	4.6400×10^{-10}	7.4220×10^{-10}	1.0867×10^{-9}	5.4220×10^{-10}	5.9989×10^{-10}	6.6610×10^{-10}	6.0072×10^{-10}	7.4891×10^{-10}

5 Comparing the Modified and Unmodified Methods for Various Initial Conditions

5.1 Number of iterations comparisson for different starting points x_0

Table 6: $x_0 = [-0.5, 1.0]$

Method	$\alpha = 10.0$		$\alpha = 5.0$		$\alpha = 1.0$		$\alpha = 0.5$	
	$\Delta t = 0.2$	$\Delta t = 0.5$	$\Delta t = 0.2$	$\Delta t = 0.5$	$\Delta t = 0.2$	$\Delta t = 0.5$	$\Delta t = 0.2$	$\Delta t = 0.5$
SE1	7	7	11	-	-	-	-	-
M. SE1	7	7	8	8	-	-	-	-
SE2	-	-	-	-	-	-	-	-
M. SE2	-	-	-	-	-	-	-	-
MidPoint	-	12	-	-	-	-	-	-
M. MidPoint	-	-	-	-	-	-	-	-

Table 7: $x_0 = [-0.4, 1.0]$

Method	$\alpha = 10.0$		$\alpha = 5.0$		$\alpha = 1.0$		$\alpha = 0.5$	
	$\Delta t = 0.2$	$\Delta t = 0.5$	$\Delta t = 0.2$	$\Delta t = 0.5$	$\Delta t = 0.2$	$\Delta t = 0.5$	$\Delta t = 0.2$	$\Delta t = 0.5$
SE1	7	7	8	7	-	-	-	-
M. SE1	7	7	8	8	-	-	-	-
SE2	-	-	-	-	-	-	-	-
M. SE2	-	-	-	-	-	-	-	-
MidPoint	11	-	-	-	-	-	-	-
M. MidPoint	-	-	-	-	-	-	-	-

Table 8: $x_0 = [-0.3, 1.0]$

Method	$\alpha = 10.0$		$\alpha = 5.0$		$\alpha = 1.0$		$\alpha = 0.5$	
	$\Delta t = 0.2$	$\Delta t = 0.5$	$\Delta t = 0.2$	$\Delta t = 0.5$	$\Delta t = 0.2$	$\Delta t = 0.5$	$\Delta t = 0.2$	$\Delta t = 0.5$
SE1	7	7	10	-	-	-	-	-
M. SE1	7	7	10	-	-	-	-	-
SE2	-	-	-	-	-	-	-	-
M. SE2	-	-	-	-	-	-	-	-
MidPoint	9	-	-	-	-	-	-	-
M. MidPoint	-	-	-	-	-	-	-	-

Table 9: $x_0 = [-0.2, 1.0]$

Method	$\alpha = 10.0$		$\alpha = 5.0$		$\alpha = 1.0$		$\alpha = 0.5$	
	$\Delta t = 0.2$	$\Delta t = 0.5$	$\Delta t = 0.2$	$\Delta t = 0.5$	$\Delta t = 0.2$	$\Delta t = 0.5$	$\Delta t = 0.2$	$\Delta t = 0.5$
SE1	8	-	-	7	-	-	-	-
M. SE1	8	7	-	-	-	243	-	-
SE2	-	-	-	-	-	-	-	-
M. SE2	-	-	-	-	-	-	-	-
MidPoint	17	-	-	-	-	-	-	-
M. MidPoint	-	-	-	-	-	-	-	-

Table 10: $x_0 = [-0.1, 1.0]$

Method	$\alpha = 10.0$		$\alpha = 5.0$		$\alpha = 1.0$		$\alpha = 0.5$	
	$\Delta t = 0.2$	$\Delta t = 0.5$	$\Delta t = 0.2$	$\Delta t = 0.5$	$\Delta t = 0.2$	$\Delta t = 0.5$	$\Delta t = 0.2$	$\Delta t = 0.5$
SE1	8	10	9	-	-	-	-	-
M. SE1	8	7	8	7	-	-	-	-
SE2	-	-	-	-	-	-	-	-
M. SE2	-	-	-	-	-	-	-	-
MidPoint	-	10	-	-	-	-	-	-
M. MidPoint	-	-	-	-	-	-	-	-

Table 11: $x_0 = [0.0, 1.0]$

Method	$\alpha = 10.0$		$\alpha = 5.0$		$\alpha = 1.0$		$\alpha = 0.5$	
	$\Delta t = 0.2$	$\Delta t = 0.5$	$\Delta t = 0.2$	$\Delta t = 0.5$	$\Delta t = 0.2$	$\Delta t = 0.5$	$\Delta t = 0.2$	$\Delta t = 0.5$
SE1	7	-	8	12	-	-	-	-
M. SE1	7	7	8	7	-	-	-	-
SE2	-	-	-	-	-	-	-	-
M. SE2	-	-	-	-	-	-	-	-
MidPoint	16	12	-	-	-	-	-	-
M. MidPoint	-	-	-	-	-	-	-	-

Table 12: $x_0 = [0.1, 1.0]$

Method	$\alpha = 10.0$		$\alpha = 5.0$		$\alpha = 1.0$		$\alpha = 0.5$	
	$\Delta t = 0.2$	$\Delta t = 0.5$	$\Delta t = 0.2$	$\Delta t = 0.5$	$\Delta t = 0.2$	$\Delta t = 0.5$	$\Delta t = 0.2$	$\Delta t = 0.5$
SE1	7	8	8	9	-	-	-	-
M. SE1	7	10	7	7	-	-	-	-
SE2	-	-	-	-	-	-	-	-
M. SE2	-	-	-	-	-	-	-	-
MidPoint	10	8	-	9	-	-	-	-
M. MidPoint	-	-	-	-	-	-	-	-

5.2 Comparison of the taken path and control

Table 13: $x_0 = [0.2, 1.0]$

Method	$\alpha = 10.0$		$\alpha = 5.0$		$\alpha = 1.0$		$\alpha = 0.5$	
	$\Delta t = 0.2$	$\Delta t = 0.5$	$\Delta t = 0.2$	$\Delta t = 0.5$	$\Delta t = 0.2$	$\Delta t = 0.5$	$\Delta t = 0.2$	$\Delta t = 0.5$
SE1	8	8	8	10	-	-	-	-
M. SE1	7	7	10	7	-	-	-	-
SE2	-	-	-	-	-	-	-	-
M. SE2	-	-	-	-	-	-	-	-
MidPoint	-	8	-	10	-	-	-	-
M. MidPoint	-	-	-	-	-	-	-	-

Table 14: $x_0 = [0.3, 1.0]$

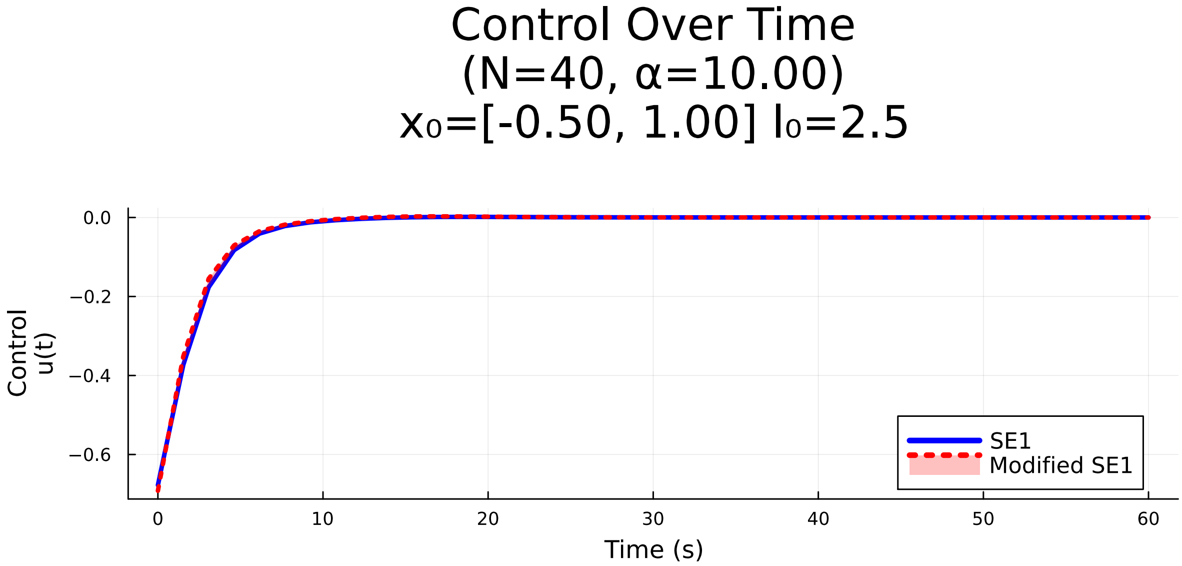
Method	$\alpha = 10.0$		$\alpha = 5.0$		$\alpha = 1.0$		$\alpha = 0.5$	
	$\Delta t = 0.2$	$\Delta t = 0.5$	$\Delta t = 0.2$	$\Delta t = 0.5$	$\Delta t = 0.2$	$\Delta t = 0.5$	$\Delta t = 0.2$	$\Delta t = 0.5$
SE1	7	9	8	9	-	-	-	-
M. SE1	7	7	8	7	-	-	-	-
SE2	-	-	-	-	-	-	-	-
M. SE2	-	-	-	-	-	-	-	-
MidPoint	11	9	-	9	-	-	-	-
M. MidPoint	-	-	-	-	-	-	-	-

Table 15: $x_0 = [0.4, 1.0]$

Method	$\alpha = 10.0$		$\alpha = 5.0$		$\alpha = 1.0$		$\alpha = 0.5$	
	$\Delta t = 0.2$	$\Delta t = 0.5$	$\Delta t = 0.2$	$\Delta t = 0.5$	$\Delta t = 0.2$	$\Delta t = 0.5$	$\Delta t = 0.2$	$\Delta t = 0.5$
SE1	7	10	8	12	-	-	-	-
M. SE1	7	7	8	7	-	-	-	212
SE2	-	-	-	-	-	-	-	-
M. SE2	-	-	-	-	-	-	-	-
MidPoint	-	10	11	12	-	-	-	-
M. MidPoint	-	-	-	-	-	-	-	-

Table 16: $x_0 = [0.5, 1.0]$

Method	$\alpha = 10.0$		$\alpha = 5.0$		$\alpha = 1.0$		$\alpha = 0.5$	
	$\Delta t = 0.2$	$\Delta t = 0.5$	$\Delta t = 0.2$	$\Delta t = 0.5$	$\Delta t = 0.2$	$\Delta t = 0.5$	$\Delta t = 0.2$	$\Delta t = 0.5$
SE1	7	9	11	-	-	-	-	-
M. SE1	7	7	8	7	-	-	-	-
SE2	-	-	-	-	-	-	-	-
M. SE2	-	-	-	-	-	-	-	-
MidPoint	-	9	-	-	-	-	-	-
M. MidPoint	-	-	-	-	-	-	-	-



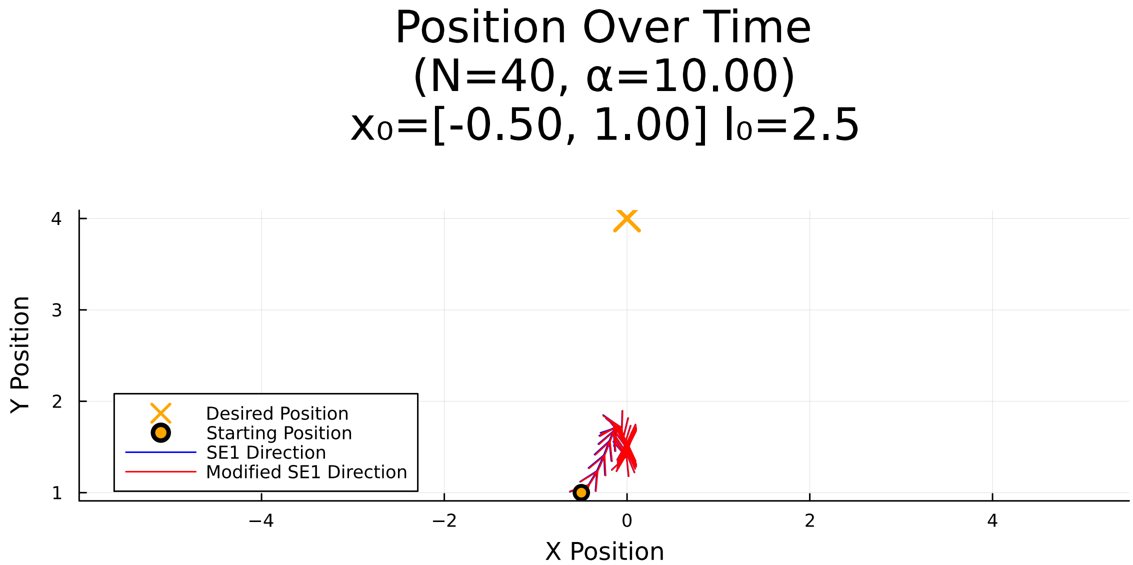
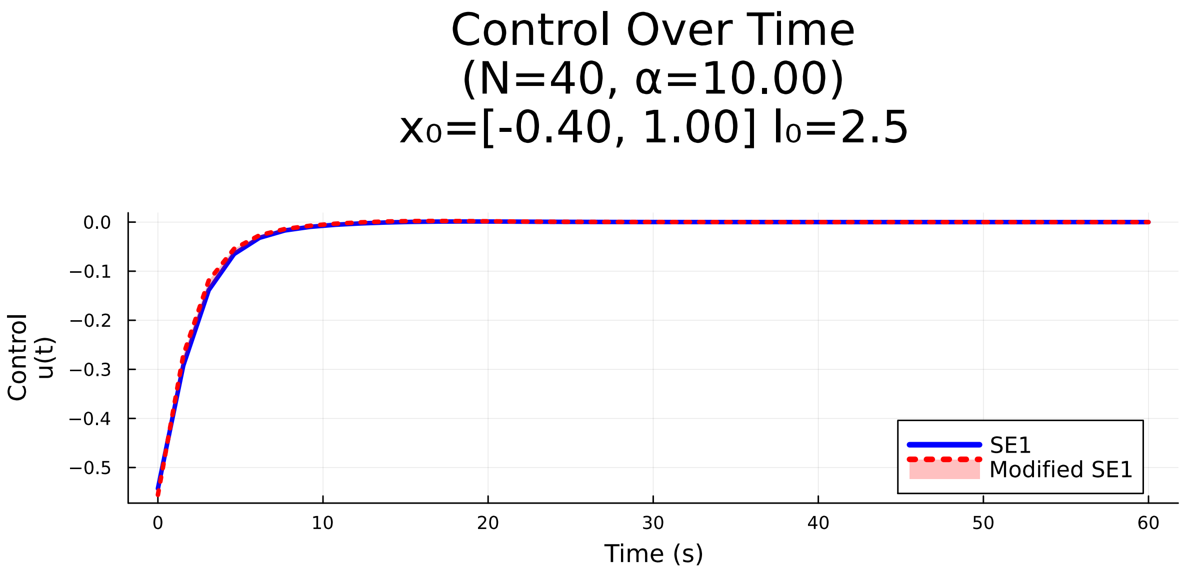


Figure 6: Comparison of SE1 and Modified SE1 for $\alpha = 10.0$, $N = 40$, $l_0 = 2.5$, and $\boldsymbol{x}_0 = [-0.5, 1.0]$



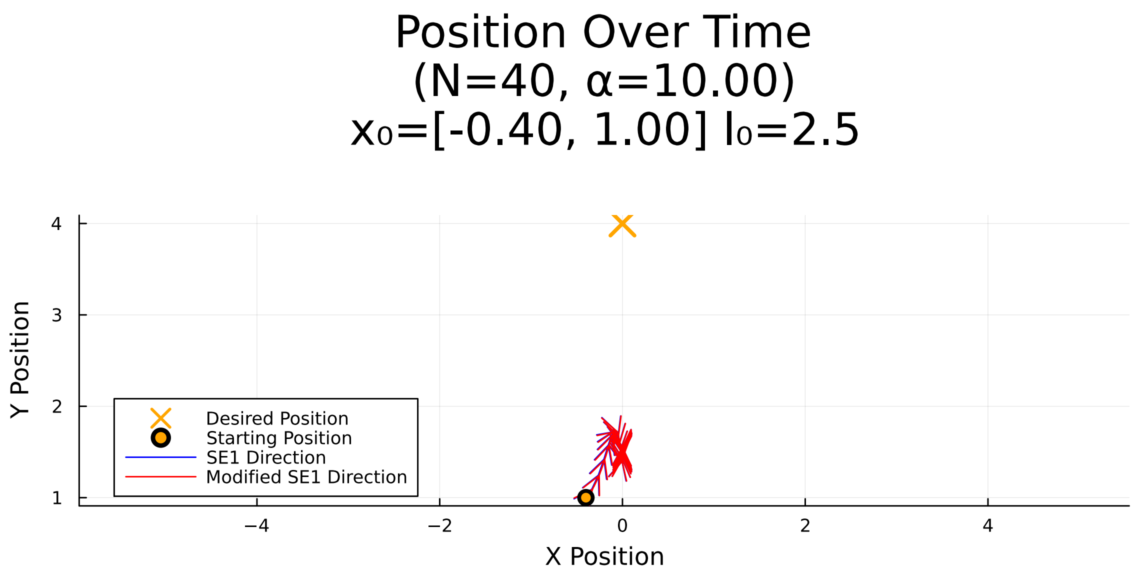


Figure 7: Comparison of SE1 and Modified SE1 for $\alpha = 10.0$, $N = 40$, $l_0 = 2.5$, and $\boldsymbol{x}_0 = [-0.4, 1.0]$

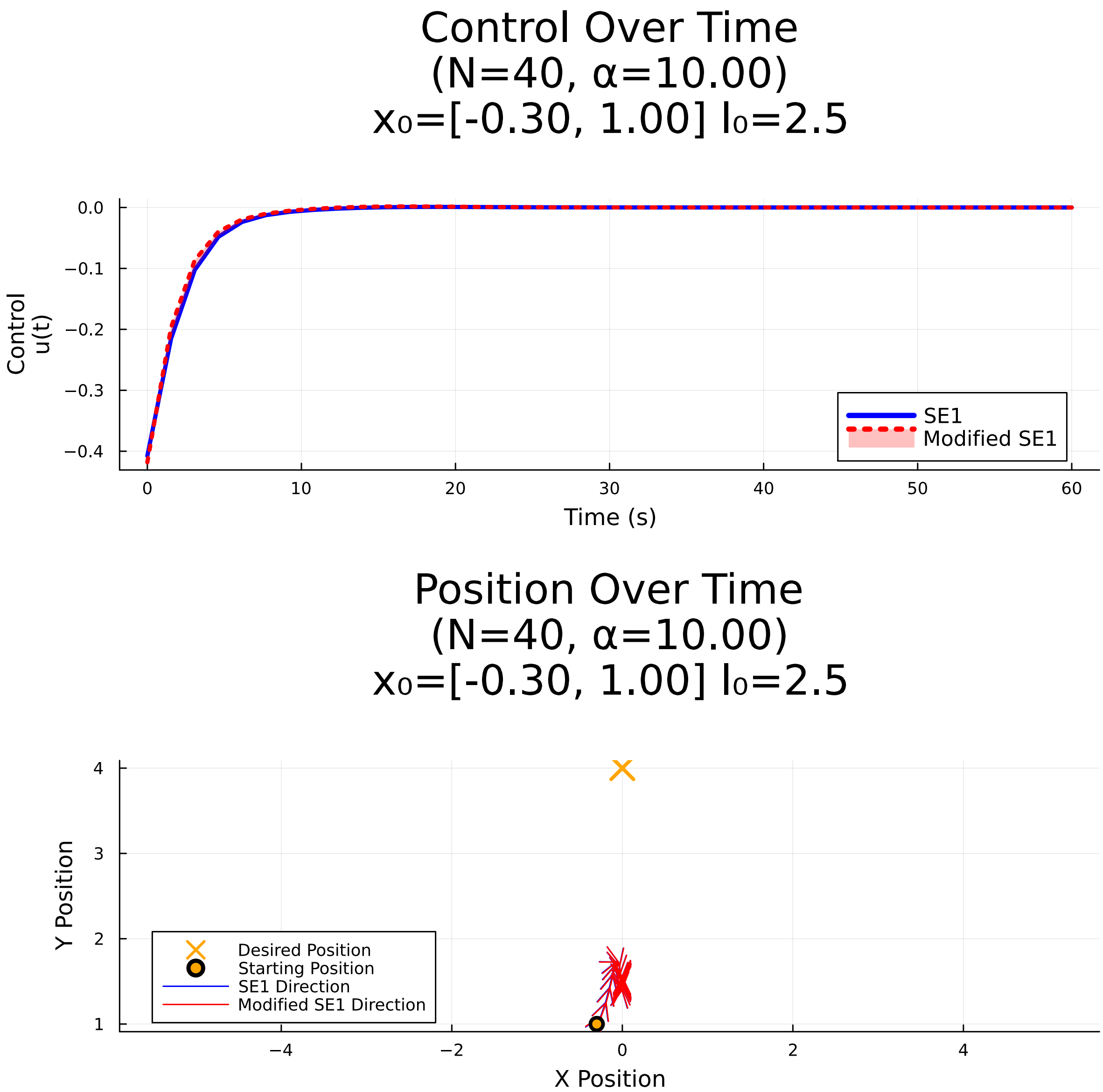
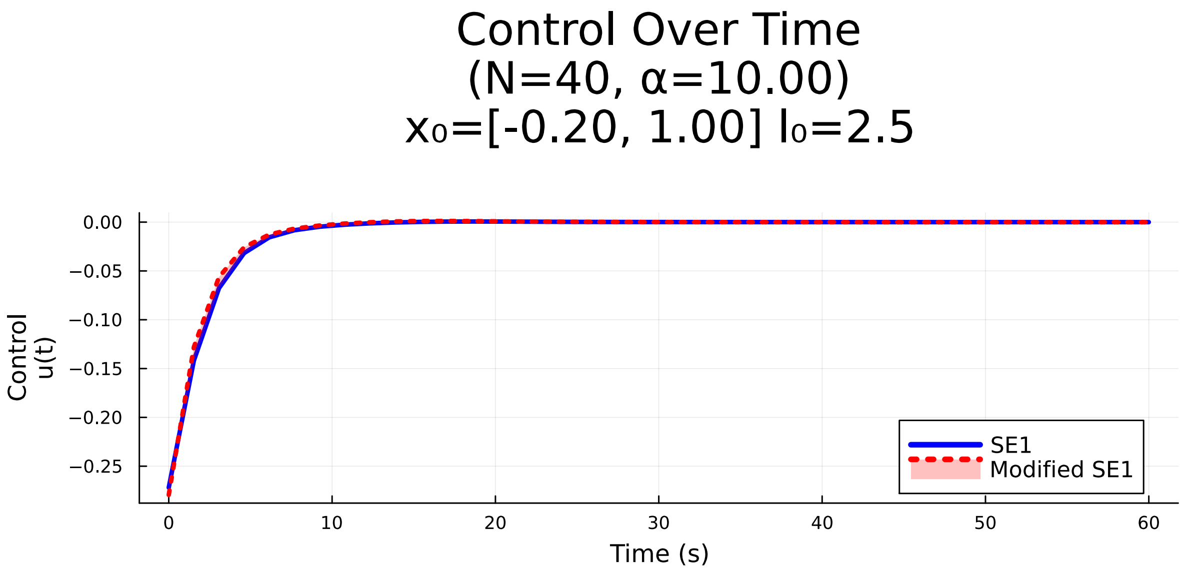


Figure 8: Comparison of SE1 and Modified SE1 for $\alpha = 10.0$, $N = 40$, $l_0 = 2.5$, and $\boldsymbol{x}_0 = [-0.3, 1.0]$



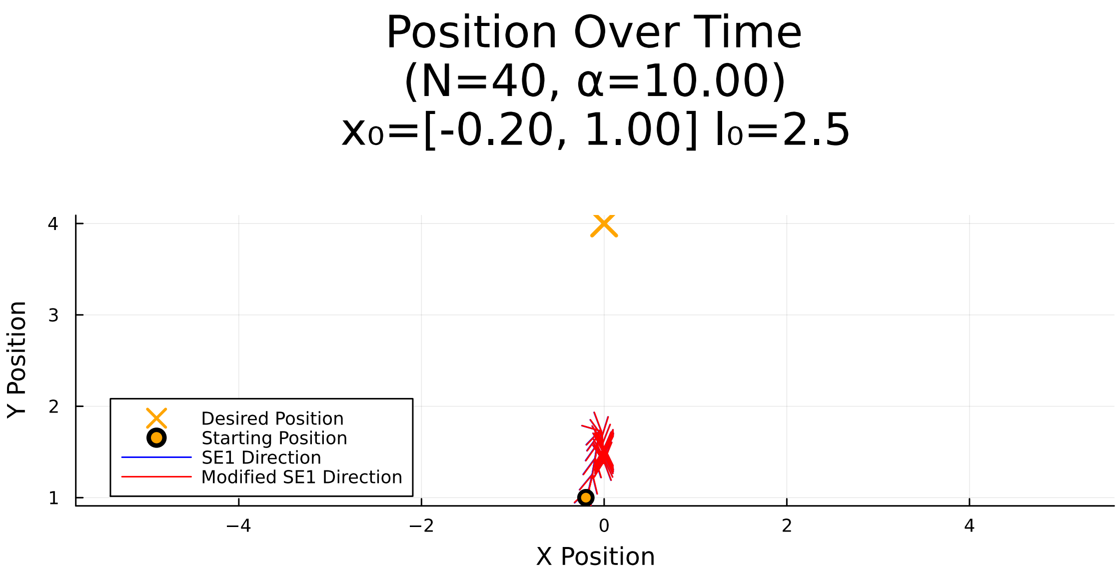
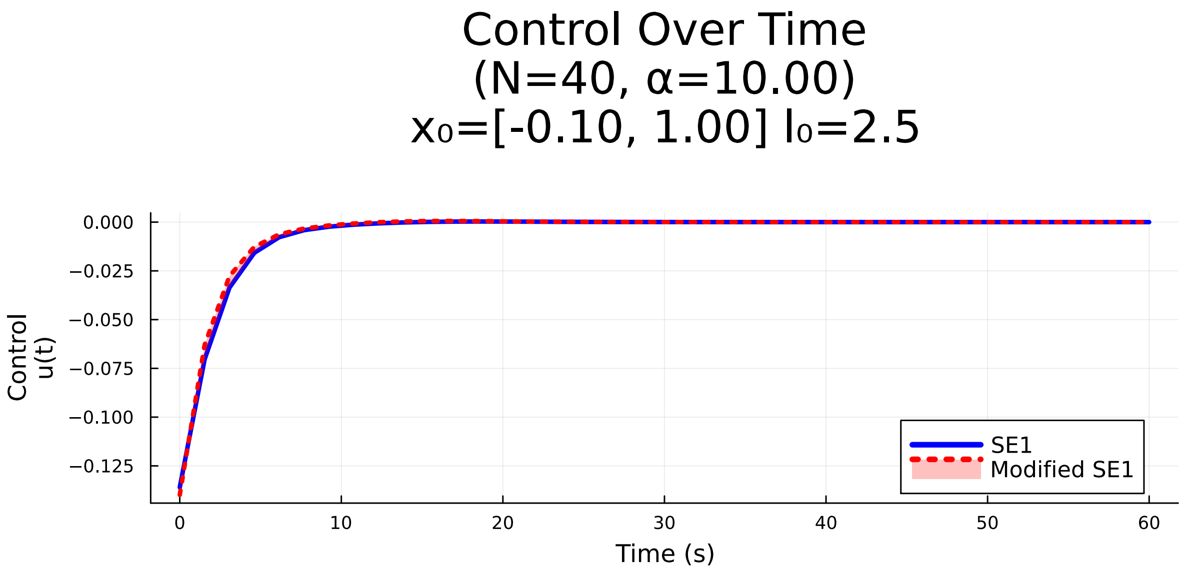


Figure 9: Comparison of SE1 and Modified SE1 for $\alpha = 10.0$, $N = 40$, $l_0 = 2.5$, and $\boldsymbol{x}_0 = [-0.2, 1.0]$



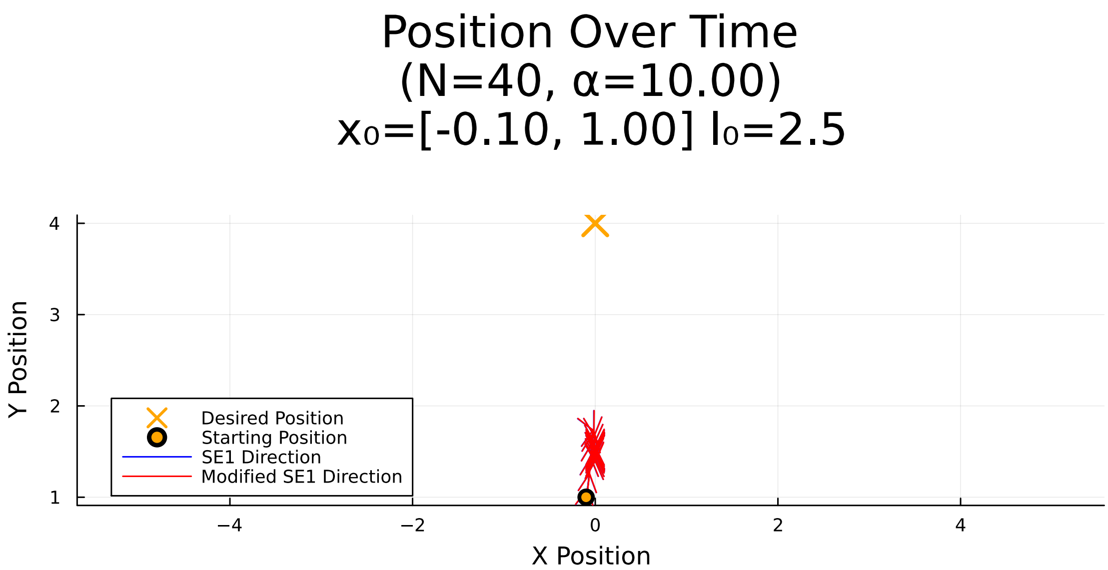
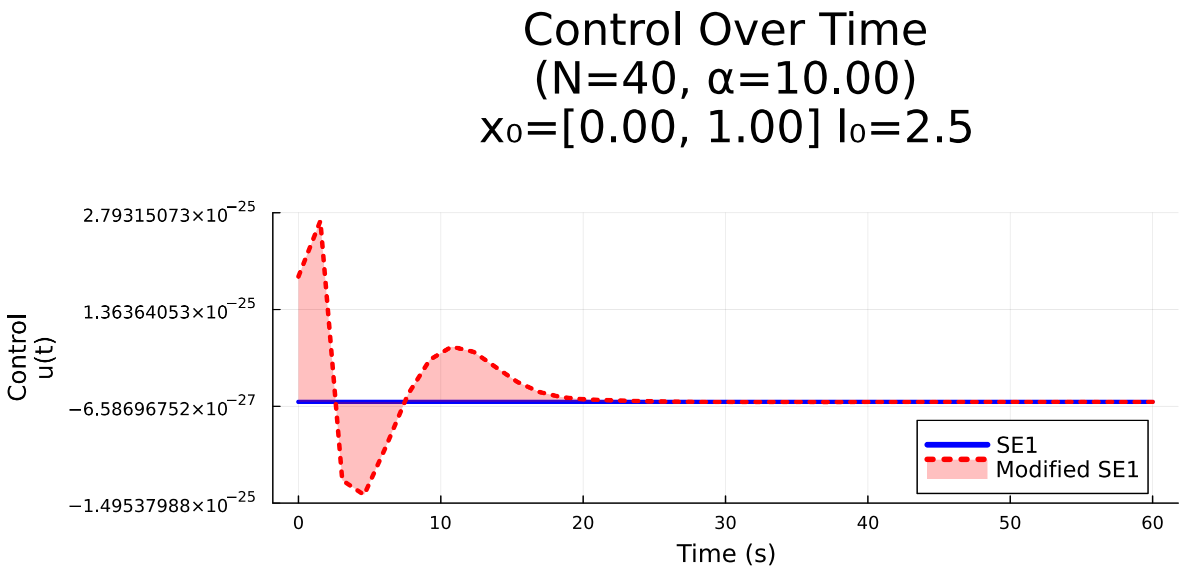


Figure 10: Comparison of SE1 and Modified SE1 for $\alpha = 10.0$, $N = 40$, $l_0 = 2.5$, and $\boldsymbol{x}_0 = [-0.1, 1.0]$



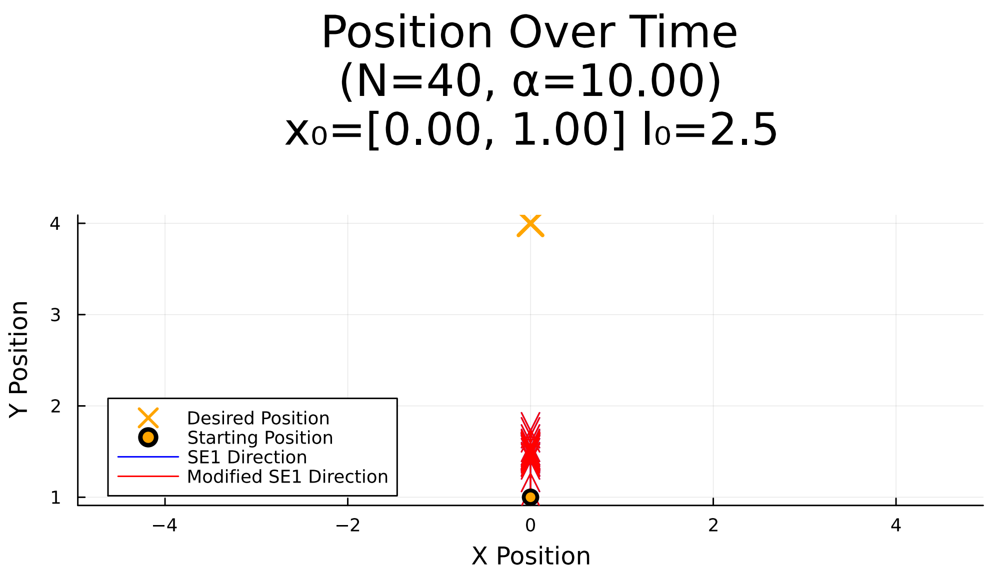


Figure 11: Comparison of SE1 and Modified SE1 for $\alpha = 10.0$, $N = 40$, $l_0 = 2.5$, and $\boldsymbol{x}_0 = [0.0, 1.0]$

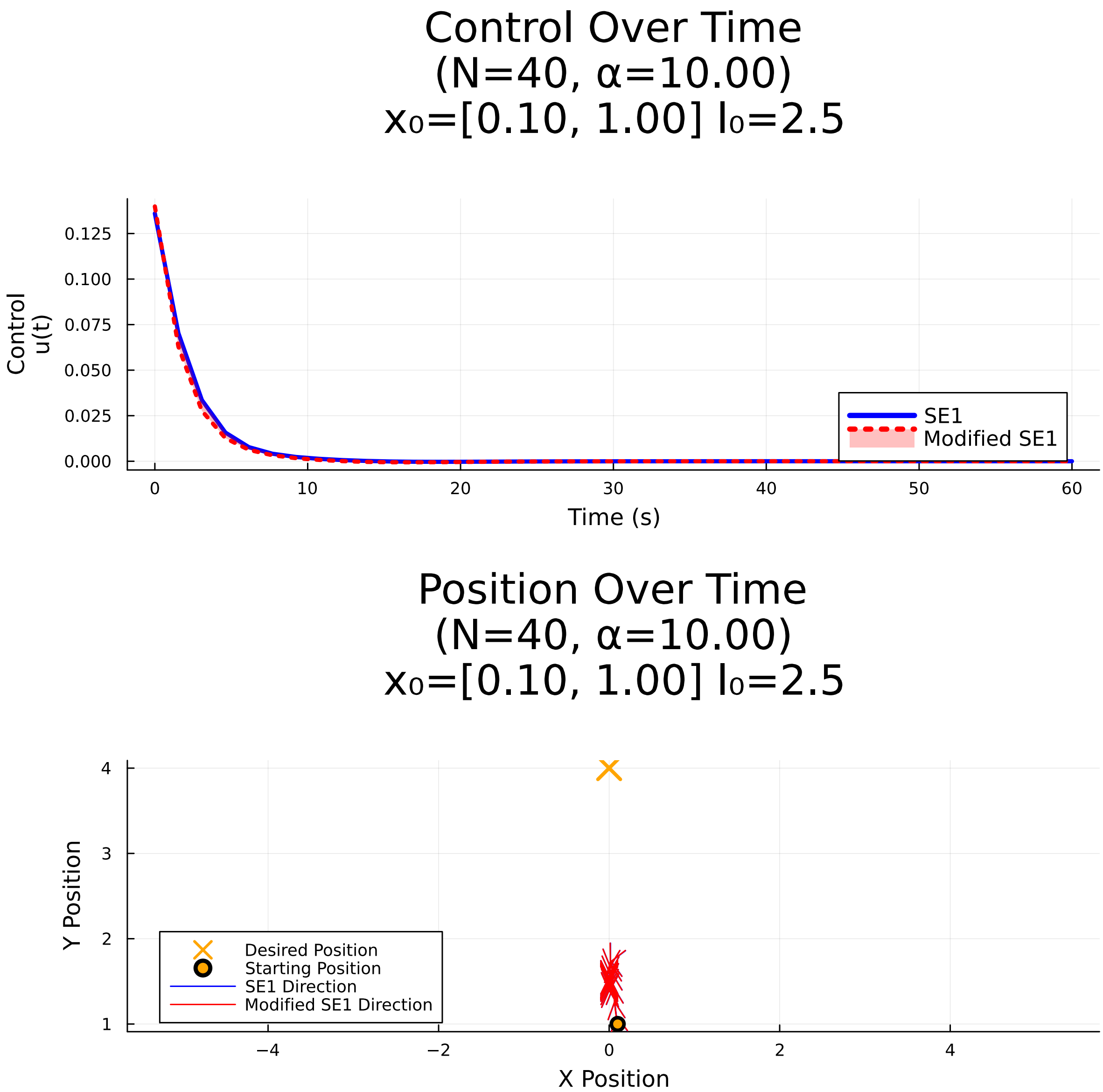


Figure 12: Comparison of SE1 and Modified SE1 for $\alpha = 10.0$, $N = 40$, $l_0 = 2.5$, and $\boldsymbol{x}_0 = [0.1, 1.0]$

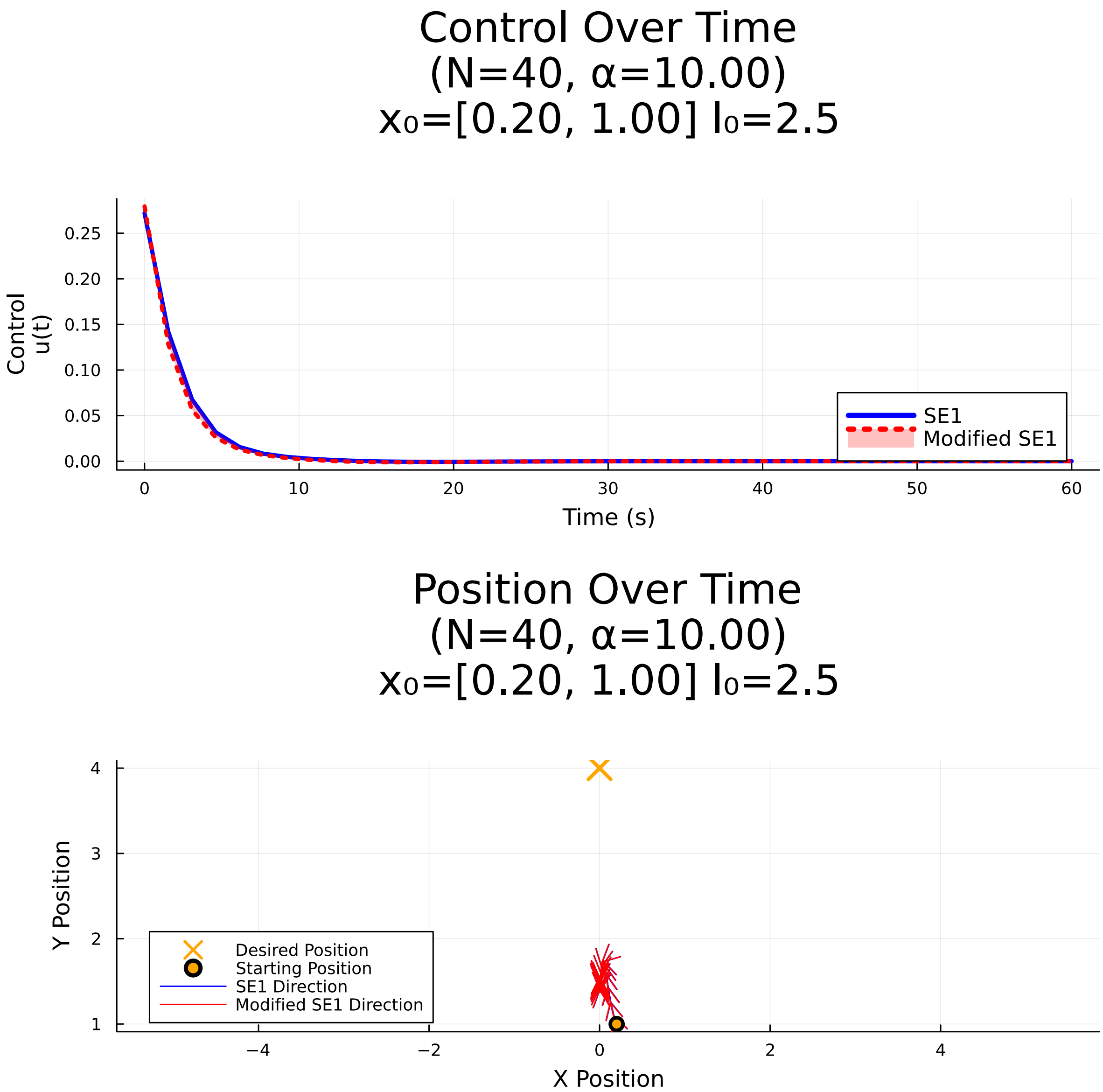


Figure 13: Comparison of SE1 and Modified SE1 for $\alpha = 10.0$, $N = 40$, $l_0 = 2.5$, and $\boldsymbol{x}_0 = [0.2, 1.0]$

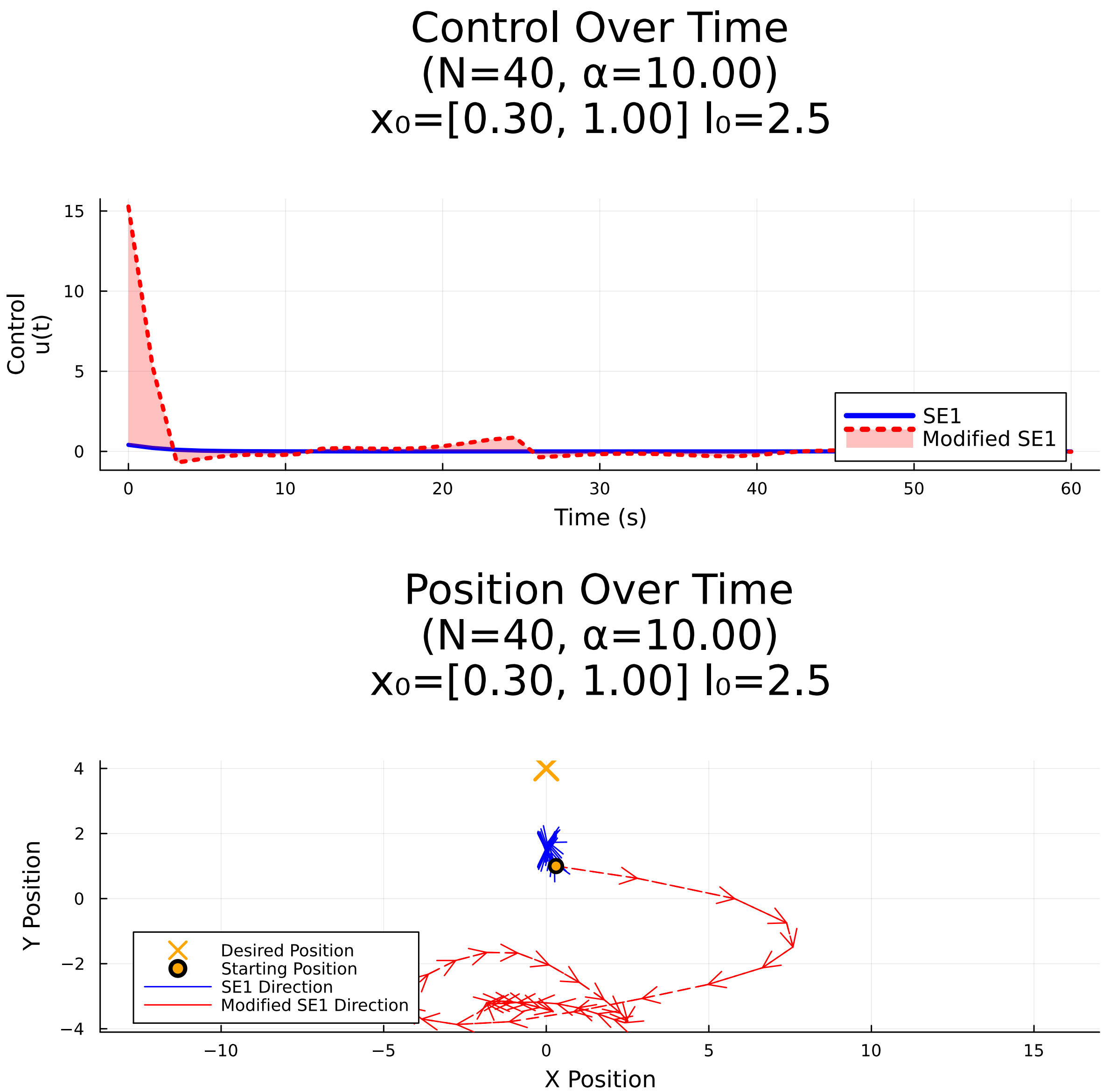


Figure 14: Comparison of SE1 and Modified SE1 for $\alpha = 10.0$, $N = 40$, $l_0 = 2.5$, and $\boldsymbol{x}_0 = [0.3, 1.0]$

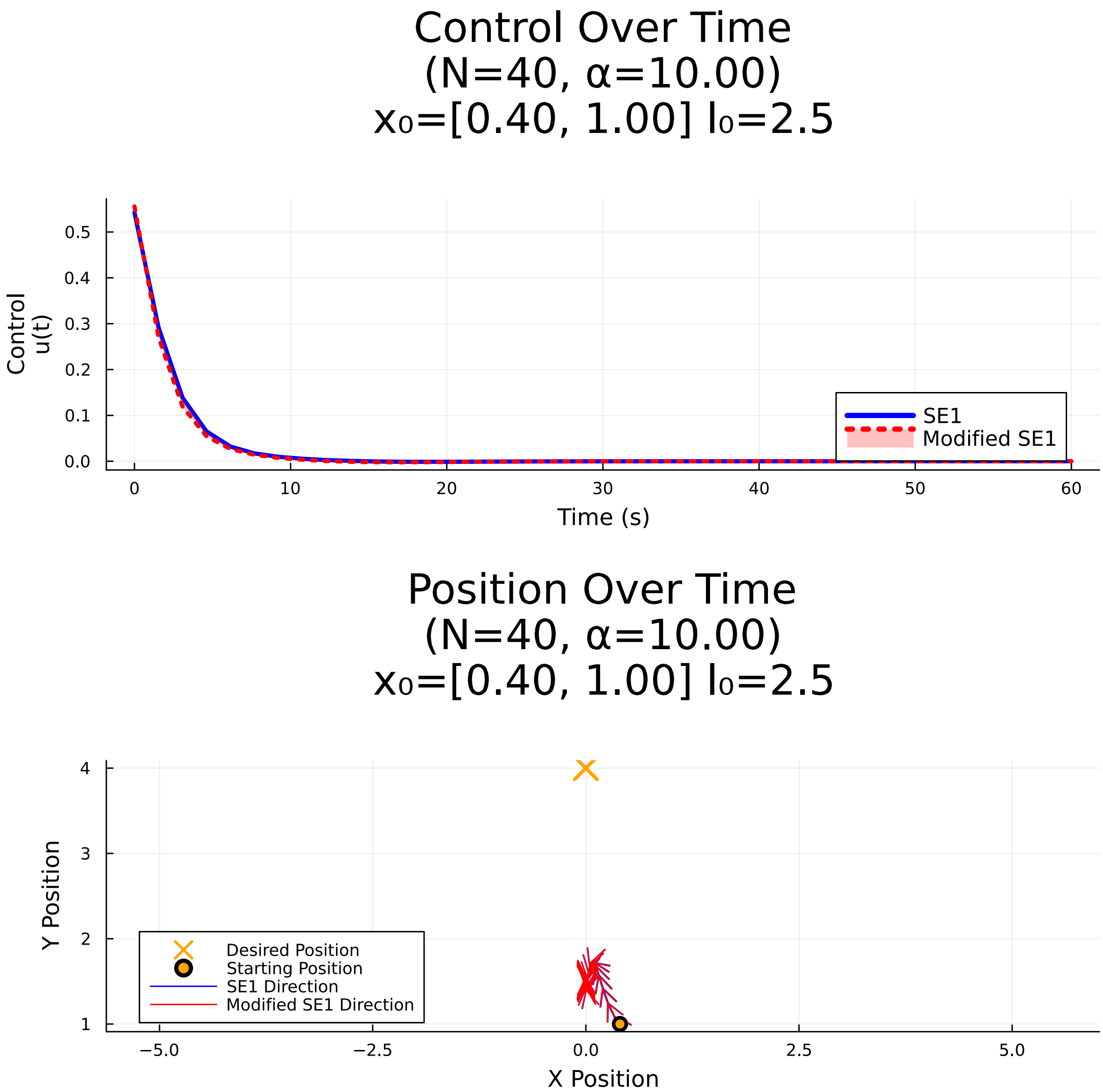


Figure 15: Comparison of SE1 and Modified SE1 for $\alpha = 10.0$, $N = 40$, $l_0 = 2.5$, and $\boldsymbol{x}_0 = [0.4, 1.0]$

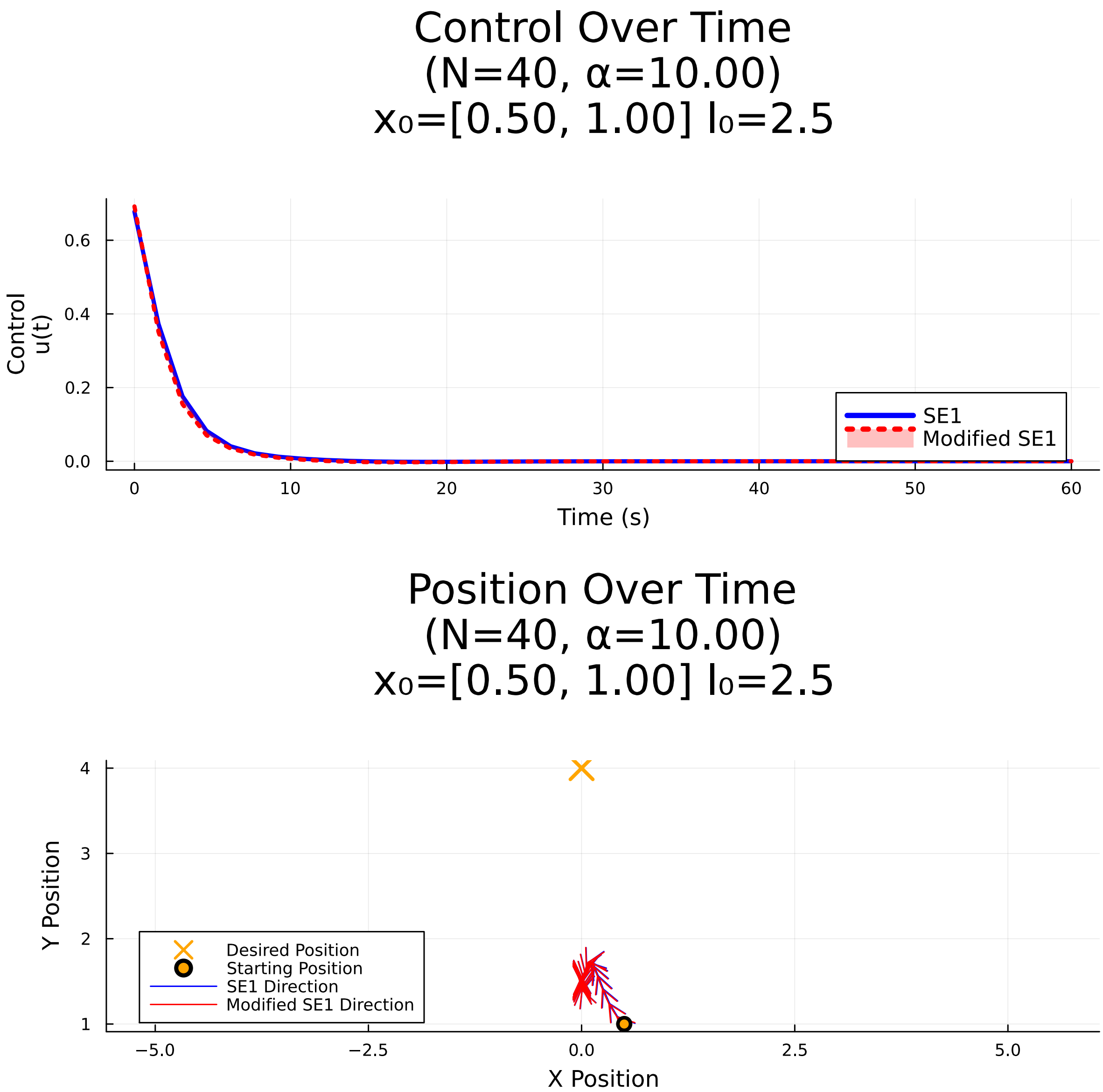
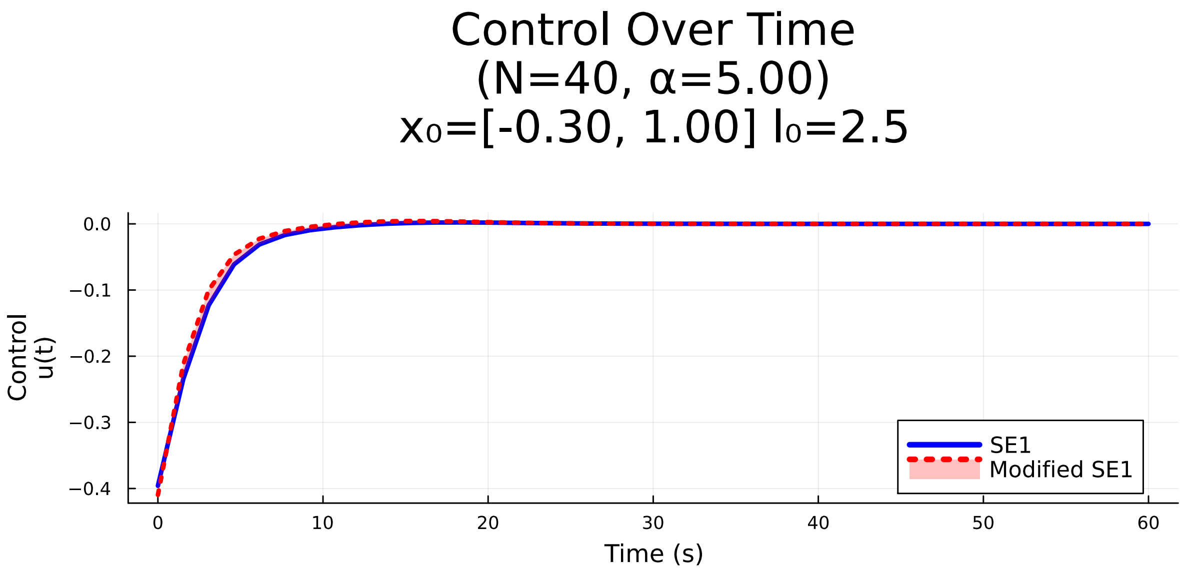


Figure 16: Comparison of SE1 and Modified SE1 for $\alpha = 10.0$, $N = 40$, $l_0 = 2.5$, and $\boldsymbol{x}_0 = [0.5, 1.0]$



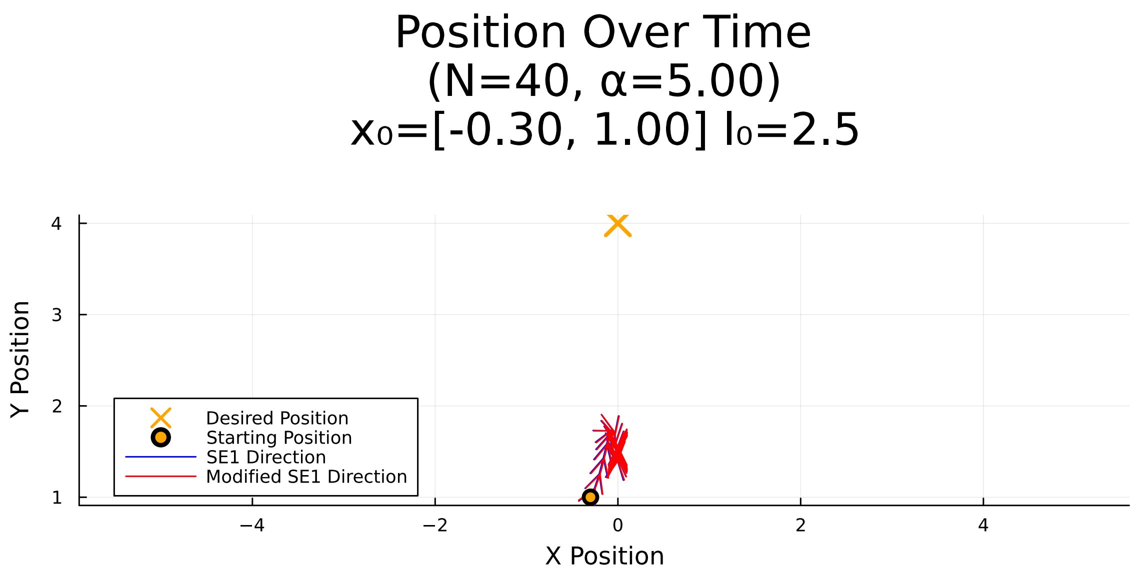


Figure 17: Comparison of SE1 and Modified SE1 for $\alpha = 5.0$, $N = 40$, $l_0 = 2.5$, and $\boldsymbol{x}_0 = [-0.3, 1.0]$

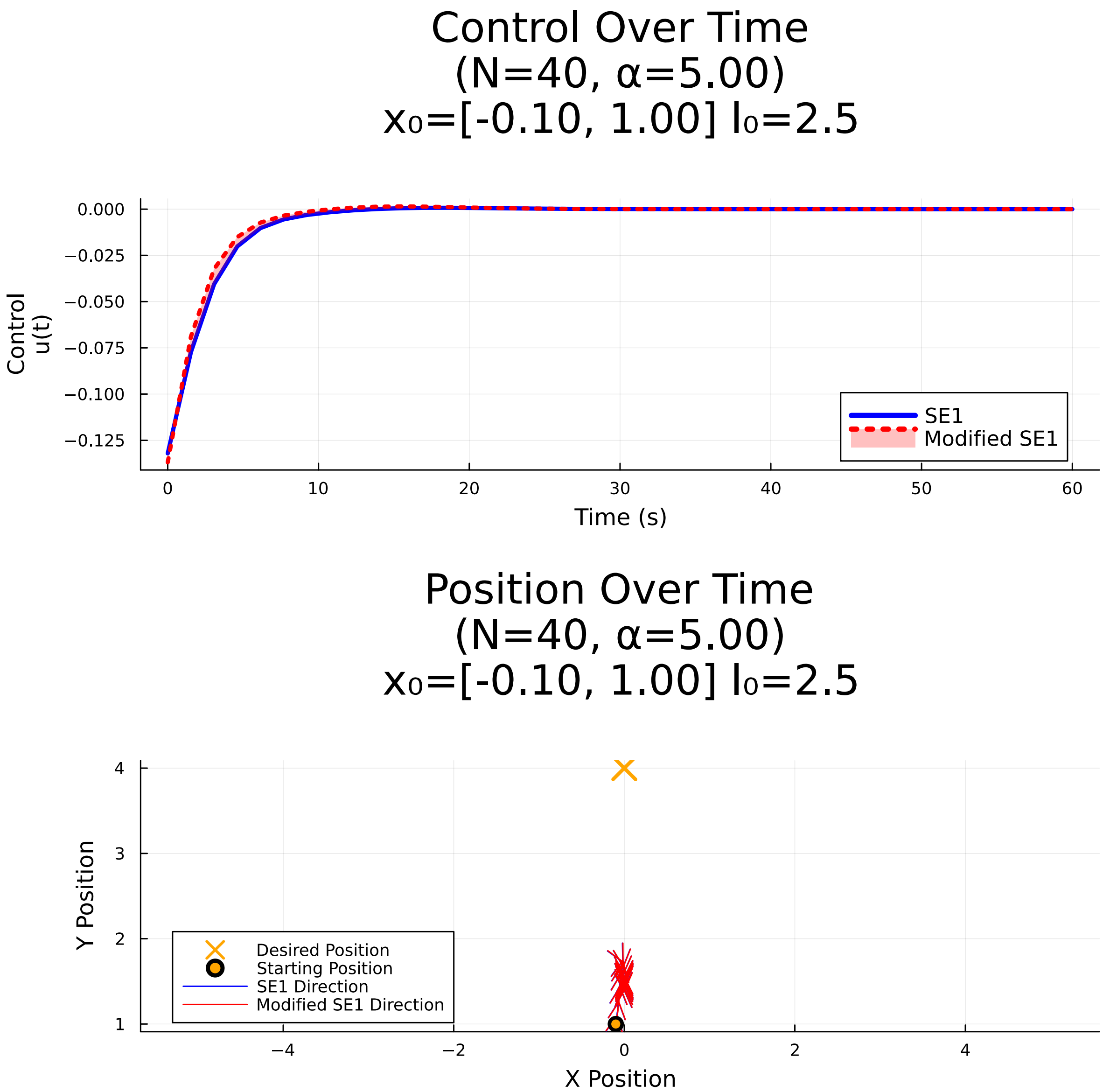


Figure 18: Comparison of SE1 and Modified SE1 for $\alpha = 5.0$, $N = 40$, $l_0 = 2.5$, and $\boldsymbol{x}_0 = [-0.1, 1.0]$

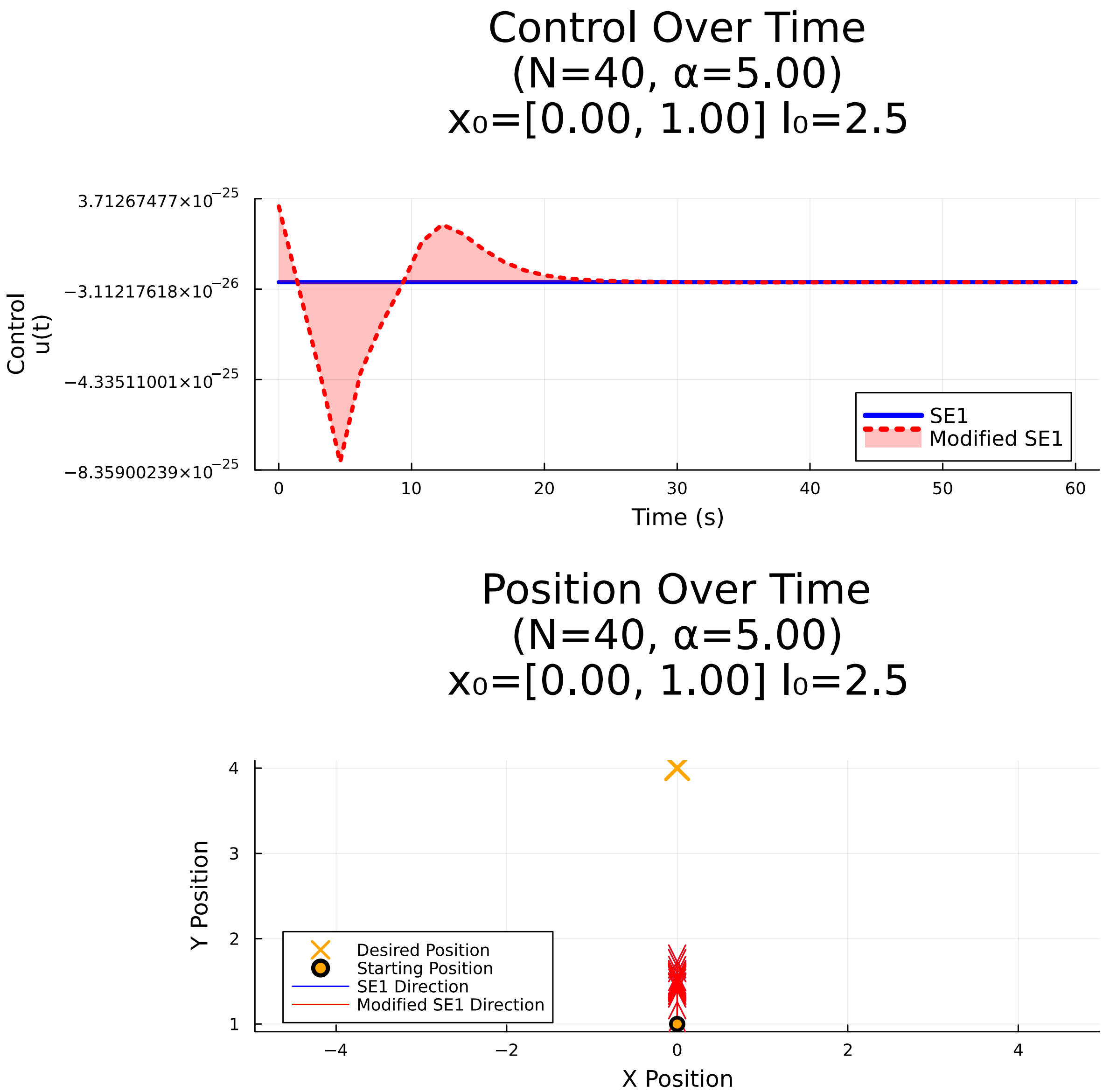


Figure 19: Comparison of SE1 and Modified SE1 for $\alpha = 5.0$, $N = 40$, $l_0 = 2.5$, and $\boldsymbol{x}_0 = [0.0, 1.0]$

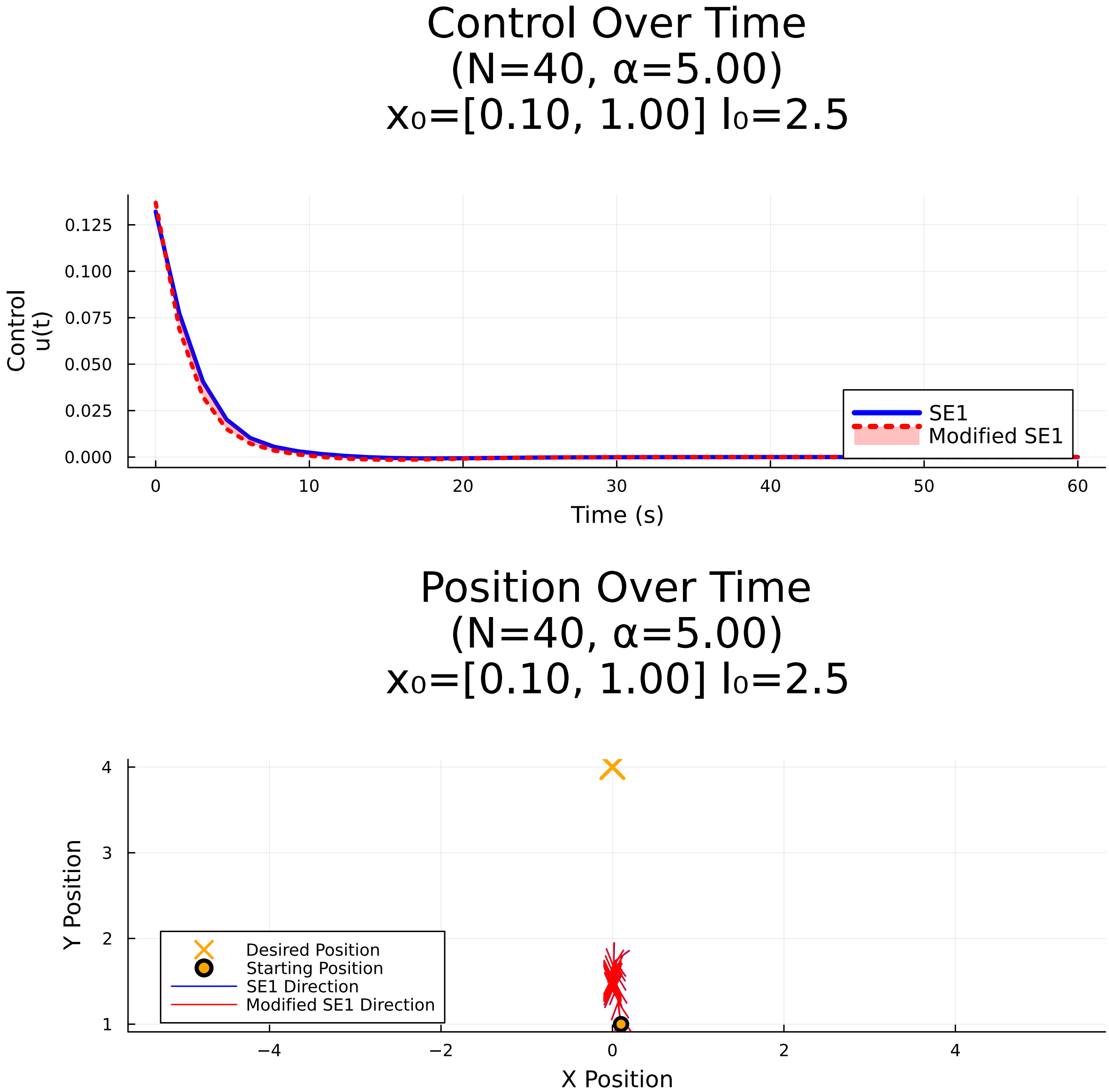


Figure 20: Comparison of SE1 and Modified SE1 for $\alpha = 5.0$, $N = 40$, $l_0 = 2.5$, and $\boldsymbol{x}_0 = [0.1, 1.0]$

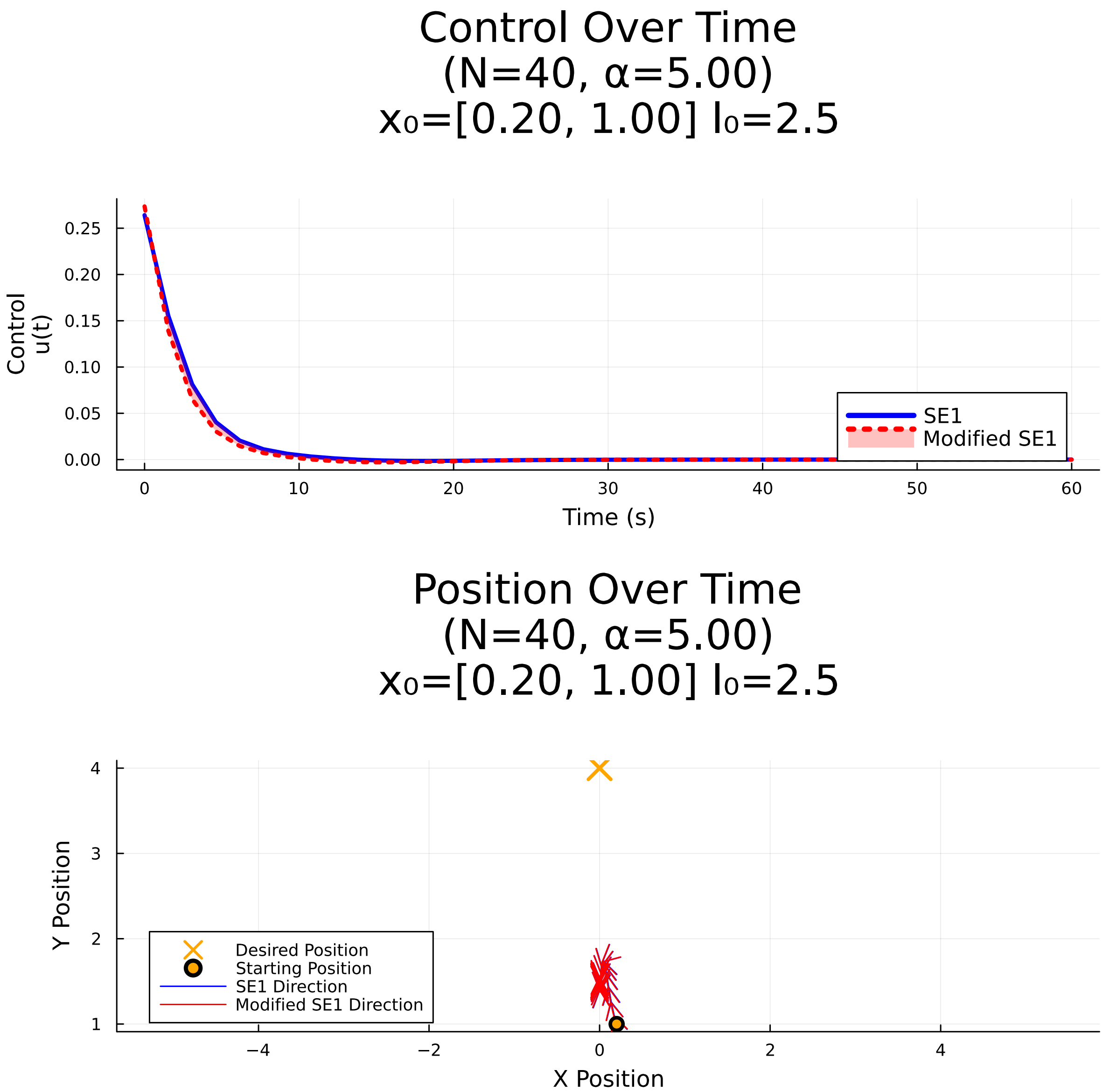


Figure 21: Comparison of SE1 and Modified SE1 for $\alpha = 5.0$, $N = 40$, $l_0 = 2.5$, and $\boldsymbol{x}_0 = [0.2, 1.0]$

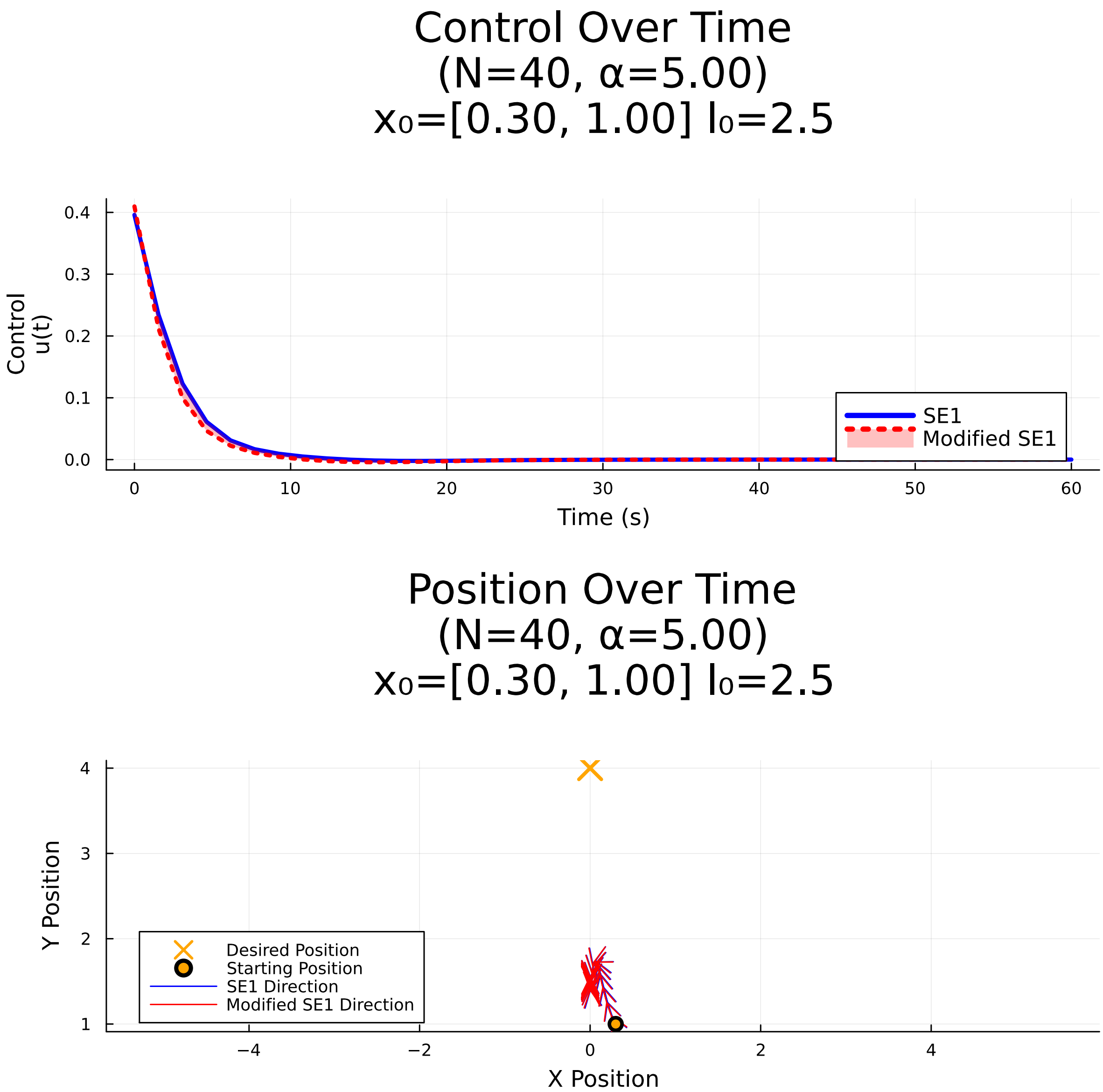


Figure 22: Comparison of SE1 and Modified SE1 for $\alpha = 5.0$, $N = 40$, $l_0 = 2.5$, and $\boldsymbol{x}_0 = [0.3, 1.0]$

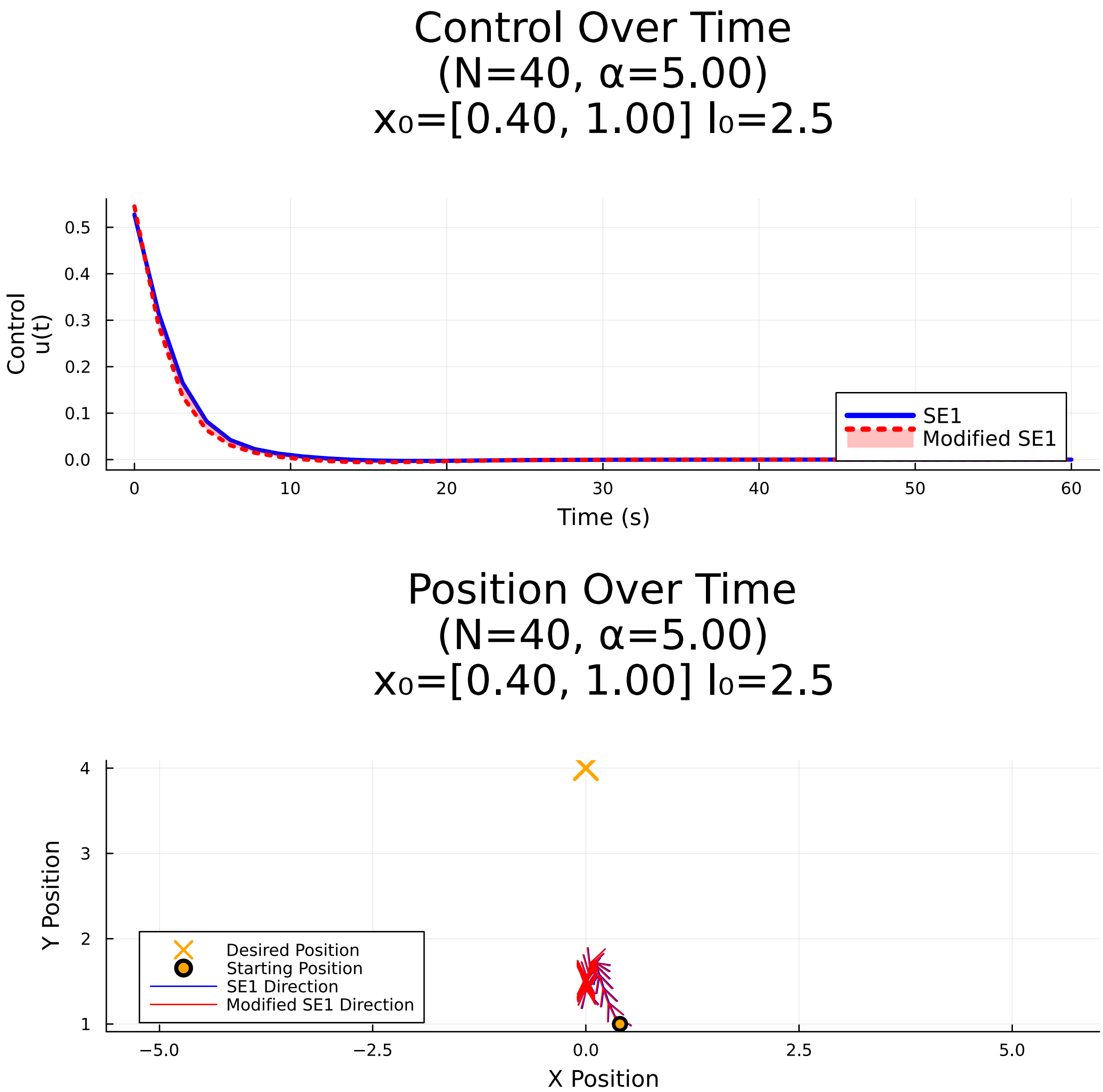
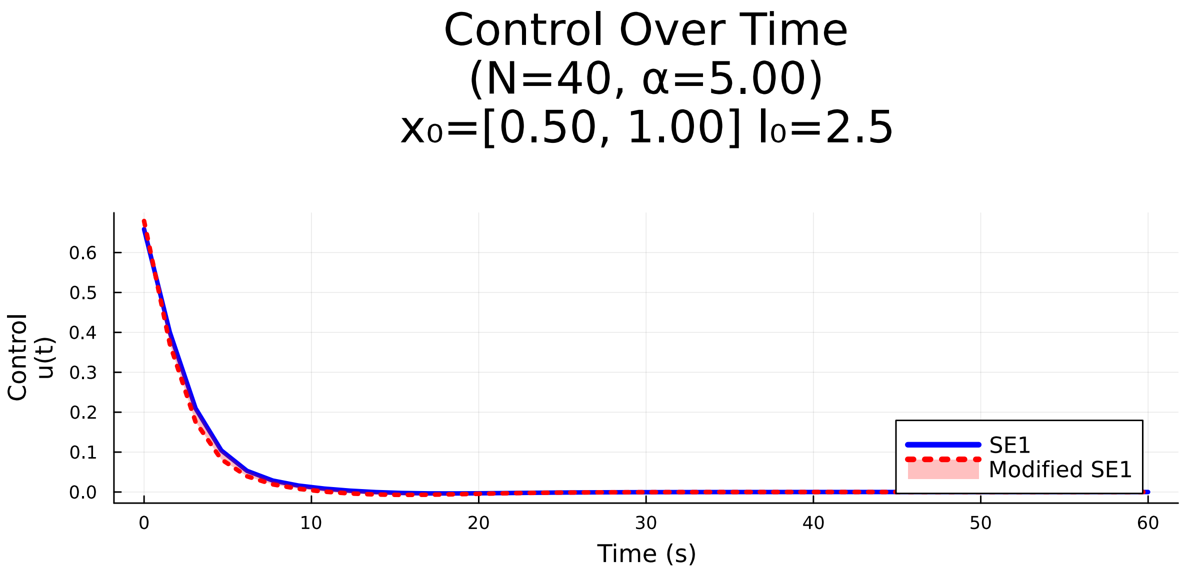


Figure 23: Comparison of SE1 and Modified SE1 for $\alpha = 5.0$, $N = 40$, $l_0 = 2.5$, and $\boldsymbol{x}_0 = [0.4, 1.0]$



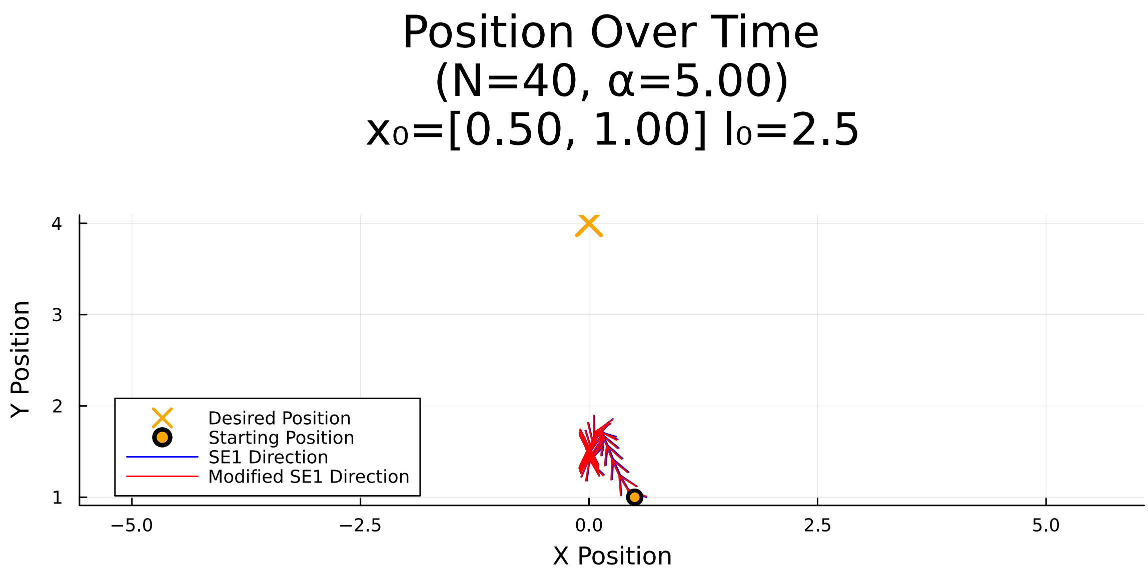
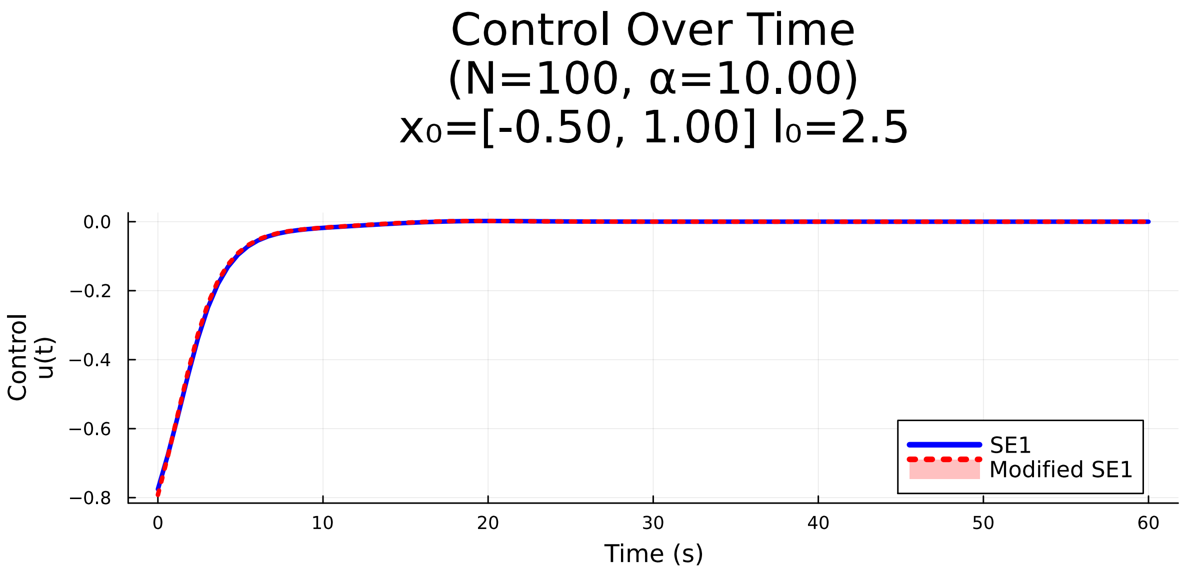


Figure 24: Comparison of SE1 and Modified SE1 for $\alpha = 5.0$, $N = 40$, $l_0 = 2.5$, and $\boldsymbol{x}_0 = [0.5, 1.0]$



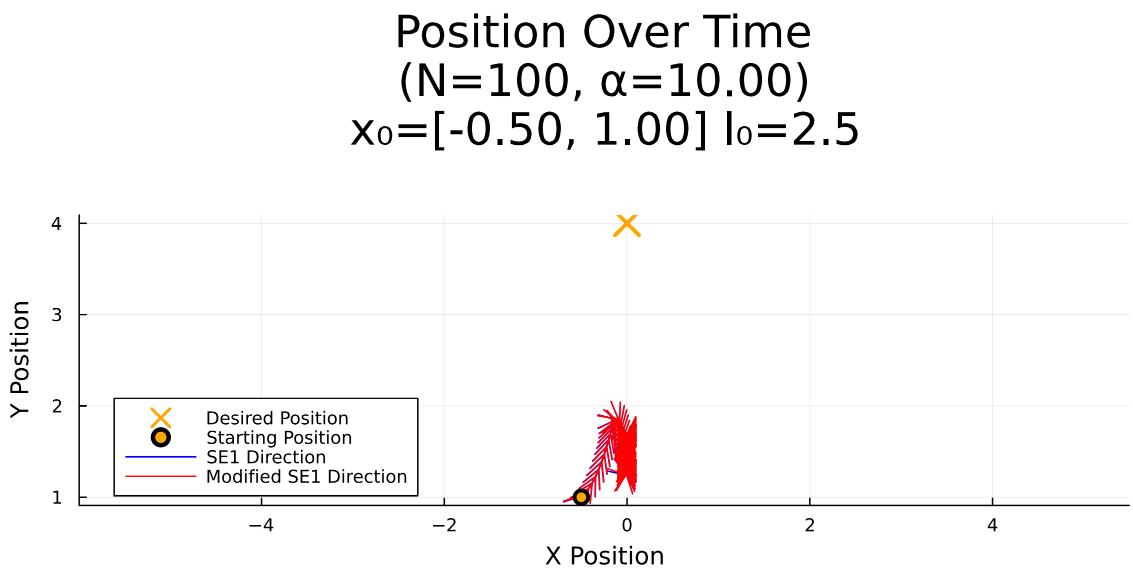


Figure 25: Comparison of SE1 and Modified SE1 for $\alpha = 10.0$, $N = 100$, $l_0 = 2.5$, and $\boldsymbol{x}_0 = [-0.5, 1.0]$

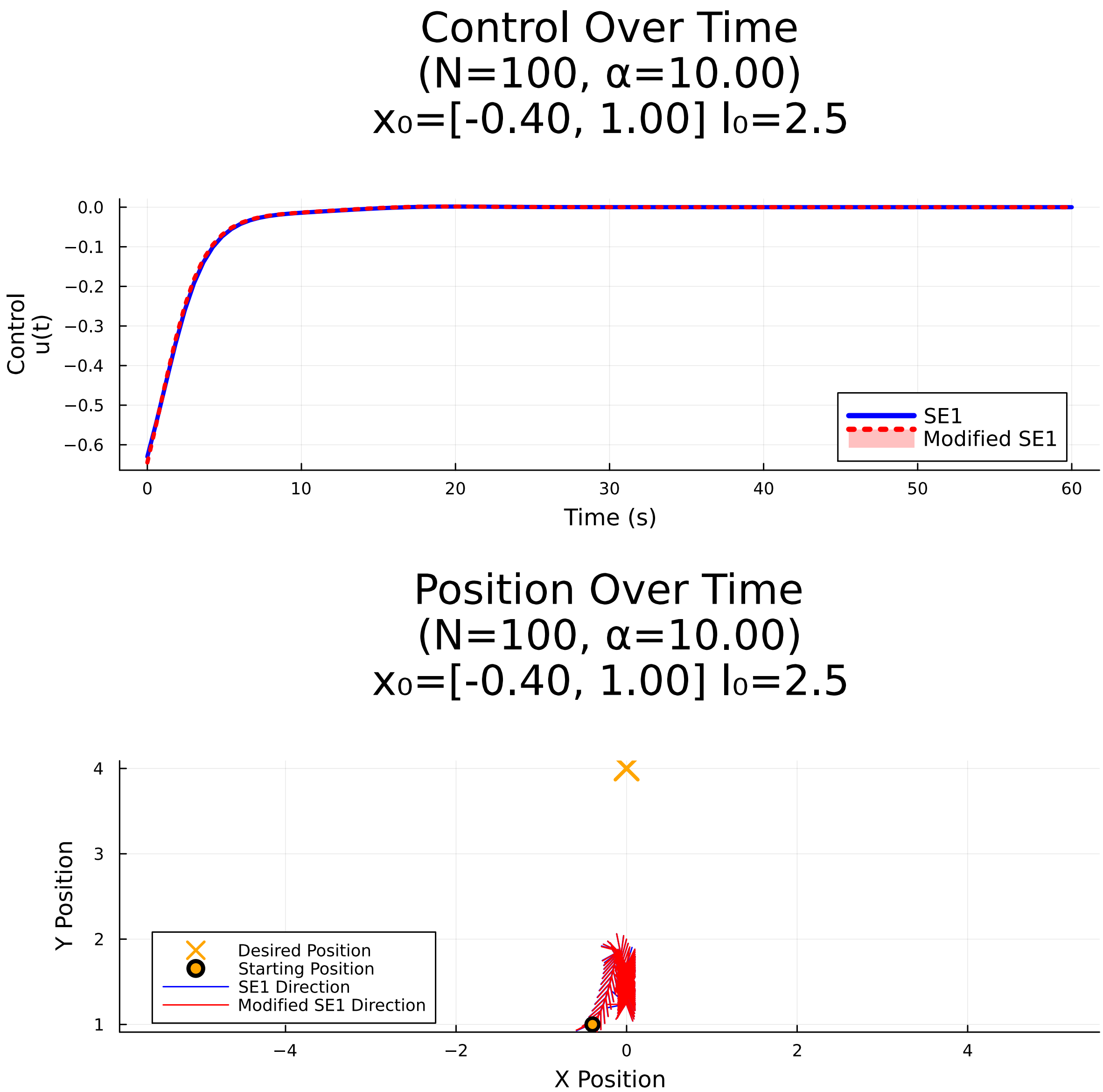


Figure 26: Comparison of SE1 and Modified SE1 for $\alpha = 10.0$, $N = 100$, $l_0 = 2.5$, and $\boldsymbol{x}_0 = [-0.4, 1.0]$

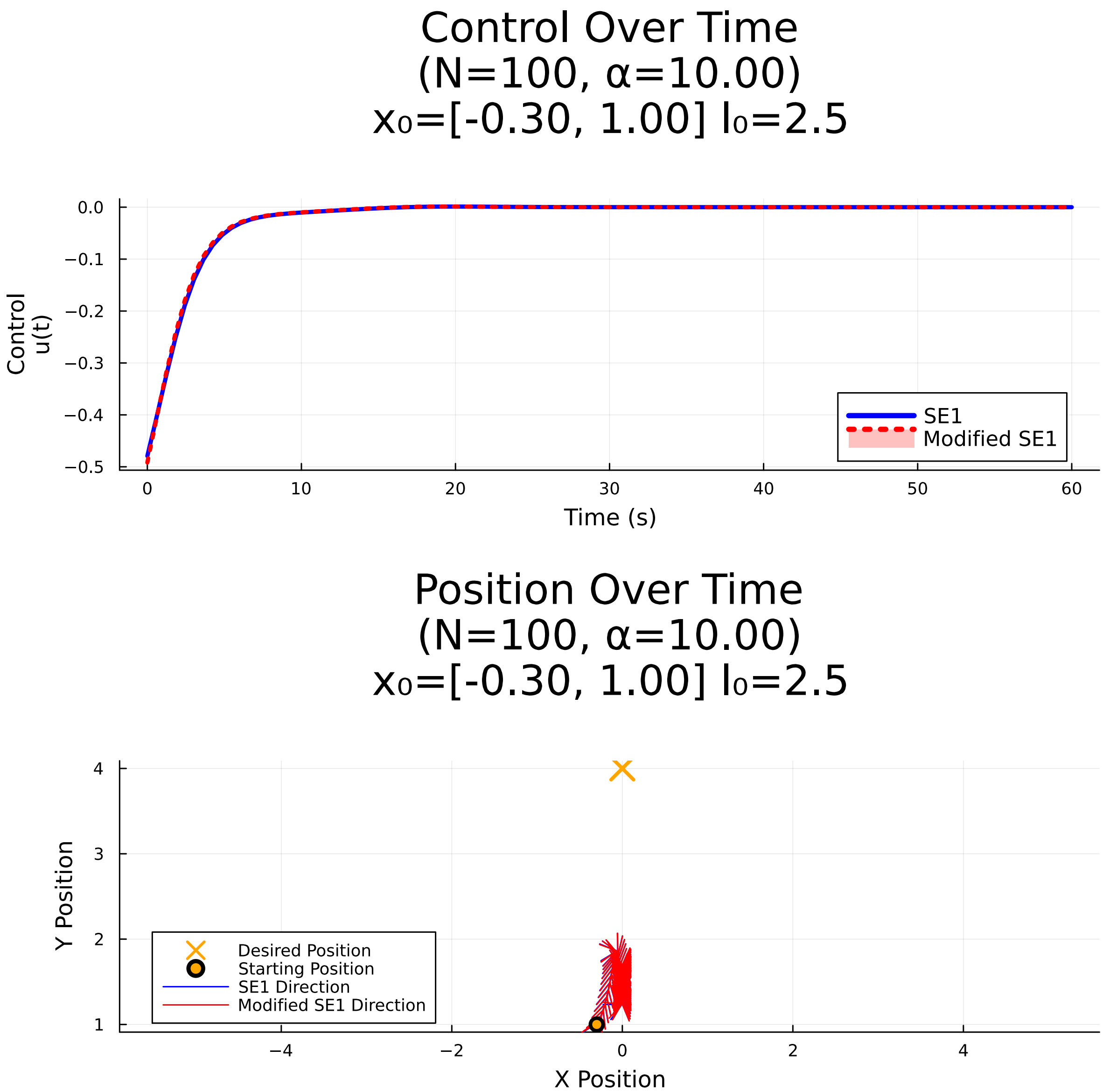


Figure 27: Comparison of SE1 and Modified SE1 for $\alpha = 10.0$, $N = 100$, $l_0 = 2.5$, and $\boldsymbol{x}_0 = [-0.3, 1.0]$

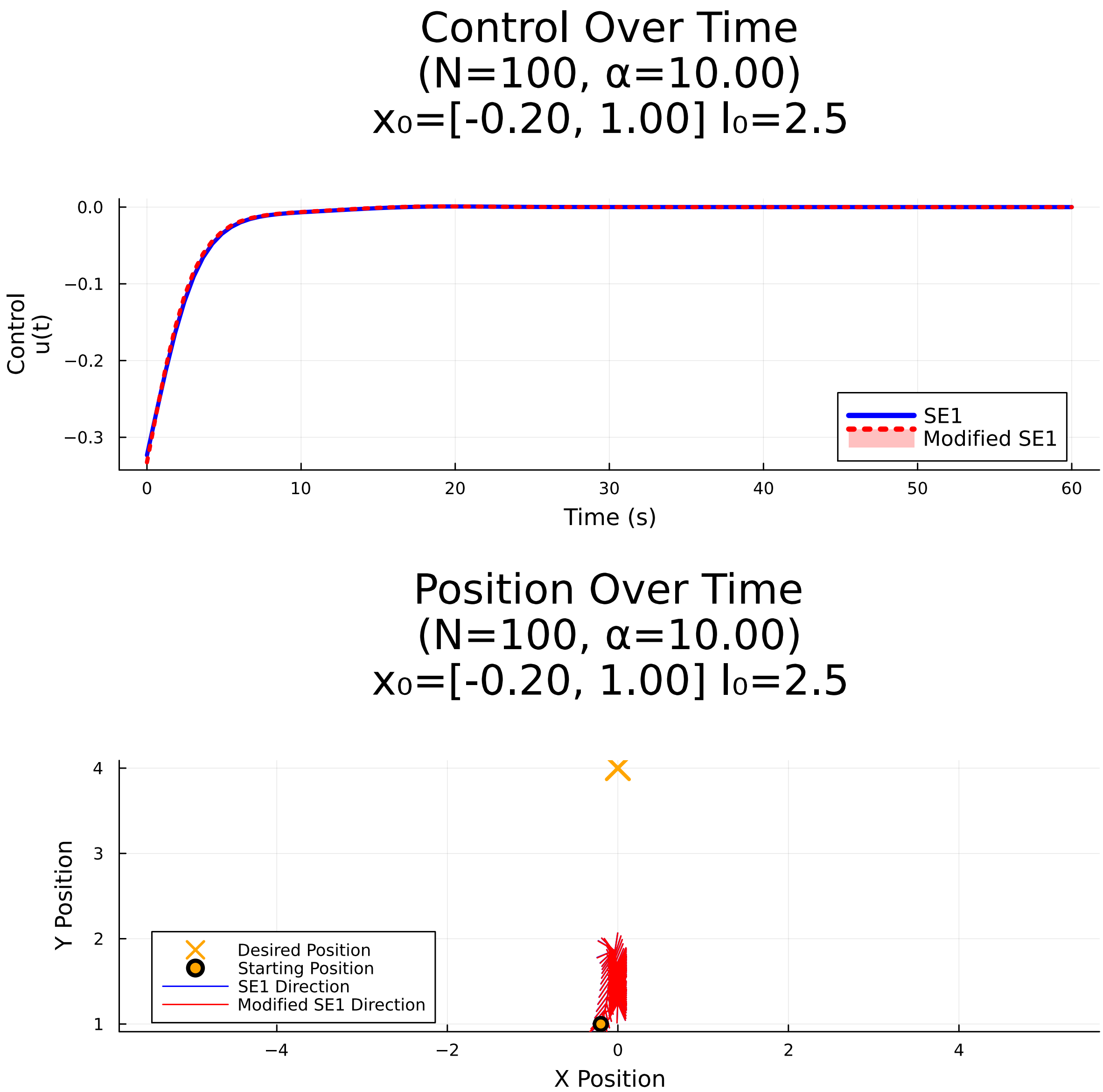


Figure 28: Comparison of SE1 and Modified SE1 for $\alpha = 10.0$, $N = 100$, $l_0 = 2.5$, and $\boldsymbol{x}_0 = [-0.2, 1.0]$

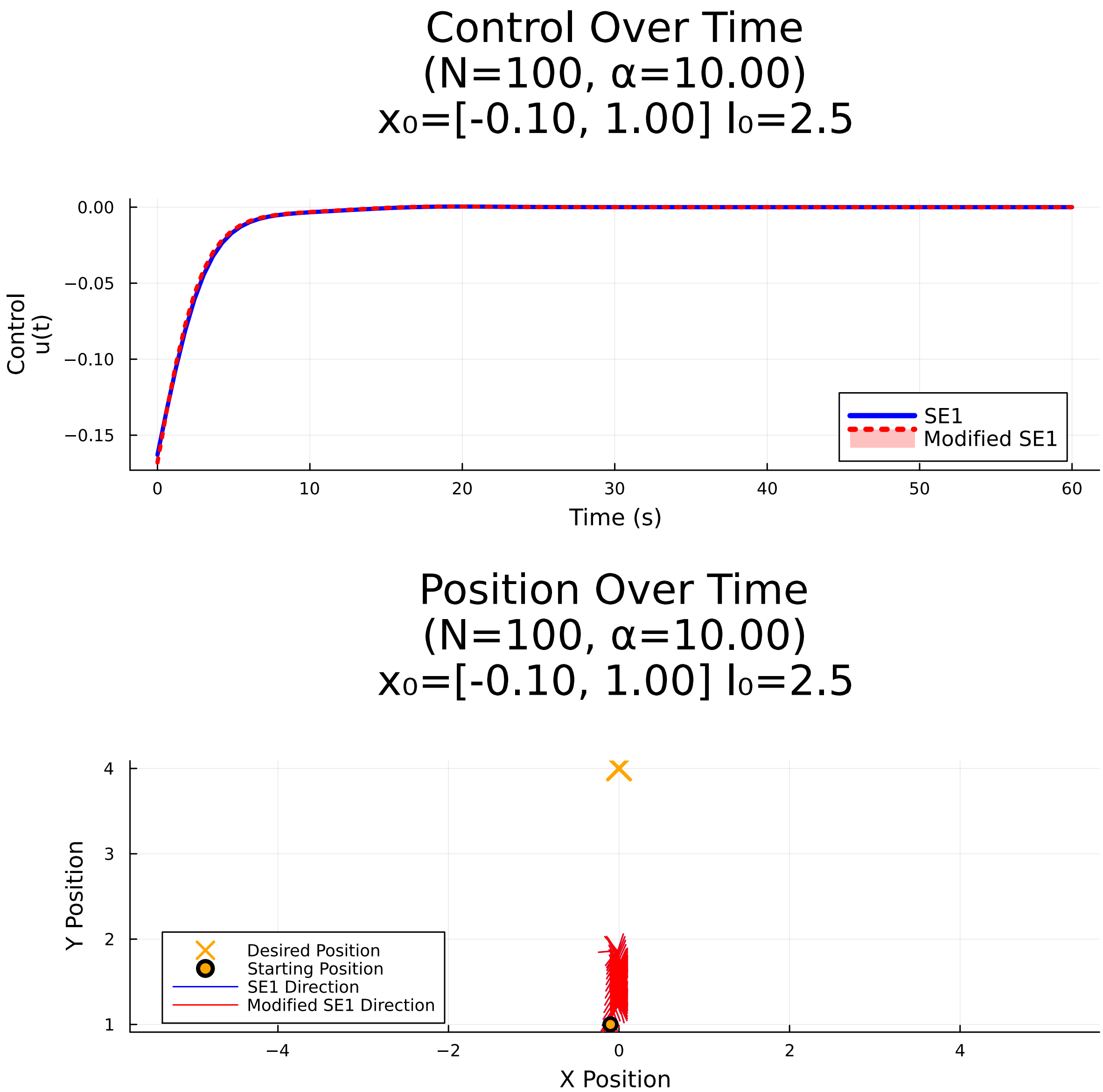


Figure 29: Comparison of SE1 and Modified SE1 for $\alpha = 10.0$, $N = 100$, $l_0 = 2.5$, and $\boldsymbol{x}_0 = [-0.1, 1.0]$

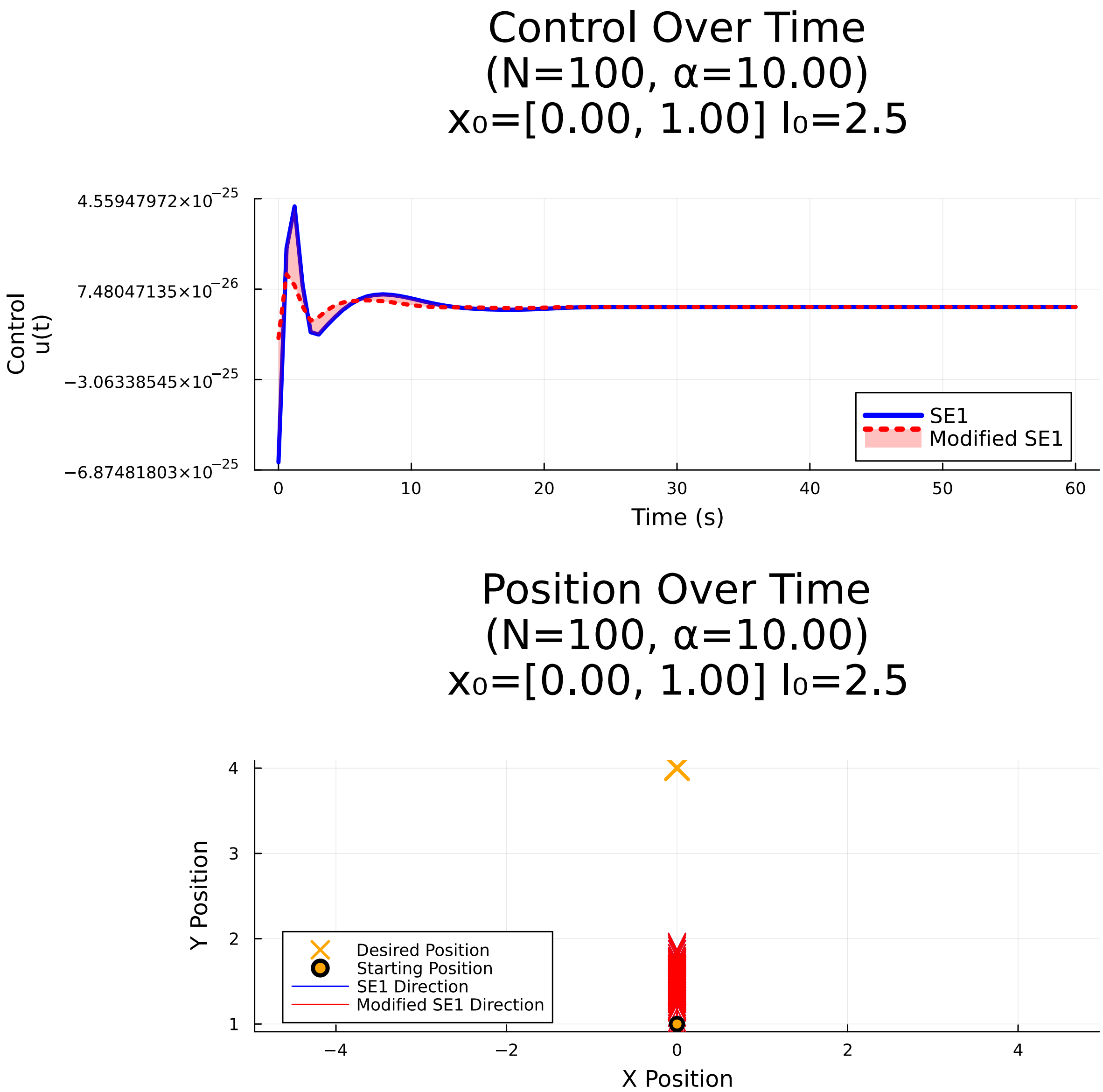


Figure 30: Comparison of SE1 and Modified SE1 for $\alpha = 10.0$, $N = 100$, $l_0 = 2.5$, and $\boldsymbol{x}_0 = [0.0, 1.0]$

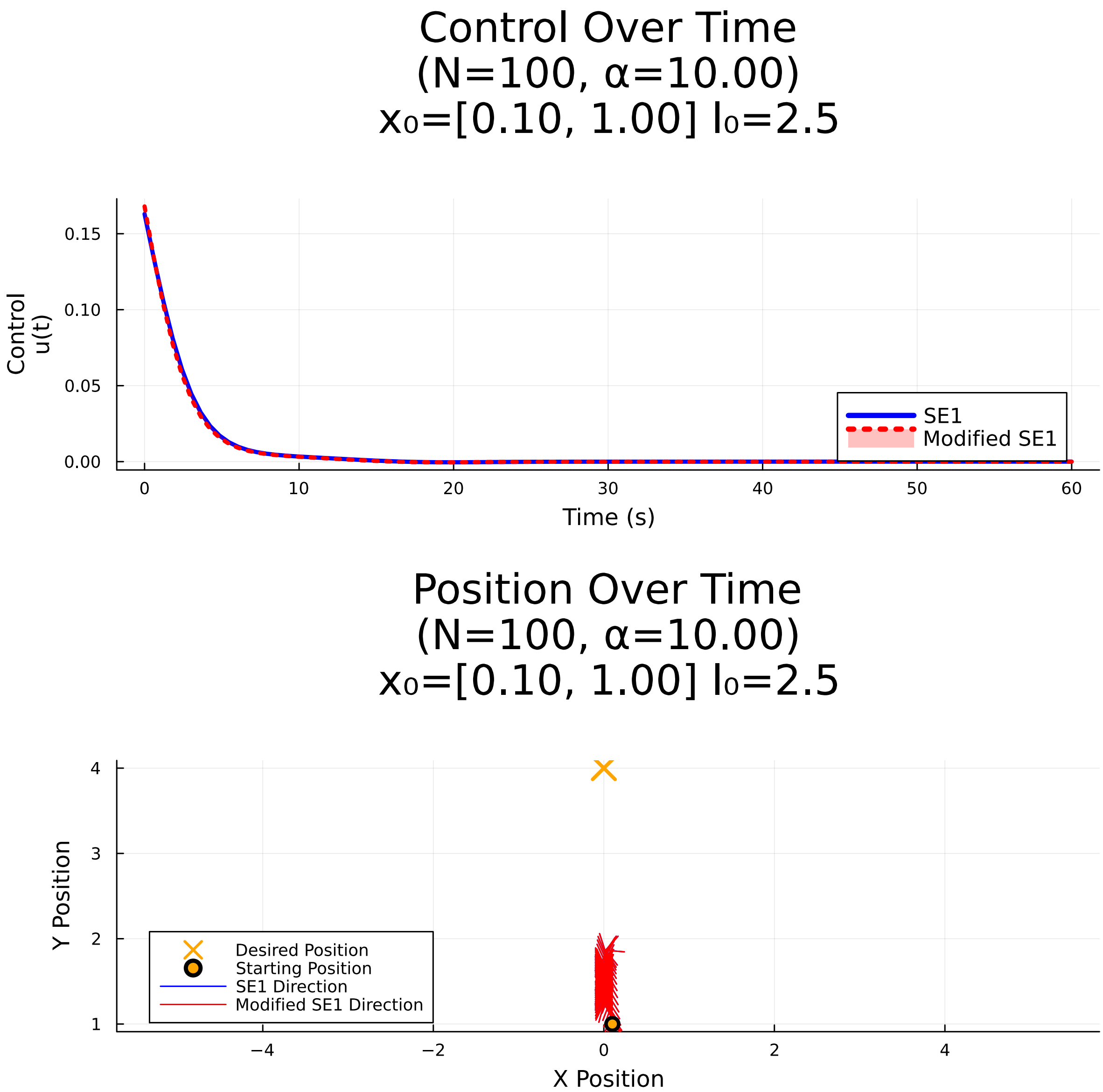


Figure 31: Comparison of SE1 and Modified SE1 for $\alpha = 10.0$, $N = 100$, $l_0 = 2.5$, and $\boldsymbol{x}_0 = [0.1, 1.0]$

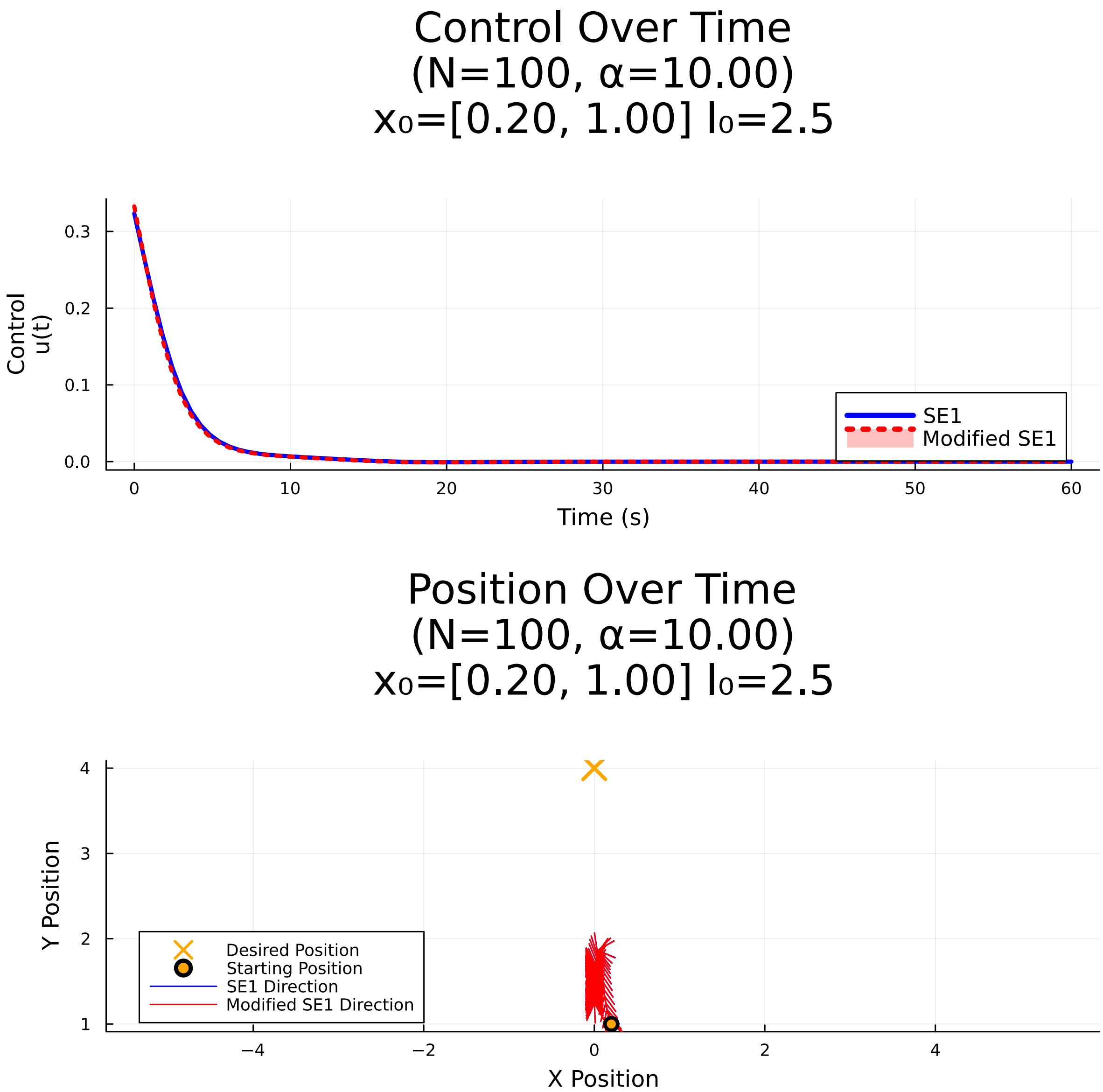


Figure 32: Comparison of SE1 and Modified SE1 for $\alpha = 10.0$, $N = 100$, $l_0 = 2.5$, and $\boldsymbol{x}_0 = [0.2, 1.0]$

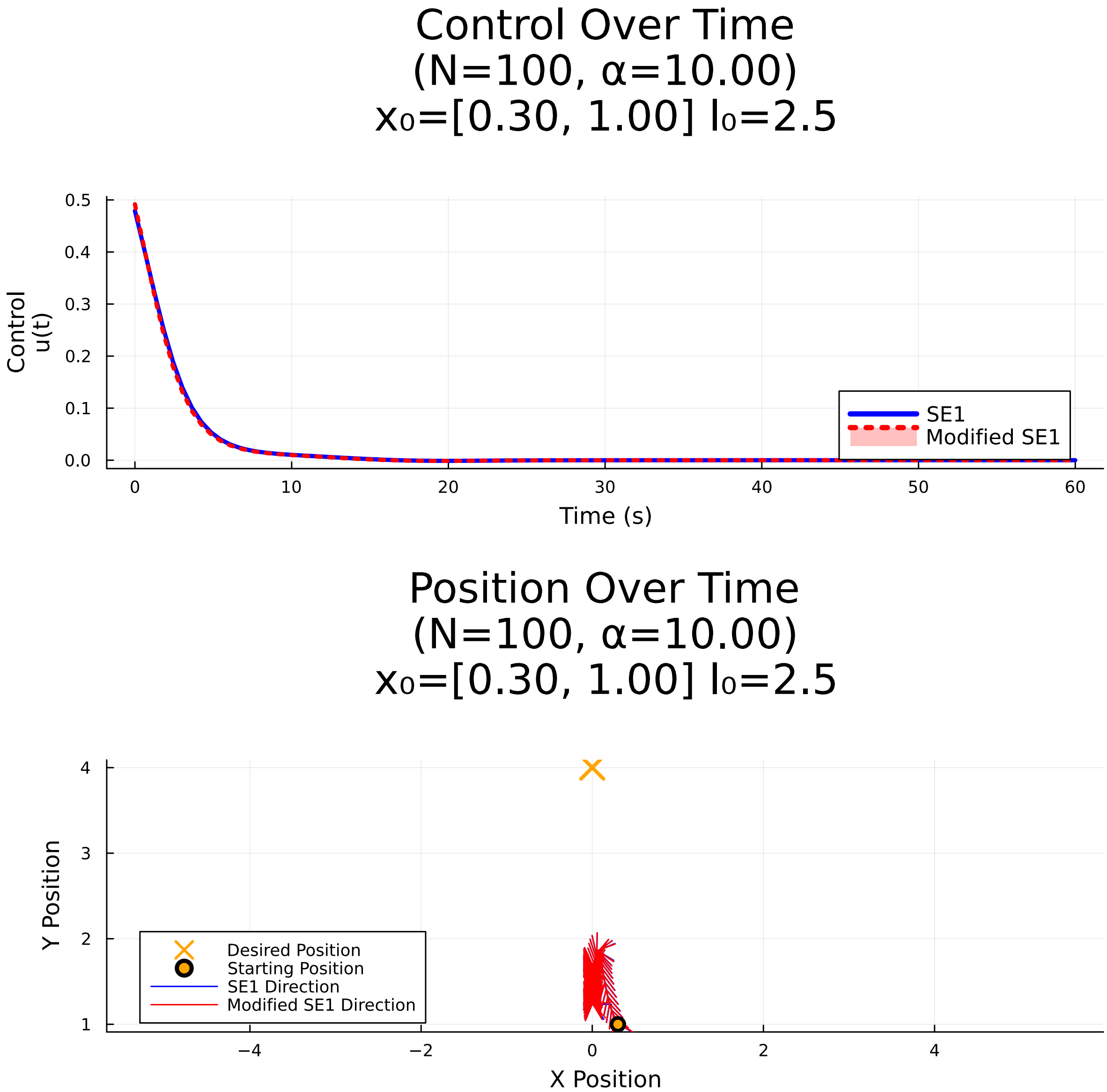


Figure 33: Comparison of SE1 and Modified SE1 for $\alpha = 10.0$, $N = 100$, $l_0 = 2.5$, and $\boldsymbol{x}_0 = [0.3, 1.0]$

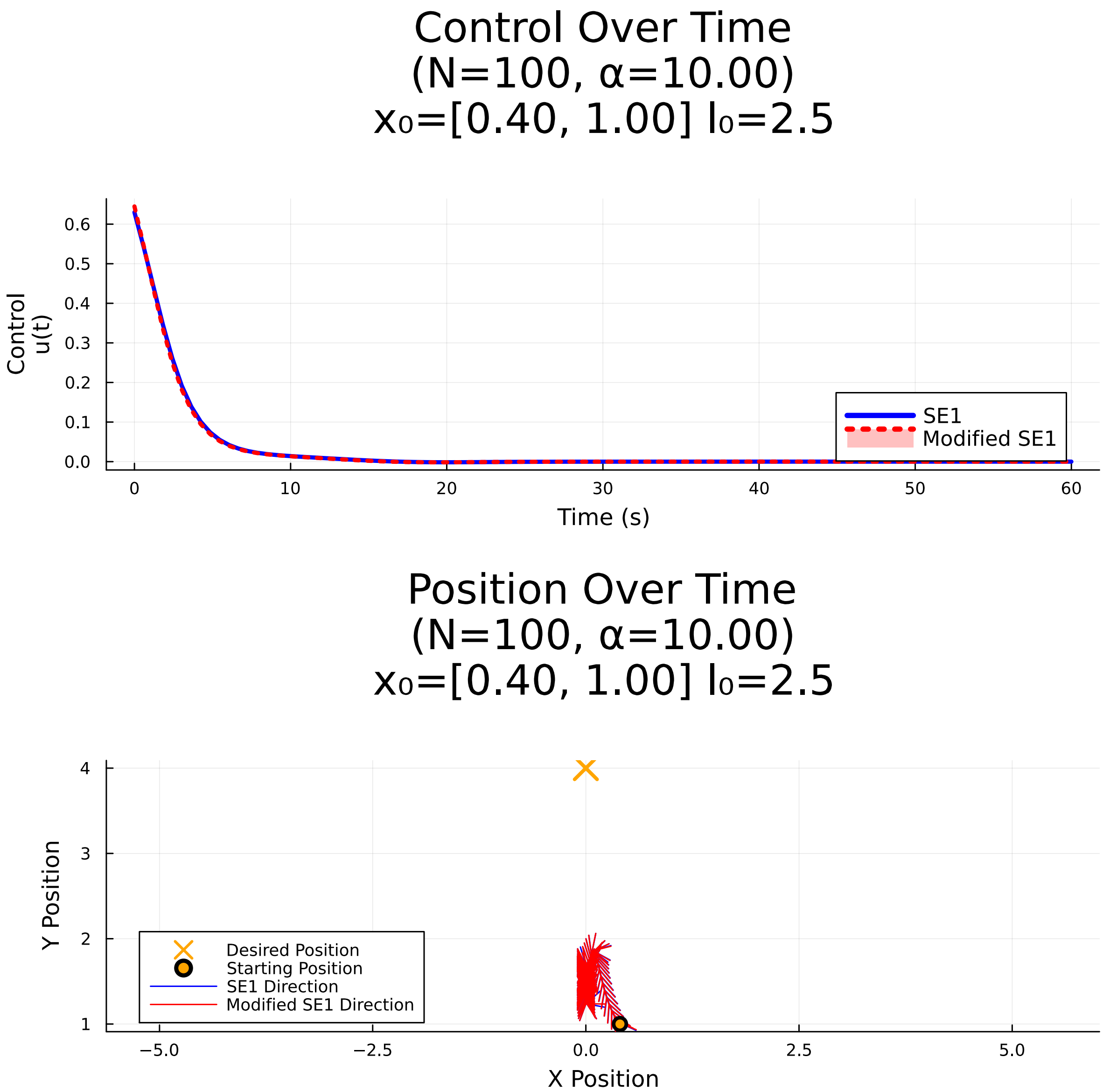


Figure 34: Comparison of SE1 and Modified SE1 for $\alpha = 10.0$, $N = 100$, $l_0 = 2.5$, and $\boldsymbol{x}_0 = [0.4, 1.0]$

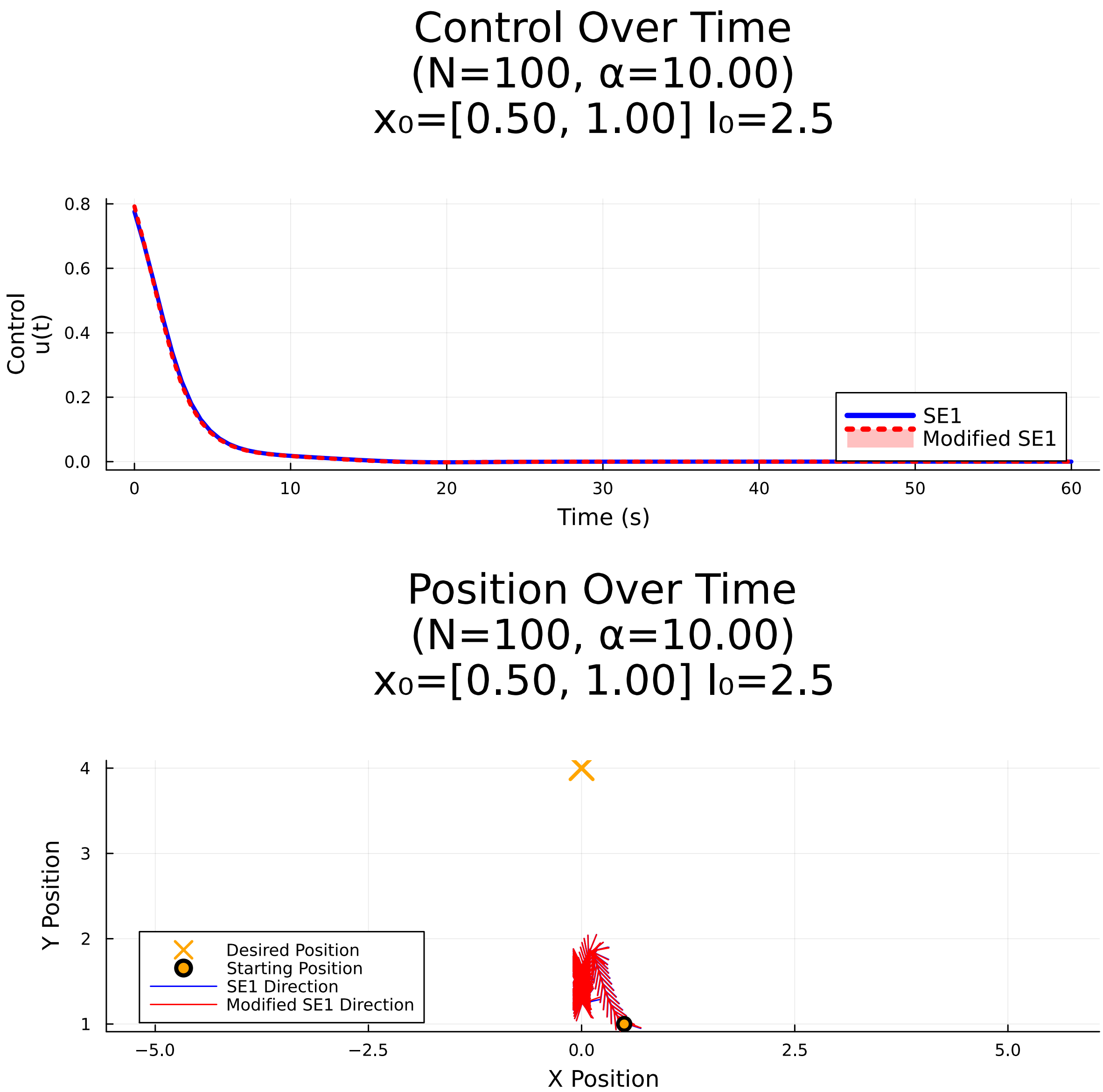


Figure 35: Comparison of SE1 and Modified SE1 for $\alpha = 10.0$, $N = 100$, $l_0 = 2.5$, and $\boldsymbol{x}_0 = [0.5, 1.0]$

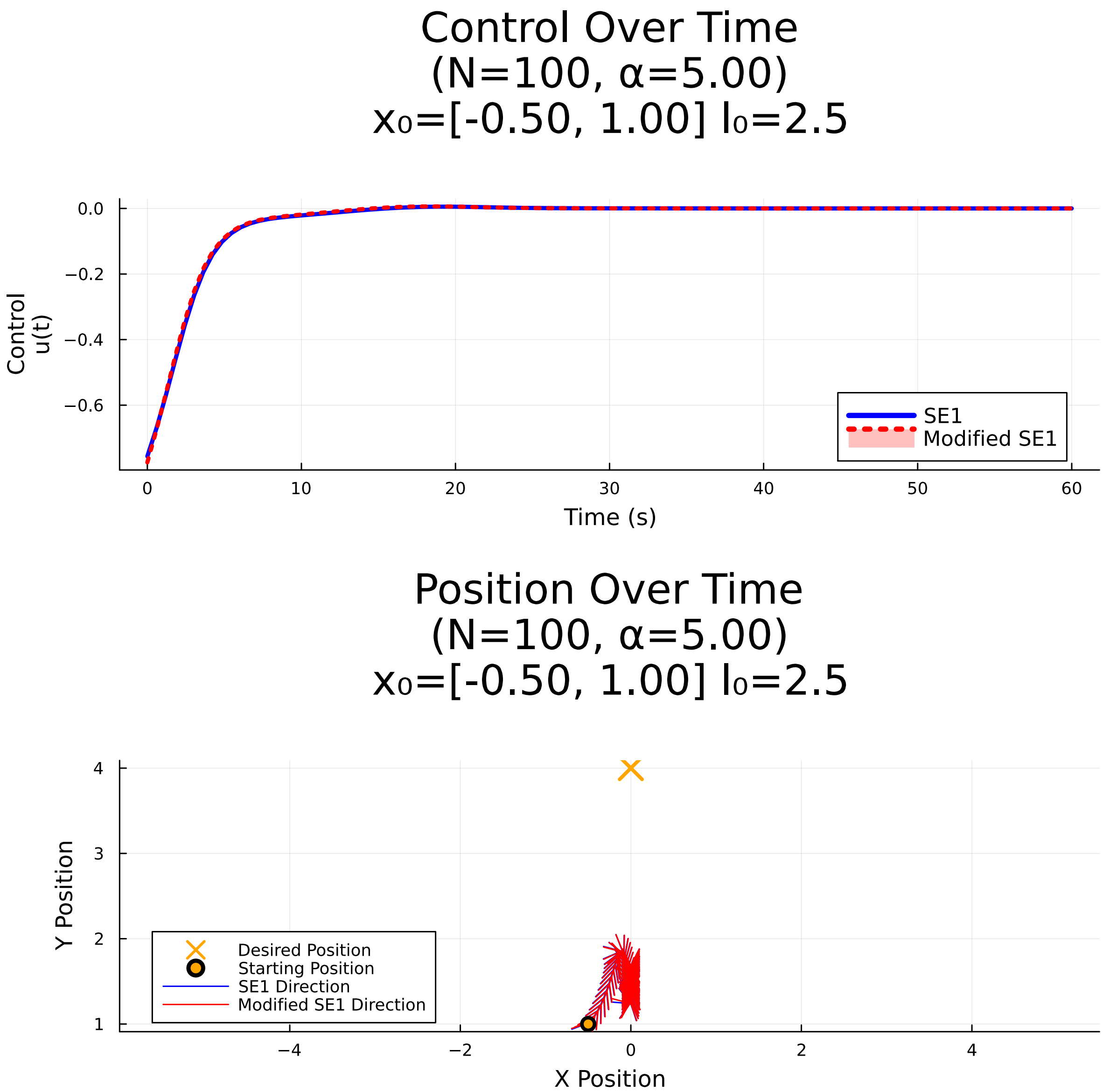


Figure 36: Comparison of SE1 and Modified SE1 for $\alpha = 5.0$, $N = 100$, $l_0 = 2.5$, and $\boldsymbol{x}_0 = [-0.5, 1.0]$

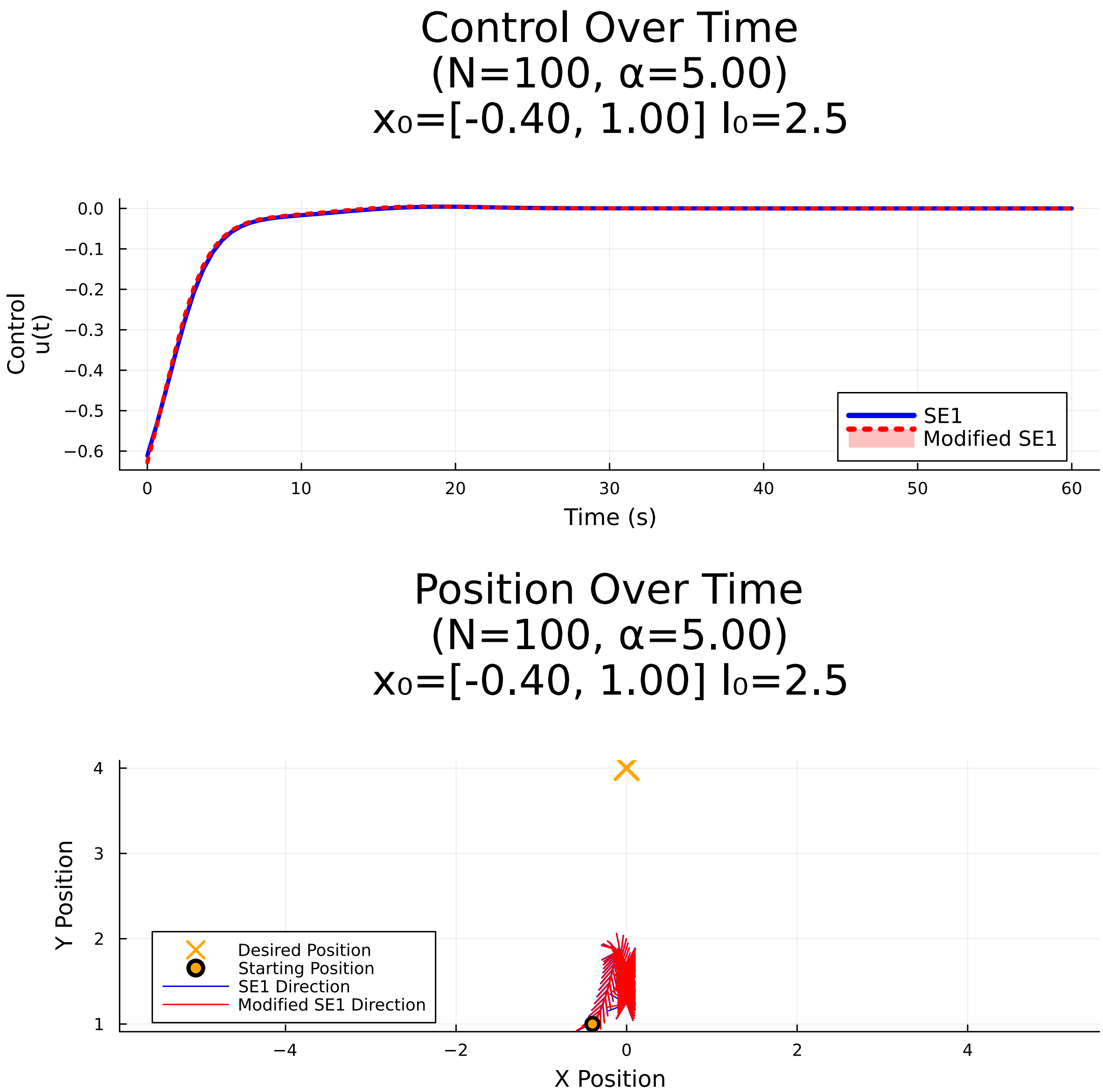


Figure 37: Comparison of SE1 and Modified SE1 for $\alpha = 5.0$, $N = 100$, $l_0 = 2.5$, and $\boldsymbol{x}_0 = [-0.4, 1.0]$

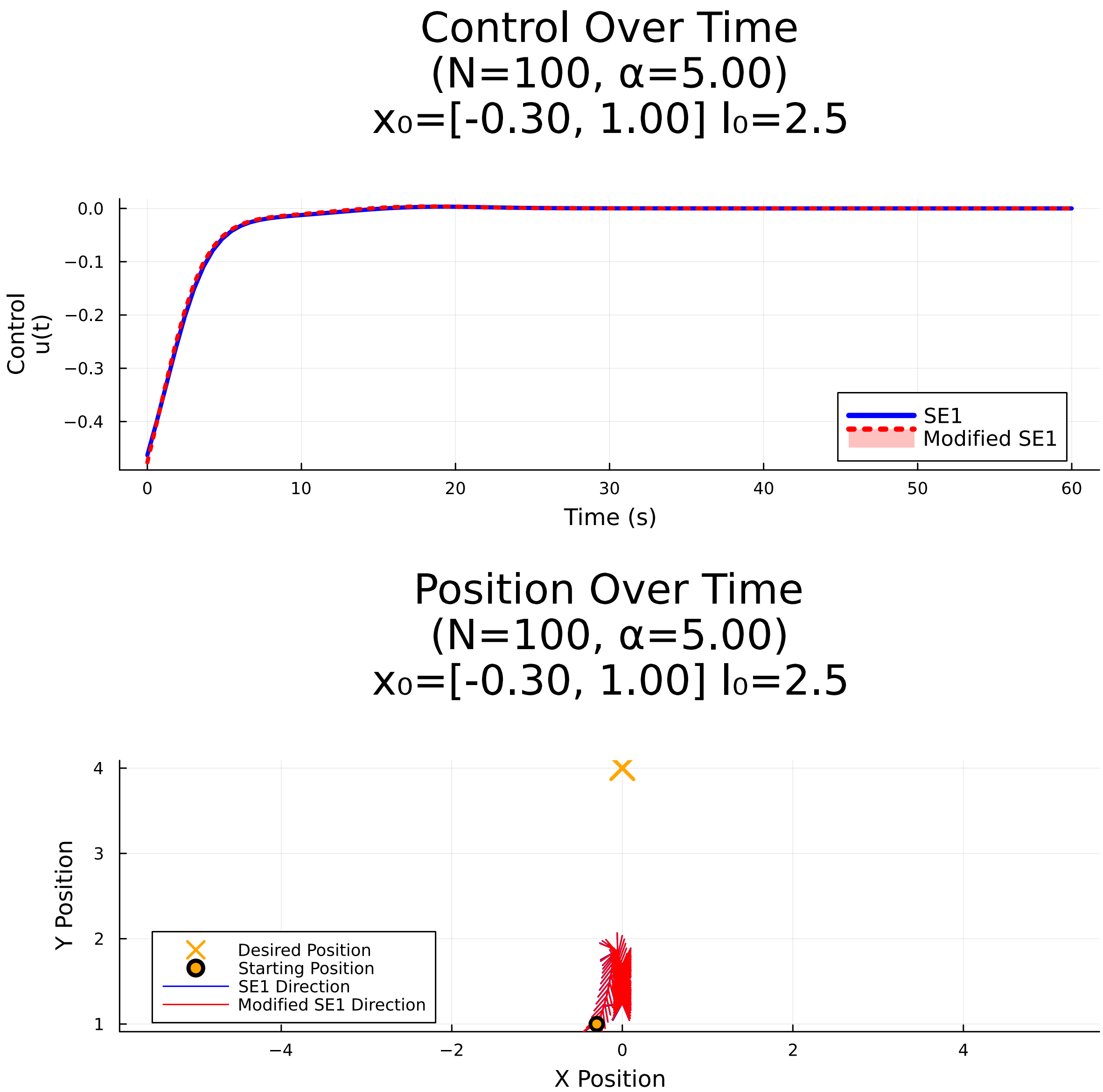


Figure 38: Comparison of SE1 and Modified SE1 for $\alpha = 5.0$, $N = 100$, $l_0 = 2.5$, and $\boldsymbol{x}_0 = [-0.3, 1.0]$

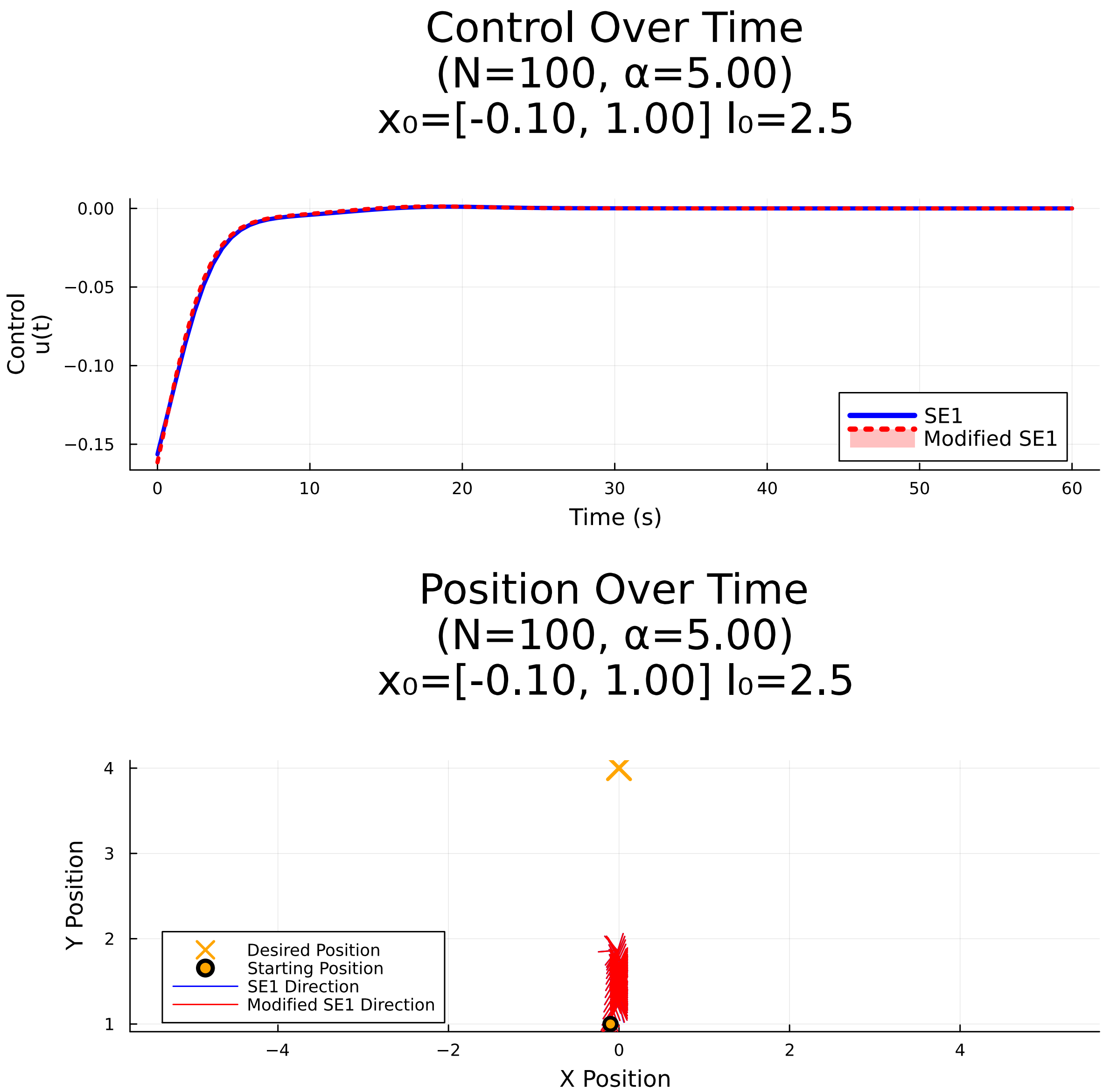


Figure 39: Comparison of SE1 and Modified SE1 for $\alpha = 5.0$, $N = 100$, $l_0 = 2.5$, and $\boldsymbol{x}_0 = [-0.1, 1.0]$

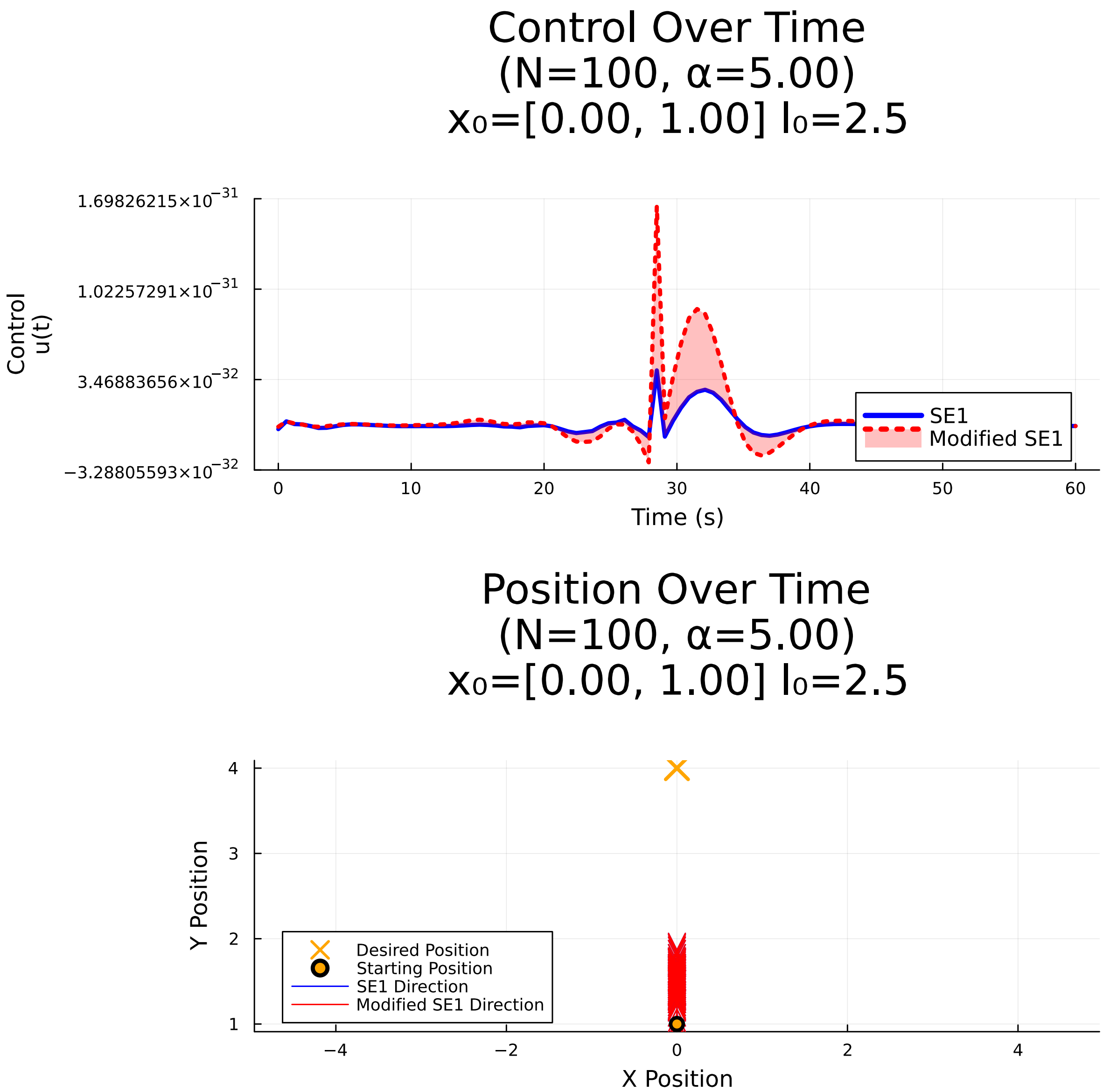


Figure 40: Comparison of SE1 and Modified SE1 for $\alpha = 5.0$, $N = 100$, $l_0 = 2.5$, and $\boldsymbol{x}_0 = [0.0, 1.0]$

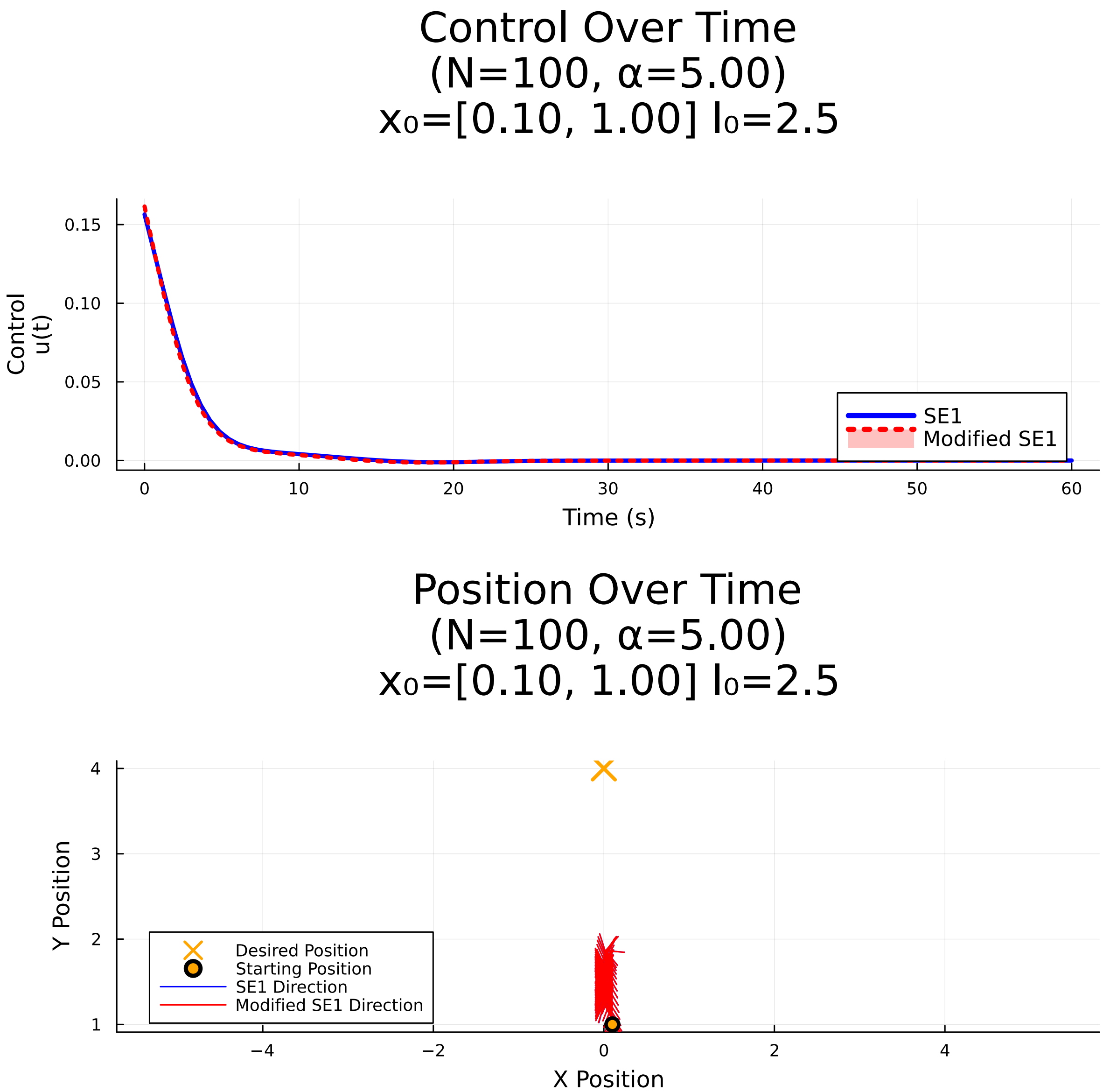


Figure 41: Comparison of SE1 and Modified SE1 for $\alpha = 5.0$, $N = 100$, $l_0 = 2.5$, and $\boldsymbol{x}_0 = [0.1, 1.0]$

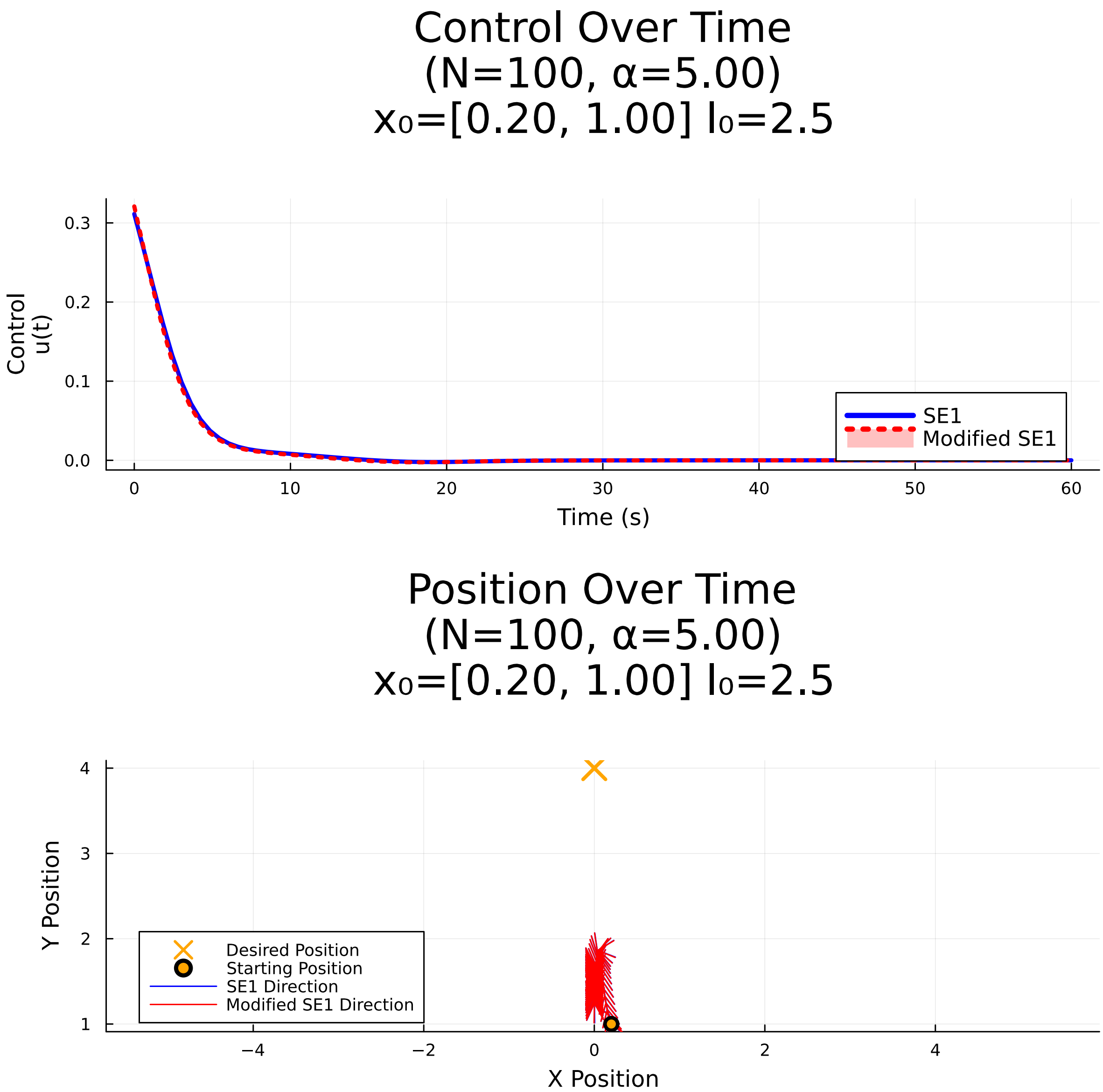


Figure 42: Comparison of SE1 and Modified SE1 for $\alpha = 5.0$, $N = 100$, $l_0 = 2.5$, and $\boldsymbol{x}_0 = [0.2, 1.0]$

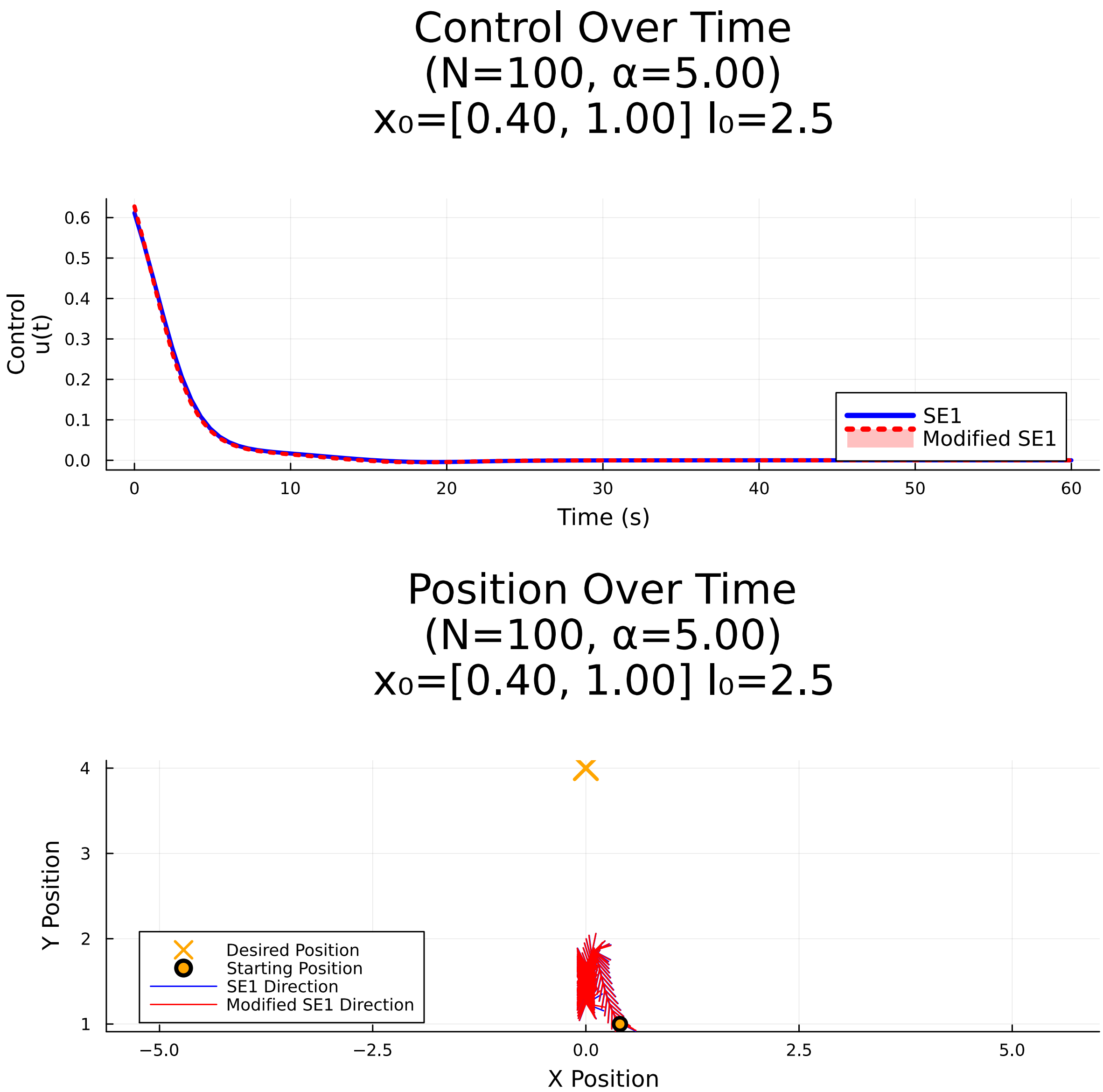


Figure 43: Comparison of SE1 and Modified SE1 for $\alpha = 5.0$, $N = 100$, $l_0 = 2.5$, and $\boldsymbol{x}_0 = [0.4, 1.0]$

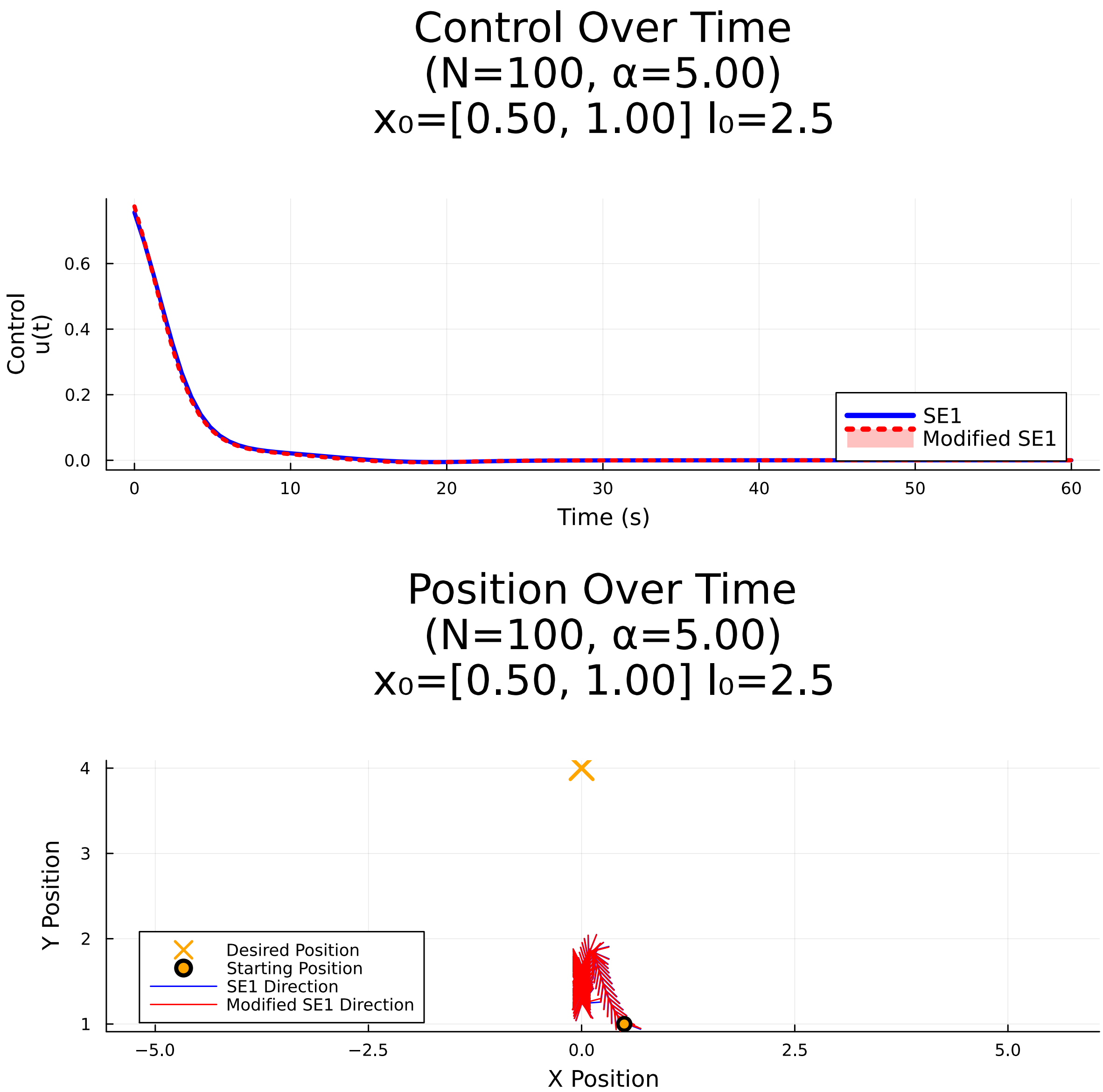


Figure 44: Comparison of SE1 and Modified SE1 for $\alpha = 5.0$, $N = 100$, $l_0 = 2.5$, and $\boldsymbol{x}_0 = [0.5, 1.0]$

Temporary page!

LaTeX was unable to guess the total number of pages correctly. As there was some unprocessed data that should have been added to the final page this extra page has been added to receive it.

If you rerun the document (without altering it) this surplus page will go away, because LaTeX now knows how many pages to expect for this document.